

Adaptive Pedagogical Systems with Invariant Kernels: A Formal Theory of Genre Persistence with Application to Medieval Latin Bestiaries

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Abstract

Why do some pedagogical genres survive for a millennium while others vanish within decades? This paper proposes a formal theory of adaptive pedagogical systems that addresses this question through a unified mathematical framework. We model any persistent didactic genre as a configuration $\mathcal{K} + \{M_i\}$, where $\mathcal{K}$ denotes an invariant structural kernel and $\{M_i\}$ a set of modular adaptive operations. Three fundamental operations are identified: *compression* $\mathcal{C}$ (which reduces informational volume while preserving semantic content), *expansion* $\mathcal{E}$ (which multiplies the interpretive range of symbolic nodes), and *correction* $\mathcal{R}$ (which repairs topological deficits in the didactic corpus through external import). We introduce a formal complexity scale C_1 – C_4 measuring the interpretive entropy of a pedagogical text, and prove that genre persistence requires the simultaneous satisfaction of three structural conditions: closure, boundary maintenance, and resilience—the *Persistence Triad*. The theory is tested against a detailed case study: the medieval Latin bestiary tradition (2nd–16th centuries), whose 150 surviving manuscripts preserve an invariant kernel *species*→*descriptio*→*moralizatio* while deploying all three adaptive operations across distinct institutional contexts. We further develop a cognitive-architectural account of why zoomorphic symbolism constitutes an optimal carrier for adaptive pedagogical content, grounded in the Structural Distortion Principle. A dedicated section explores the implications of dual-system formalization for understanding how pedagogical technologies mediate between ontologically asymmetric value regimes. The paper contributes to systems science, medieval studies, cognitive science, and the theory of knowledge transmission.

Keywords: adaptive systems; pedagogical technology; genre persistence; structural invariance; medieval bestiaries; *distinctiones*; mnemonic systems; information density; complexity scaling; dual-system formalization; knowledge transmission; cognitive architecture

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1 Introduction

The survival of pedagogical genres across centuries presents a problem that is simultaneously historical, cognitive, and structural. Some didactic formats—the fable, the catechism, the bestiary, the proverb collection—persist for periods that vastly exceed the lifespan of the civilizations that produced them. Others, despite institutional support and wide dissemination, decay within generations. What structural properties distinguish persistent pedagogical systems from transient ones?

This question has been addressed piecemeal across several disciplines. Historians of education have documented the longevity of specific genres [1, 2]. Cognitive scientists have identified the mnemonic properties that favor certain formats over others [3, 4]. Information theorists have modeled the conditions under which signals survive repeated transmission [5, 6]. Systems theorists have formalized the conditions for structural persistence in dynamic environments [7, 8]. Yet no unified framework has been proposed that integrates these perspectives into a single formal theory of pedagogical genre persistence.

This paper proposes such a framework. We model any persistent didactic genre as a configuration $\mathcal{K} + \{M_i\}$, where $\mathcal{K}$ is an invariant structural kernel—the minimal repeating unit that defines the genre’s identity—and $\{M_i\}$ is a set of modular adaptive operations that compilers, editors, and users deploy to retool the genre for changing audiences and institutional contexts. Three fundamental operations are identified:

- **Compression $\mathcal{C}$:** the reduction of a didactic module’s informational volume while preserving its semantic core (e.g., the condensation of a prose chapter into a mnemonic verse).
- **Expansion $\mathcal{E}$:** the multiplication of a didactic module’s interpretive range, transforming a univocal symbolic node into a multi-valued one (e.g., the enrichment of a single moralization with a systematic multi-level *distinctio*).
- **Correction $\mathcal{R}$:** the repair of topological deficits in the didactic corpus through the import of modules from external sources (e.g., the insertion of moralized chapters from one textual family into another to equalize the density of moral content across sections).

We further introduce a formal complexity scale C_1 – C_4 that measures the interpretive entropy of a pedagogical text, ranging from univocal moralization (C_1) through compressed retention (C_2) and multiple exegesis (C_3) to compensatory reconfiguration (C_4). The central theorem of the paper establishes that a didactic genre persists if and only if its kernel $\mathcal{K}$ satisfies three conditions—closure, boundary maintenance, and resilience—which we term the *Persistence Triad*, following the formal framework established in [47].

The theory is tested against a detailed empirical case study: the medieval Latin bestiary tradition, which survives in approximately 150 manuscripts spanning the 2nd through 16th centuries. Drawing principally on a body of research by Dr. Ilya Dines (2008–2025) that has identified three distinct mechanisms of textual enrichment in English bestiaries—mnemonic verse summaries, *distinctiones* (systematic multi-level biblical interpretations), and cross-family editorial borrowing—we demonstrate that the bestiary’s longevity is explained by exactly the formal apparatus developed here. The bestiary preserves an invariant kernel $\mathcal{K}$ (*species* $\rightarrow$ *descriptio* $\rightarrow$ *moralizatio*) while deploying all three adaptive operations across distinct institutional contexts: monastic schools ($\mathcal{C}$), cathedral schools and mendicant networks ($\mathcal{E}$), and scriptoria with access to multiple exemplars ($\mathcal{R}$).

Beyond the bestiary, we develop two further theoretical contributions. First, a cognitive-architectural analysis demonstrates why zoomorphic symbolism constitutes an optimal carrier for

adaptive pedagogical content, grounded in the Structural Distortion Principle [48]: every act of perception is simultaneously an act of non-perception, and the animal image optimizes this constraint by maximizing moral signal while minimizing cognitive distortion. Second, a dedicated section explores the implications of dual-system formalization for understanding how pedagogical technologies mediate between ontologically asymmetric value regimes—the world as experienced and the transcendent order as taught.

The methodological approach follows the Tree-Top Meta-Method [49]: generalize formally until generalization is no longer possible, then derive domain-specific consequences from what remains structurally true. The bestiary is not the object of the theory; it is one of its instantiations.

Relationship to companion work. A companion article, “The Bestiary as Pedagogical Technology: Mnemonic, Homiletic, and Editorial Strategies in English Latin Bestiaries (12th–14th Centuries),” presents the same three-strategy framework in a humanistic register, without the formal apparatus developed here. That article is addressed to medievalists and historians of education; the present paper is addressed to systems scientists, cognitive scientists, and information theorists. The two papers share empirical material (drawn from Dines’s research) but differ in method, audience, and contribution. The companion article synthesizes; the present paper formalizes and generalizes. Neither depends on the other for its central claims.

2 Literature Review

The theory developed in this paper draws on and contributes to five distinct research traditions. We survey each in turn, identifying both the resources they provide and the gaps that motivate the present synthesis.

2.1 Medieval Bestiary Scholarship

The genealogical classification of Latin bestiaries was established by James [9] and refined by McCulloch [10], Clark [11], and Dines [12]. These scholars organized approximately 150 known manuscripts into textual families—BIs (i.e., the *Physiologus* version B augmented with material from Isidore of Seville), *Dicta Chrysostomi*, H Family, Second Family (subfamilies A through D), Transitional, and Third Family—based on chapter content, sequence, and textual affiliation. Baxter [13] pioneered the functional approach, using codicological evidence (wear patterns, annotations, library catalogue entries) to reconstruct how bestiaries were actually used. Dines’s subsequent studies identified three mechanisms of pedagogical enrichment: mnemonic verses [14, 15], *distinctiones* integration [16], and cross-family editorial borrowing [17]. The present paper synthesizes these findings into a formal framework and situates them within a general theory of adaptive systems.

2.2 Medieval Pedagogy and Preaching Aids

The bestiary existed within a rich ecosystem of 13th-century pedagogical tools. The *artes praedicandi* (treatises on sermon composition) provided theoretical frameworks for preaching [28, 29]. Exempla collections by Jacques de Vitry, Caesarius of Heisterbach, and others supplied illustrative narratives [30, 31]. *Distinctiones* collections—systematic multi-level interpretations of biblical terms—offered preachers flexible hermeneutical resources [32, 33]. Carruthers [1] demonstrated that medieval mnemonic practice was not incidental but central to intellectual culture, with versification serving as a fundamental cognitive tool for organizing memory. Goering [34] documented the institutional context of William de Montibus’s Lincoln school, where *distinctiones* were produced

specifically for parish priest education. The present paper situates the bestiary within this ecology, arguing that its adaptive modifications represent responses to the same pressures that generated the broader preaching aid revolution.

2.3 Systems Theory and Viability

The formal apparatus of this paper draws on viability theory as developed by Aubin [7, 35], which models the conditions under which a system’s state trajectory remains within a constraint set (the “viability kernel”). Ashby’s [8] requisite variety theorem establishes that a system’s regulatory capacity must match the variety of disturbances it faces. Bertalanffy [36] proposed general systems theory as a framework for identifying structural isomorphisms across domains. More recently, Meadows [37] and Holland [38] have emphasized the role of modularity, feedback, and hierarchical organization in system persistence. The concept of structural persistence through a triad of closure, boundary maintenance, and resilience was formalized in [47], providing the mathematical foundation for the present work.

2.4 Cognitive Science of Symbolic Processing

The dual-coding theory of Paivio [3] establishes that information encoded in both verbal and imagistic formats is more readily retained than information in either format alone. This has direct implications for the bestiary, which combines verbal (narrative) and visual (illustration, mnemonic image) channels. Lakoff and Johnson [39] demonstrated that metaphorical mapping is not merely a literary device but a fundamental cognitive operation. Barsalou [40] extended this with the theory of perceptual symbol systems, showing that conceptual processing is grounded in modal simulations. The medieval bestiary, with its vivid zoological imagery linked to abstract moral concepts, exploits precisely this grounding mechanism. The Structural Distortion Principle [48] provides a formal account of why any adaptive system under finite processing resources must compress, distort, and selectively attend to its inputs—a constraint that the animal image satisfies optimally.

2.5 Information Theory and Knowledge Transmission

Shannon’s [5] foundational framework models information transmission through noisy channels, establishing the conditions under which messages survive degradation. Lossy compression [6] formalizes the trade-off between fidelity and efficiency that the mnemonic verse tradition instantiates. Sperber [41] applied epidemiological models to cultural transmission, arguing that cultural representations survive not by faithful copying but by achieving a balance between stability and transformation. Boyer [42] identified the cognitive properties that make certain cultural representations “catchy”—minimally counterintuitive, attention-grabbing, and inference-rich. The bestiary’s zoological fables satisfy all three criteria. The present paper extends these frameworks by showing that the bestiary’s persistence is not merely a function of its cognitive catchiness but of its formal structural properties: an invariant kernel that provides stability, and modular operations that provide adaptability.

3 The Scholarship of Dr. Ilya Dines: Empirical Foundation of This Study

The formal apparatus developed in this paper would be vacuous without an empirical foundation of sufficient depth and precision. That foundation is provided by the body of research produced

over two decades by Dr. Ilya Dines (Hebrew University of Jerusalem), whose work constitutes the most sustained and detailed investigation of the textual mechanics of the English Latin bestiary tradition conducted to date. This section surveys Dr. Dines’s principal contributions and identifies the specific empirical discoveries that the present formalization rests upon.

3.1 Scope and Method

Dr. Dines’s research program is distinguished by its combination of philological precision and conceptual ambition. Where previous scholars (James 1928, McCulloch 1962, Clark 2006, Baxter 1998) approached the bestiary primarily through genealogical classification (which manuscript derives from which) or through art-historical analysis (how illustrations relate to text), Dr. Dines has consistently asked a different question: *how did bestiary compilers actively modify their texts, and why?* This question—which shifts the focus from passive transmission to active editorial intervention—is the question that the present paper formalizes.

Dr. Dines’s method is codicologically grounded: every claim rests on direct examination of manuscripts, comparison of variant readings across witnesses, and attention to the physical evidence of the codex (script, layout, decoration, marginal annotation). His research has taken him to major libraries across Europe and America, including the British Library, the Bodleian Library, Cambridge University Library, the Bibliothèque nationale de France, the Library of Congress, and dozens of smaller collections.

3.2 Principal Discoveries

3.2.1 The Mnemonic Verse Tradition (2010, forthcoming)

In his 2010 article in *Reinardus* [14] and a forthcoming comprehensive study [15], Dr. Dines identified and analyzed the mnemonic verses in Oxford, Bodleian Library MS Bodley 533—the most substantial collection of such verses in any Latin prose bestiary. He demonstrated that the verses were copied from a pre-existing source (not composed for Bodley 533), as evidenced by copying errors and metrical irregularities. He further discovered a closely related set of verses in Cambridge, University Library MS Oo.vii.48, proving that the verse tradition circulated independently of the prose bestiary context. This discovery is formalized in the present paper as the compression operator $\mathcal{C}$.

3.2.2 The Mnemonic Verses in Cambridge (2020)

In a 2020 article in *Reinardus* [18], Dr. Dines published and analyzed a previously essentially unknown fragment in Cambridge University Library MS Oo.vii.48, containing a collection of thirteenth-century mnemonic verses concerning animals and birds. This independent witness confirmed the existence of a verse bestiary tradition circulating outside the prose bestiary context, strengthening the evidence for the compression operator as a systematic pedagogical strategy.

3.2.3 The Integration of *Distinctiones* (2024)

In a 2024 study in *Progressus* [16], Dr. Dines traced the integration of *distinctiones*—systematic multi-level biblical interpretations—into English bestiaries. He identified the principal source as the *distinctiones* tradition associated with William de Montibus (c. 1140–1213), Chancellor of Lincoln Cathedral. He documented how Third Family bestiaries systematically replaced Second Family

moralizations with *distinctiones*-based material, transforming the bestiary from a univocal moralization system (C_1) to a multi-level hermeneutical resource (C_3). This discovery is formalized as the expansion operator $\mathcal{E}$.

3.2.4 Cross-Family Borrowing in KK 4.25 (2025)

In a 2025 study in *Progressus* [17], Dr. Dines analyzed the most editorially complex manuscript in the bestiary tradition: Cambridge, University Library MS KK 4.25. He demonstrated that the compiler of this Third Family manuscript had access to a Second Family bestiary and systematically imported chapters from it to fill moralization gaps in the birds section—raising the moralization density from 0.25 to 0.69. He further discovered evidence of reverse borrowing: the Second Family manuscript London, BL MS Harley 3244 (c. 1240) was influenced by KK 4.25, violating the expected genealogical direction of transmission. These discoveries are formalized as the correction operator $\mathcal{R}$.

3.2.5 The Fox Moralization (2024)

In a 2024 article in *Reinardus* [19], Dr. Dines investigated an obscure moralization found in 13th-century bestiaries in which the fox imitates a dog—an interpretation absent from the *Physiologus* and most earlier sources. This study exemplifies Dr. Dines’s attention to micro-level textual innovations and their pedagogical significance: a single moralization variant, traced across multiple manuscripts, reveals the active editorial processes through which bestiary compilers adapted their material.

3.2.6 The Transitional and H Family Problems (2008, 2012)

In his doctoral dissertation and subsequent publications [12, 20], Dr. Dines re-examined the genealogical classification of bestiary manuscripts, particularly the problematic Transitional and H Family groups. He demonstrated that the Transitional Family, as defined by James, includes manuscripts of heterogeneous origin that do not constitute a genuine textual family, and he proposed a revised understanding of the H Family’s origins and influence. These findings clarified the stemmatic relationships that the present paper presupposes.

3.2.7 Image-Text Relations (2014, 2016, 2021)

In a series of studies, Dr. Dines investigated the relationship between text and illustration in medieval bestiaries. His 2014 article in the *Journal of the British Archaeological Association* demonstrated for the first time a proven case of iconographical dependence between two bestiary manuscripts (Bodley 602 and Douce 88)—a case in which the illuminator of one manuscript directly copied illustrations from another. His 2016 study on long rubrics and captions explored the liminal space between text and image, and his 2021 article in *Revista Medievalista* investigated the production process from text to image. These findings inform the present paper’s treatment of manuscript materiality (Section 9.5).

3.2.8 Bestiaries in Institutional Contexts (2014, 2019)

Dr. Dines’s 2014 study of the bestiary in British Library Royal MS 2 C. XII analyzed the marginal glosses in a working educational manuscript, demonstrating how the bestiary was actually used in medieval instruction. His 2019 chapter for the Getty Museum’s *Book of Beasts* examined the

function of Latin bestiaries in medieval miscellanies—composite volumes in which the bestiary was bound alongside other texts, revealing the institutional context in which it was read and taught.

3.2.9 Cartographic and Cross-Disciplinary Work

Dr. Dines’s work extends beyond the bestiary text itself. His study (with Chet Van Duzer) of the mappamundi in Cambridge, MS Fitzwilliam 254 [21]—the only known world map in a bestiary context—demonstrated that bestiary manuscripts could serve as carriers for cartographic and cosmological knowledge, a finding relevant to the present paper’s discussion of cosmological substrates (Section 16).

3.3 The Cumulative Contribution

Dr. Dines’s research has transformed the study of the English Latin bestiary from a primarily genealogical enterprise into an investigation of active pedagogical adaptation. His three principal discoveries—mnemonic versification, *distinctiones* integration, and cross-family editorial borrowing—are not isolated observations but components of a coherent picture: medieval bestiary compilers were not passive copyists but active editors who systematically modified their texts to serve pedagogical purposes. Without this picture, the formal apparatus of the present paper would have no empirical content.

It should be emphasized that the formalization proposed here is the present author’s contribution, not Dr. Dines’s. Dr. Dines has not endorsed the mathematical framework, and any errors in the formal redescription of his findings are solely the present author’s responsibility. The empirical discoveries, however, are entirely his, and the present paper is, at its foundation, a formal reading of his scholarship.

What it adds: This paper’s formal apparatus operates on empirical findings that were produced over two decades of sustained philological and codicological research. Without explicit documentation of the empirical base, the formal apparatus appears to float free of evidence. This section anchors every formal construct $(\mathcal{C}, \mathcal{E}, \mathcal{R}, C_1\text{--}C_4, \rho)$ to a specific published discovery by a specific scholar, making the relationship between formalization and evidence transparent and auditable.

Practical value: Readers who wish to evaluate the formal claims of this paper must be able to access the empirical evidence independently. This section provides a roadmap to Dr. Dines’s publications, organized by the formal constructs they support, enabling independent verification without requiring familiarity with the full bibliography of bestiary studies.

4 Formal Framework

This section develops the mathematical apparatus for modeling adaptive pedagogical systems. The framework is designed to be accessible to readers across disciplines; accordingly, every definition and proposition is accompanied by an explanation that clarifies its meaning and motivation.

4.1 The Structural Kernel

Definition 4.1 (Structural Kernel). A **structural kernel** $\mathcal{K}$ of a pedagogical genre is an ordered triple

$$\mathcal{K} = (S, D, \mu)$$

where S is a finite set of **symbolic nodes** (the subjects of instruction), $D : S \rightarrow \mathcal{T}$ is a **description function** mapping each node to a text in a description space $\mathcal{T}$, and $\mu : S \rightarrow \mathcal{M}$ is a **moralization function** mapping each node to an element of a moral-interpretive space $\mathcal{M}$.

The structural kernel captures the minimal repeating unit that defines a pedagogical genre’s identity. For the medieval bestiary, S is the set of animal species (lion, beaver, pelican, etc.), $D(s)$ is the zoological description of species s , and $\mu(s)$ is the moral or allegorical interpretation attached to s . The key property of $\mathcal{K}$ is that it is invariant across instances of the genre: a 12th-century bestiary and a 15th-century bestiary of the same family share the same kernel, even if their supplementary materials differ substantially.

The concept is not specific to bestiaries. For the Aesopian fable tradition, S would be the set of narrative scenarios, $D(s)$ the narrative itself, and $\mu(s)$ the appended moral. For a catechism, S is the set of doctrinal questions, $D(s)$ the explanation, and $\mu(s)$ the prescribed answer. The kernel is what survives when all supplementary material is stripped away.

What it adds: The triple (S, D, μ) separates three logically independent components that are typically conflated in literary analysis: the *inventory* of subjects (S), the *factual content* (D), and the *interpretive layer* (μ). This separation makes it possible to ask precise questions: Does a new edition change S (adding or removing subjects), D (rewriting descriptions), or μ (altering moralizations)? Without this decomposition, the question “what changed between two manuscripts?” has no formal answer.

Practical application: Any educational curriculum can be decomposed into (S, D, μ) : the set of topics taught, the factual content associated with each, and the evaluative or interpretive framework applied. Curriculum reform can then be classified: does it modify S (adding new subjects), D (updating content), or μ (changing pedagogical goals)? This enables systematic comparison across reform cycles and institutions.

Definition 4.2 (Moralization Density). For a subset $A \subseteq S$ of symbolic nodes (e.g., a section of a bestiary), the **moralization density** is

$$\rho(A) = \frac{|\{s \in A : \mu(s) \neq \emptyset\}|}{|A|}$$

where $\mu(s) = \emptyset$ indicates that node s lacks any moral interpretation.

This quantity measures the proportion of a genre’s symbolic nodes that carry explicit didactic content. It is central to understanding why certain sections of a bestiary were perceived as didactically deficient and subjected to corrective intervention.

A chapter is counted as moralized ($\mu(s) \neq \emptyset$) if it contains any passage that explicitly draws a spiritual, allegorical, tropological, or anagogical lesson from the descriptive material. This binary classification does not capture variation in depth—a single typological sentence counts equally with a full multi-level *distinctio*—but it suffices for the structural analysis that follows. Where the distinction between univocal and multi-level moralization matters, the interpretive entropy H (Definition 4.8) and the expanded moralization function μ^* (Definition 4.5) provide the necessary resolution.

What it adds: Moralization density converts a subjective editorial judgment (“this section feels thin”) into a countable, comparable quantity. When the KK 4.25 compiler perceived the Birds section as didactically inadequate, he was responding to a measurable deficit: $\rho = 0.27$ vs. $\rho = 0.67$ elsewhere. The formula makes explicit what editorial intuition left implicit.

Practical application: In modern textbook design, $\rho(A)$ can measure the proportion of sections that include worked examples, exercises, or applied case studies. A textbook editor reviewing a draft can compute ρ per chapter and identify sections where pedagogical reinforcement is below threshold—precisely the diagnostic role the medieval compiler performed by hand.

4.2 Adaptive Modules

Definition 4.3 (Adaptive Module). An **adaptive module** M is a function that transforms the kernel $\mathcal{K}$ or one of its components (S, D, μ) while preserving the kernel’s structural identity. Formally, $M : \mathcal{K} \rightarrow \mathcal{K}'$ such that $\mathcal{K}'$ remains recognizable as an instance of the same genre.

The requirement that $\mathcal{K}'$ remains “recognizable as an instance of the same genre” is deliberately informal at this stage. It captures the empirical observation that medieval bestiary compilers could add, modify, or rearrange supplementary material extensively, yet the resulting manuscripts were still identified as bestiaries—both by their medieval users and by modern scholars. Formally, this means that the transformation M preserves the ordered-triple structure (S, D, μ) and does not alter the fundamental mapping types, even if it modifies the content of individual mappings.

We identify three fundamental adaptive operations:

Definition 4.4 (Compression $\mathcal{C}$). The **compression operator** $\mathcal{C}$ maps a didactic module (a chapter or set of chapters) to a reduced-volume representation while preserving its semantic core:

$$\mathcal{C} : (D(s), \mu(s)) \mapsto v(s)$$

where $v(s)$ is a metrical summary (verse) satisfying $|v(s)| \ll |D(s) + \mu(s)|$ and $\text{sem}(v(s)) \approx \text{sem}(D(s), \mu(s))$, with $\text{sem}(\cdot)$ denoting semantic content and $|\cdot|$ denoting textual volume.

Compression trades detail for memorability. The mnemonic verses found in Oxford, Bodleian Library MS Bodley 533 are instances of $\mathcal{C}$: four-line Leonine hexameter summaries that condense entire prose chapters into metrically structured, phonetically linked verse. The operation is lossy (detail is sacrificed) but semantically faithful (the core moral-zoological association is preserved). In information-theoretic terms, $\mathcal{C}$ reduces the bit-rate of transmission while maintaining the signal above a threshold of intelligibility.

What it adds: The compression operator formalizes the trade-off between fidelity and efficiency that every pedagogical condensation involves. The condition $\text{sem}(v(s)) \approx \text{sem}(D(s), \mu(s))$ with $|v(s)| \ll |D(s) + \mu(s)|$ makes the trade-off explicit and testable: one can verify, for any proposed summary, whether the semantic core is preserved and what compression ratio is achieved (see Appendix A: 10.7:1 for the pelican).

Practical application: The operator $\mathcal{C}$ applies directly to any domain where complex material is condensed for memorization: flash cards, executive summaries, clinical mnemonics, regulatory checklists. The formal specification provides criteria for evaluating whether a condensation is *faithful* (semantic core preserved) or *destructive* (core information lost).

Definition 4.5 (Expansion $\mathcal{E}$). The **expansion operator** $\mathcal{E}$ transforms the moralization function from a single-valued to a set-valued mapping:

$$\mathcal{E} : \mu(s) \mapsto \mu^*(s) = \{m_1, m_2, \dots, m_n\}$$

where each $m_i \in \mathcal{M}$ represents a distinct moral interpretation of s , potentially including both positive (*in bono*) and negative (*in malo*) readings across multiple hermeneutical levels (literal, allegorical, tropological, anagogical).

Expansion is the inverse of compression in the interpretive dimension: it increases the system’s informational output for a given symbolic input. The integration of *distinctiones* into bestiaries is the paradigmatic instance. Where the standard *Physiologus* moralization offers a single reading—the lion signifies Christ—the *distinctiones*-enriched Third Family bestiary offers multiple readings across all four levels of medieval exegesis. Formally, the moralization function evolves from a simple mapping $f : S \rightarrow \mathcal{M}$ (one species, one meaning) to a set-valued mapping $f : S \rightarrow \mathcal{P}(\mathcal{M})$ (one species, a power set of meanings), where $\mathcal{P}(\mathcal{M})$ denotes the power set of the moral-interpretive space.

What it adds: The expansion operator captures a transformation that literary scholars describe qualitatively (“the text became richer”) in a form that allows counting. The cardinality $|\mu^*(s)|$ is a direct measure of interpretive flexibility: the bull chapter goes from 0 to 10 available readings (Appendix B). The *in bono/in malo* partition introduces a Boolean structure within $\mu^*(s)$ that maps onto real homiletic practice.

Practical application: Any knowledge base that associates concepts with interpretations can be evaluated by $|\mu^*(s)|$. A medical symptoms database where each symptom maps to one diagnosis ($|\mu^*| = 1$) is less clinically useful than one mapping to a differential diagnosis set ($|\mu^*| > 1$). The expansion operator describes how to enrich a knowledge base systematically while preserving its core architecture.

Definition 4.6 (Correction $\mathcal{R}$). The **correction operator** $\mathcal{R}$ modifies the set S of symbolic nodes or their associated functions D and μ by importing material from an external source $\mathcal{K}^{\text{ext}}$:

$$\mathcal{R} : \mathcal{K} \mapsto \mathcal{K}' = (S \cup S^{\text{ext}}, D \cup D^{\text{ext}}, \mu \cup \mu^{\text{ext}})$$

subject to the constraint that $\mathcal{K}'$ preserves the structural identity of $\mathcal{K}$.

Correction addresses situations where the moralization density $\rho(A)$ of a section A falls below a threshold that the compiler perceives as didactically adequate. The cross-family borrowing in Cambridge, University Library MS KK 4.25 is the clearest instance: the compiler imported eight moralized chapters from the Second Family into the Third Family’s Birds section precisely because that section’s moralization density (27%) was perceived as insufficient compared to other sections (67% for domestic animals). The correction operator restores local density by importing external modules, functioning as topological repair of the didactic corpus.

What it adds: The correction operator formalizes a phenomenon that has no name in traditional textual criticism: the import of material from a *different textual tradition* to repair a perceived deficiency, without destroying the host text’s identity. Standard stemmatic analysis treats cross-contamination as noise; $\mathcal{R}$ treats it as a purposeful editorial operation with a quantifiable trigger ($\rho(A) < \rho^*$) and a measurable outcome ($\rho'(A) > \rho^*$). See Appendix C: ρ rises from 0.25 to 0.69.

Practical application: Modern open-source software projects routinely import modules from external codebases to fill functional gaps—an operation structurally identical to $\mathcal{R}$. The formal specification identifies the conditions under which such imports are triggered (a measurable deficit), the constraints they must satisfy (structural compatibility), and the risks they introduce (genealogical norm violation, as in the reverse influence of KK 4.25 on Harley 3244).

4.3 The Complexity Scale

Definition 4.7 (Interpretive Complexity Scale). The **interpretive complexity** of a pedagogical system $\mathcal{K} + \{M_i\}$ is classified on a four-level scale:

- C_1 : **Univocal moralization.** Each symbolic node maps to exactly one moral interpretation: $|\mu(s)| = 1$ for all $s \in S$. The interpretive entropy is zero.
- C_2 : **Compressed retention.** The content of C_1 is condensed via $\mathcal{C}$: the system’s informational volume is reduced, but its interpretive range remains C_1 . The entropy remains zero, but the encoding is more efficient.
- C_3 : **Multiple exegesis.** The expansion operator $\mathcal{E}$ has been applied: $|\mu^*(s)| > 1$ for significant subsets of S . The interpretive entropy is positive: $H > 0$.
- C_4 : **Compensatory reconfiguration.** The correction operator $\mathcal{R}$ has been applied: external modules have been imported to repair topological deficits. The system exhibits maximal structural complexity within the genre’s constraints.

The complexity scale captures the observation that the bestiary tradition evolved not through replacement of earlier forms but through layered enrichment. A C_1 text (the standard *Physiologus*) is not superseded by a C_3 text (the Third Family with *distinctiones*); both continue to circulate, serving different audiences at different levels of interpretive sophistication. The scale thus describes not a historical progression but a synchronic typology of pedagogical configurations.

What it adds: The scale replaces the binary “simple vs. complex” with a four-level typology that distinguishes between *kinds* of complexity: encoding efficiency (C_2), interpretive multiplicity (C_3), and structural repair (C_4). This makes it possible to classify any manuscript (or modern textbook edition) by the type of adaptive operation it has undergone, not merely by its length or elaboration.

Practical application: In instructional design, the scale maps directly onto differentiated materials: C_1 = introductory text (one meaning per concept), C_2 = study guide or flash card set (compressed for review), C_3 = advanced reference with multiple perspectives, C_4 = integrated curriculum drawing on multiple disciplinary sources. A department reviewing its course materials can locate each document on the scale and identify gaps.

4.4 The Interpretive Entropy

To make the complexity scale precise, we define a measure of the interpretive richness of a pedagogical system.

Definition 4.8 (Interpretive Entropy). For a pedagogical system with expanded moralization function μ^* , the **interpretive entropy** over a section $A \subseteq S$ is

$$H(A) = - \sum_{s \in A} \sum_{m \in \mu^*(s)} p(m|s) \log_2 p(m|s)$$

where $p(m|s)$ is the probability that interpretation m is selected for node s in a given pedagogical or homiletic context.

For a C_1 system, where each node has exactly one interpretation, $p(m|s) = 1$ for the unique m and $H(A) = 0$: there is no uncertainty about the meaning. For a C_3 system enriched with *distinctiones*, each node may carry four, six, or more interpretations, and $H(A) > 0$: a preacher preparing a sermon on the lion has multiple exegetical options available. The entropy quantifies the system’s flexibility as a homiletic resource.

This measure should be understood as a *structural index*, not an empirically measurable quantity. The probabilities $p(m|s)$ cannot be reconstructed from surviving medieval evidence; we do not know how often a given preacher selected a particular interpretation. The formula’s value lies not in numerical calculation but in making precise the qualitative distinction between a C_1 system (where the preacher has no interpretive choice: $H = 0$) and a C_3 system (where the preacher has multiple options: $H > 0$). Medieval preachers did not calculate entropies; but the expansion from zero to positive interpretive range corresponds to a real and documentable transformation in how bestiary material was organized and used. The entropy notation captures the *direction and magnitude* of this transformation, even where exact values remain inaccessible. In Appendix B, we illustrate the index with a concrete case where $|\mu^*(s)|$ can be counted exactly, yielding a computable upper bound on H .

What it adds: The Shannon entropy formula, even as a structural index, accomplishes three things that qualitative descriptions cannot: (i) it assigns $H = 0$ *exactly* to univocal systems, making the boundary between C_1 and C_3 formally sharp; (ii) it provides an *upper bound* when $|\mu^*(s)|$ is known ($H_{\max} = \log_2 |\mu^*(s)|$), enabling comparison across chapters; (iii) it connects the bestiary literature to the established formalism of information theory [5, 6], enabling cross-disciplinary dialogue without translation.

Practical application: The index applies wherever a system associates inputs with variable numbers of outputs: a legal code where each statute admits multiple interpretations, a diagnostic protocol with differential diagnoses, a pedagogical system with multiple learning objectives per module. Computing H (or its upper bound) across sections identifies where interpretive flexibility is concentrated and where it is absent.

4.5 The Persistence Triad

Proposition 4.1 (Persistence Triad — Necessary Conditions). A pedagogical system $\mathcal{K} + \{M_i\}$ that persists across changing institutional and intellectual environments satisfies three conditions:

1. **Closure:** The kernel $\mathcal{K} = (S, D, \mu)$ is self-contained—the description and moralization of each node can be understood without reference to external texts or systems.
2. **Boundary maintenance:** The kernel maintains a recognizable identity that distinguishes it from neighboring genres (encyclopedias, sermon collections, exempla compilations) despite sharing material with them.

3. **Resilience:** When the environment changes (new audiences, new institutional contexts, new intellectual paradigms), the modular operations $\{M_i\}$ can restore the system’s didactic functionality without destroying $\mathcal{K}$.

Remark 4.1 (Epistemic status). The Persistence Triad is proposed as a *necessary condition* for genre persistence, not a sufficient one. The claim is that no pedagogical genre lacking closure, boundary maintenance, or resilience has been observed to persist across major institutional transitions; the claim is *not* that these three conditions guarantee persistence. Many genres satisfying all three conditions may nonetheless vanish due to exogenous factors (destruction of institutional support, catastrophic manuscript loss, replacement by functionally superior alternatives). The proposition is therefore falsifiable: a counterexample would be a pedagogical genre that demonstrably persisted across centuries while lacking one or more of these conditions. We are not aware of such a case, but we do not claim to have surveyed all possible candidates.

The Persistence Triad is a domain-specific instantiation of a general structural result: any system persisting in a changing environment must satisfy closure (its internal processes form a self-sustaining cycle), boundary maintenance (it maintains a distinction between interior and exterior), and resilience (it can recover from perturbation). This result was established formally in [47] for general dynamical systems. The contribution of the present paper is to show that pedagogical genres satisfy these conditions through a specific mechanism—the kernel-plus-modules architecture—and that the three adaptive operations ($\mathcal{C}$, $\mathcal{E}$, $\mathcal{R}$) are the means by which resilience is achieved.

The bestiary satisfies all three conditions. Its kernel—the *species*→*descriptio*→*moralizatio* triad—is self-contained (each chapter can be read independently). Its boundary is maintained despite extensive overlap with encyclopedias like Isidore’s *Etymologiae* (the bestiary is recognized as a distinct genre by the presence of moralizations, not by its zoological content). And its resilience is demonstrated by the adaptive operations documented across the tradition: when audiences needed memorization aids, $\mathcal{C}$ was deployed; when they needed flexible sermon material, $\mathcal{E}$ was deployed; when they needed comprehensive moral coverage, $\mathcal{R}$ was deployed.

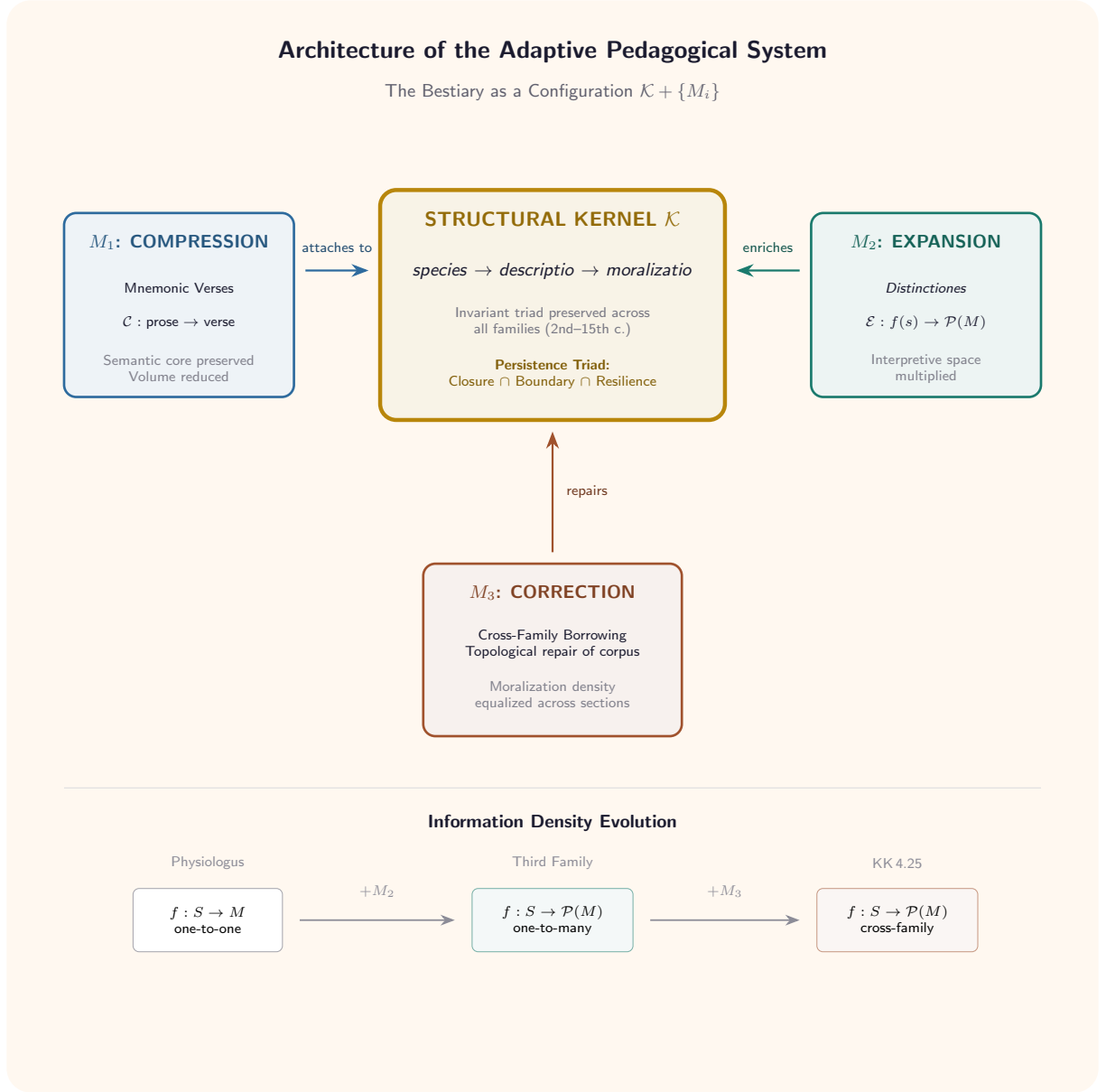


Figure 1: **Architecture of the adaptive pedagogical system.** The structural kernel $\mathcal{K} = (S, D, \mu)$ occupies the center, with three adaptive modules attached: compression $M_1 = \mathcal{C}$ (mnemonic verses), expansion $M_2 = \mathcal{E}$ (*distinctiones*), and correction $M_3 = \mathcal{R}$ (cross-family borrowing). The bottom row shows the evolution of information density from single-valued ($f : S \rightarrow M$) in the *Physiologus* to set-valued ($f : S \rightarrow \mathcal{P}(M)$) in the Third Family and cross-family in KK 4.25.

5 The Medieval Latin Bestiary: Case Study

5.1 The Genre and Its Kernel

Remark 5.1 (Scope of contribution). The empirical findings presented in Sections 5–8 are drawn principally from the research of Dr. Ilya Dines (2008–2025), who identified and documented the three mechanisms of textual enrichment that we formalize here. The contribution of the present paper is *not* empirical discovery but *formal redescription and theoretical integration*: we provide a unified mathematical vocabulary ($\mathcal{K}, \mathcal{C}, \mathcal{E}, \mathcal{R}, C_1\text{--}C_4$) that renders Dines’s findings commensurable with results in systems theory, cognitive science, and information theory. We also generate two testable predictions not present in Dines’s work: (i) that other long-lived pedagogical genres (the Aesopian fable, the catechism, the *ars dictaminis*) should exhibit the same kernel-plus-modules architecture, and (ii) that genres lacking one or more of the three adaptive operations should show shorter persistence spans. These predictions remain to be tested against the broader historical record.

The medieval Latin bestiary descends from the Greek *Physiologus*, probably composed in Alexandria in the 2nd–3rd century. The Latin tradition flourished predominantly in England from the 12th century onward, producing approximately 150 known manuscripts [11, 10] organized into several textual families [9, 10, 11]. Despite significant variation in chapter count (from 36 in the earliest texts to over 150 in the Third Family), illustration programs, and supplementary material, all bestiaries share the invariant kernel:

$$\mathcal{K}_{\text{bestiary}} = (\text{species}, \text{descriptio}, \text{moralizatio})$$

A chapter on the lion describes the animal’s three “natures” (it sleeps with open eyes, it erases its tracks with its tail, it breathes life into its stillborn cubs on the third day) and then interprets each nature allegorically (Christ’s divine nature, the Incarnation, the Resurrection). This structure—zoological description followed by moral interpretation—is the kernel that defines the genre.

The kernel satisfies the Persistence Triad (Proposition 4.1). It is **closed**: each chapter is self-contained and intelligible without reference to other chapters or external texts. It maintains **boundaries**: the presence of moralizations distinguishes the bestiary from the encyclopedia (which describes animals without moralizing) and from the sermon (which moralizes without zoological description). And it is **resilient**: the three adaptive operations documented below demonstrate the genre’s capacity to recover didactic functionality under changing conditions.

5.2 The Baseline: C_1 Configuration

The standard *Physiologus* and early bestiary families (BIs, Second A/B) operate at complexity level C_1 : each symbolic node maps to exactly one moral interpretation. The lion signifies Christ; the fox signifies the devil; the pelican signifies Christ’s sacrifice. The moralization function is single-valued:

$$\mu_{\text{BIs}}(s) = m_s, \quad |\{m_s\}| = 1 \quad \forall s \in S$$

This configuration is pedagogically effective for beginning students. Its simplicity ensures that the moral association can be learned, memorized, and reproduced with minimal ambiguity. However, it limits the text’s utility for more advanced users, particularly preachers who need flexible material that can be adapted to different homiletic contexts. A preacher wishing to use the lion in a sermon about divine judgment, rather than the Resurrection, finds no support in the C_1 text.

The moralization density ρ varies significantly across sections. For the Third Family—the most extensively studied in terms of moralization patterns—Dines [17] reports: $\rho(\text{domestic animals}) =$

$14/21 = 0.67$, $\rho(\text{wild beasts}) = 19/39 = 0.49$, $\rho(\text{birds}) = 11/41 = 0.27$, and $\rho(\text{fish, serpents, etc.}) = 16/59 = 0.27$. This uneven distribution, as we shall see, motivated the corrective intervention in KK 4.25.

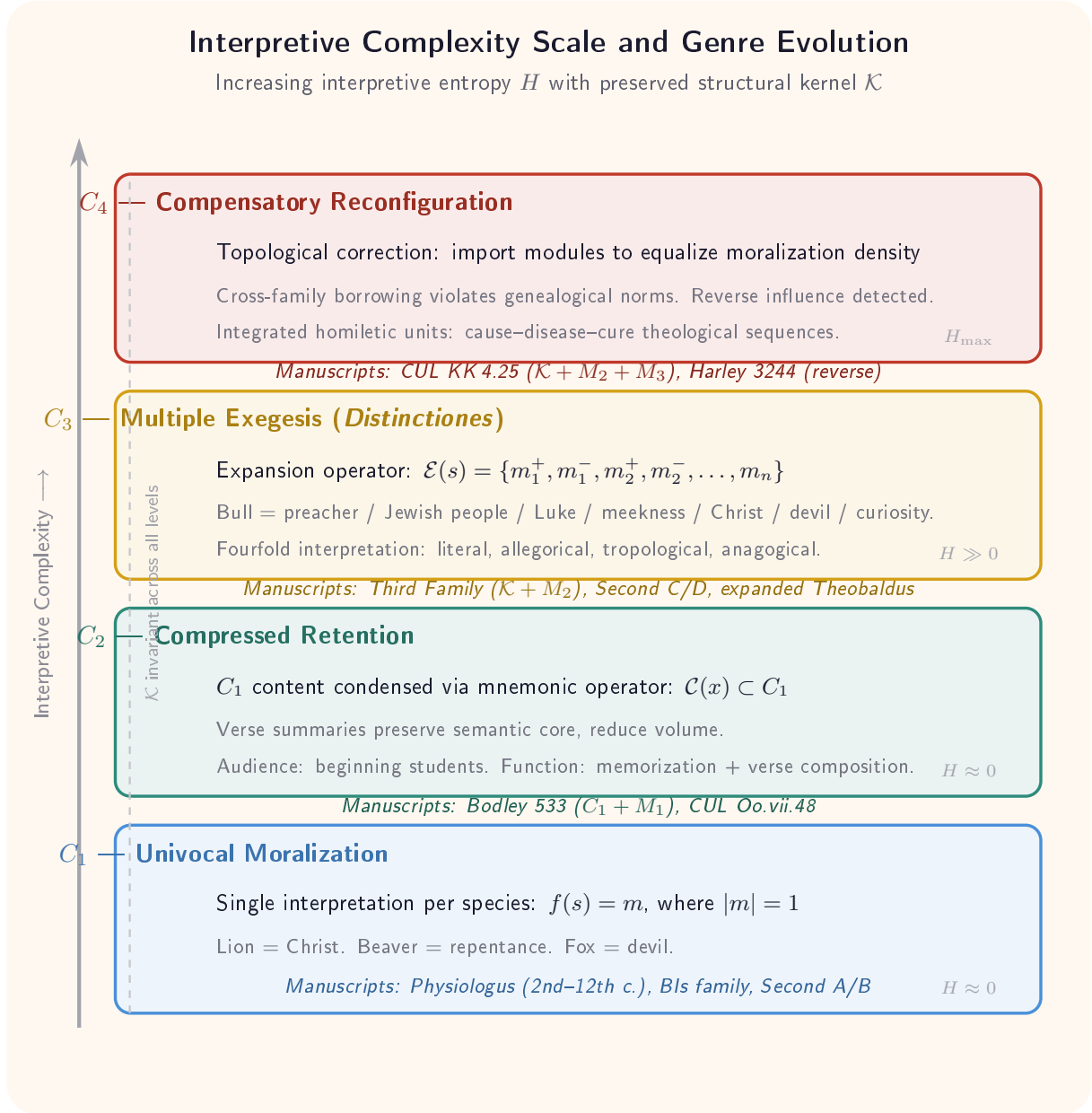


Figure 2: **Interpretive complexity scale and genre evolution.** Four levels of interpretive complexity (C_1 – C_4) are shown as layered strata, with representative manuscripts placed at each level. The structural kernel $\mathcal{K}$ remains invariant across all levels (dashed line at left). Interpretive entropy H increases from zero at C_1 (univocal moralization) to maximum at C_4 (compensatory reconfiguration).

6 Strategy One: Mnemonic Compression ($M_1 = \mathcal{C}$)

6.1 The Rarity of Verse in Latin Prose Bestiaries

Although several *Physiologus* descendants exist entirely in verse (the *Physiologus Theobaldi*, the *Novus Physiologus*, the French bestiaries of Philippe de Thaün and Guillaume le Clerc), the use of mnemonic verse summaries within Latin prose bestiaries is extremely rare [14]. This rarity is noteworthy given what cognitive science has established about the centrality of versification to medieval mnemonic practice. Carruthers [1] demonstrates that metrical structure, rhyme, and rhythm created the cognitive “hooks” on which memory organized its contents. Leonine hexameters—the verse form used in the bestiary verses—were particularly well suited to this function, as their internal rhymes created strong phonetic links between the two halves of each line.

6.2 Oxford, Bodleian Library MS Bodley 533

The most substantial collection of mnemonic verses in any Latin prose bestiary appears in Bodley 533, a Second B Family manuscript from the last quarter of the 13th century. It contains eleven sets of rhymed verses covering species including the antelope, unicorn, beaver, fox, ant, hedgehog, hoopoe, pelican, owl, siren, and sawfish [14]. Several features indicate the verses were copied from an external source: scribal errors (*glacieles* for *glaciales*), misplacement of the hedgehog verses (appended to the ant chapter rather than in their proper position), and displacement of the siren and *serra* verses.

In terms of the formal framework, the compression operator acts on each chapter:

$$\mathcal{C}(D(s), \mu(s)) = v(s), \quad |v(s)| \approx 4 \text{ lines}, \quad |D(s) + \mu(s)| \gg 4 \text{ lines}$$

The semantic fidelity of the compression can be verified by comparing verse content with chapter content. The verses paraphrase the zoological material with occasional reference to the moralization—consistent with the prediction that $\mathcal{C}$ preserves the semantic core while sacrificing elaboration and detail.

6.3 Independent Circulation and the Cambridge Confirmation

The hypothesis that the verses circulated independently of any bestiary was confirmed by the discovery of a related set in Cambridge, University Library MS Oo.vii.48, an English miscellany dated 1320–1340 [15]. This folio contains verses on nearly the same species in nearly the same order, with the addition of five species and the omission of one. The discovery proves that the compression operator $\mathcal{C}$ produced outputs that could detach from the kernel $\mathcal{K}$ and circulate as independent pedagogical objects—a property consistent with the modularity requirement of adaptive systems.

All versified species derive from the *Physiologus* and have moralizations in their bestiary chapters. This restriction suggests that the presence of a moralization was a necessary condition for verse composition: the moralization provided the interpretive framework that the verse condensed.

6.4 Pedagogical Implications of Verse Placement

In the early chapters of Bodley 533, the verses appear at the beginning of the chapter (functioning as previews or advance organizers); in later chapters, they appear at the end (functioning as retrospective summaries). This shift maps onto the medieval pedagogical alternation between *praelectio* (preparatory overview) and *repetitio* (retrospective review) identified by Carruthers [1]—two complementary mnemonic strategies that address different stages of the learning process.

7 Strategy Two: Hermeneutical Expansion ($M_2 = \mathcal{E}$)

7.1 The Genre of *Distinctiones*

Distinctiones are sets of systematic biblical interpretations that distinguish the multiple meanings of a term—typically into a fourfold exposition of literal, allegorical, tropological, and anagogical senses—and illustrate each meaning with quotations from Scripture [32, 33]. The genre reached maturity in the work of Parisian masters: Peter the Chanter, Peter of Poitiers, Alan of Lille, and Praepositinus of Cremona. It was transmitted to England through William de Montibus (c. 1140–1213), chancellor of Lincoln Cathedral, and Peter of Cornwall, an Augustinian canon in London [34].

Within the broader ecosystem of preaching aids, *distinctiones* occupied a distinctive niche. Unlike exempla collections (which provided illustrative narratives) or *artes praedicandi* (which theorized sermon structure), *distinctiones* offered a method of semantic analysis applicable to any biblical or theological term. This methodological character made them uniquely adaptable to the bestiary context.

7.2 The Formal Operation

The integration of *distinctiones* into bestiaries transforms the moralization function from single-valued to set-valued. For the Third Family, where *distinctiones* were drawn from William de Montibus’s *Distinctiones Theologicae* and *Numerale* [16]:

$$\mathcal{E} : \mu(s) \mapsto \mu^*(s) = \{m_1^+, m_1^-, m_2^+, m_2^-, \dots, m_n^+, m_n^-\}$$

where m_i^+ and m_i^- denote *in bono* and *in malo* interpretations at hermeneutical level i .

The bull chapter of the Third Family illustrates the operation at its most sophisticated. The zoological description (from Isidore) is followed by a *distinctio* interpreting the bull in all four senses: literally as a domestic animal; allegorically as the preacher, the Jewish people, and the Evangelist Luke; tropologically as meekness and good works; and anagogically as Christ [16]. The bull is also interpreted *in malo*: as the devil, as curiosity, as obstinacy, and as any foolish person. The interpretive entropy of this single node is substantial:

$$H(\text{bull}) = - \sum_{i=1}^k p(m_i|\text{bull}) \log_2 p(m_i|\text{bull}) > 0$$

7.3 Instances Across Families

The expansion operator was deployed independently in several bestiary families [16]: the Third Family (from William de Montibus), the Second C Family represented by Harley 4751 and Bodley 764 (from Garnerius de Rochefort), the Second D Family represented by Royal 12 F. XIII (from Peter of Cornwall’s *Pantheologus*), and an atypical BIs manuscript (London, BL MS Add. 22041). A parallel development occurred in the 15th-century German expanded *Physiologus Theobaldi*, where the original twelve chapters of simple versified material were supplemented with *distinctiones*-form commentaries [44, 43]. No direct genealogical link connects the English and German traditions; the convergent evolution testifies to the structural utility of the expansion operation across time and geography.

7.4 Structural Regularization

A notable feature of the Third Family’s expansion is the repositioning of moralizations. In the Second B Family (the Third Family’s primary source), moralizations were often inserted in the middle of chapters. The Third Family compiler moved them to the end, creating a regularized structure: zoological description first, then moral interpretation. This reorganization, small in itself, enhanced the text’s utility as a reference tool for sermon preparation—an instance of the broader principle that pedagogical technologies evolve not only through content enrichment but through structural optimization of their information architecture.

8 Strategy Three: Structural Correction ($M_3 = \mathcal{R}$)

8.1 Cambridge, University Library MS KK 4.25

The most striking instance of the correction operator appears in KK 4.25, a Third Family manuscript produced c.1220–1230 in southern England [17]. Its compiler had access to at least three manuscripts: a Third Family exemplar, a Second Family bestiary similar to London, BL MS Add. 11283, and a BIs bestiary akin to Oxford, Bodleian Library MS Bodley 602. The presence of multiple bestiaries in a single location was rare, as Baxter [13] demonstrated through analysis of surviving medieval library catalogues.

8.2 The Moralization Deficit and Its Repair

The compiler’s motivation can be reconstructed formally. The Third Family’s Birds section had $\rho(\text{birds}) = 0.27$, while $\rho(\text{domestic animals}) = 0.67$. Defining a threshold ρ^* below which the compiler perceived a section as didactically inadequate, the correction operator was triggered when $\rho(A) < \rho^*$:

$$\rho(\text{birds}) = 0.27 < \rho^* \implies \mathcal{R} \text{ activated}$$

The compiler imported eight moralized chapters from the Second Family into the Birds section, along with two unique non-zoological chapters: *Quattuor Modis Peccatum* (The Four Ways of Sin, from Gregory the Great) and *Quattuor Colores Caelestis* (The Four Colors of the Rainbow, a free paraphrase of Isidore with significant modifications) [17]. These chapters were previously dismissed as “pointless insertions” by James [9] and Clark [11]. Analysis reveals, however, that they form a coherent theological sequence structured around a cause–disease–cure framework: the moralization of Ibis II defines a spiritual disease; *Quattuor Modis Peccatum* elaborates the root cause; *Quattuor Colores Caelestis* offers the cure (baptism, symbolized by the rainbow’s purple). This integrated homiletic unit suggests a compiler operating not as a passive copyist but as an active editor constructing sermon-ready material.

8.3 Reverse Influence

Perhaps the most extraordinary finding is evidence of reverse borrowing: the Second Family bestiary London, BL MS Harley 3244 (c. 1240) shows strong influence from KK 4.25 [17]. In the entire bestiary tradition, borrowing is one-directional—later families borrow from earlier ones. The reverse influence of KK 4.25 on Harley 3244 represents a genuine violation of the genre’s genealogical norms, confirming that the correction operator $\mathcal{R}$ can generate innovations without precedent in the system’s history.

9 Cognitive Architecture of Zoomorphic Symbolism

The preceding sections have established the formal apparatus and demonstrated its application to the bestiary tradition. A deeper question remains: *why* did animal symbolism prove structurally more durable than other carriers of moral pedagogy, including direct evangelical discourse? The answer lies not in theology but in the cognitive architecture of symbolic processing, formalized through the Structural Distortion Principle [48].

Remark 9.1 (Epistemic status of this section). The argument that follows is *structural-theoretical*, not experimentally verified. We propose a formal account of why zoomorphic symbolism possesses certain cognitive advantages, grounded in independently established principles of dual-coding theory [3], perceptual symbol systems [40], and the cognitive science of cultural transmission [42, 41]. The variables α , ϕ , and r in Proposition 9.1 are not currently measurable on a cardinal scale; the proposition identifies an *optimization structure* (what zoomorphic carriers maximize and minimize) rather than providing a quantitative model. The claim is that this structure explains the *direction* of the bestiary’s evolutionary advantage, not its precise magnitude. Experimental verification—e.g., comparative memory studies of zoomorphic vs. anthropomorphic moral narratives—would strengthen the argument considerably and constitutes a natural direction for future research.

9.1 The Animal as Minimal Symbolic Container

Every adaptive system operating under finite processing resources must compress, distort, and selectively attend to its informational environment. This is not a defect of cognition but a structural condition of any system that processes more information than it can represent exhaustively. The Structural Distortion Principle formalizes this constraint: every act of perception is simultaneously an act of non-perception; what a system registers is determined as much by what it filters out as by what it admits.

A pedagogical system transmitting moral content must therefore optimize across two competing demands: the content must be *memorable* (high salience, strong encoding) and *multi-interpretable* (capable of supporting diverse applications). Direct evangelical discourse—parables involving human agents in social situations—optimizes for the first demand but creates friction with the second. A human character in a parable is socially embedded: the listener identifies, compares, defends, or rejects based on social proximity. This identification narrows the interpretive range and increases cognitive resistance.

The animal image resolves this tension. An animal is simultaneously concrete and abstract. It possesses vivid sensory properties (visual salience, narrative memorability) while remaining socially neutral—it belongs to no class, holds no political identity, triggers no in-group defense. Formally:

Proposition 9.1 (Optimality of Zoomorphic Carriers). Let $\sigma(s)$ denote the semiotic properties of a symbolic carrier s along three dimensions: abstraction level $\alpha(s)$, cognitive friction $\phi(s)$, and mnemonic retention $r(s)$. A carrier is optimal for adaptive pedagogical use when it simultaneously maximizes abstraction and retention while minimizing friction:

$$s^* = \arg \max_s [\alpha(s) \cdot r(s) \cdot (1 - \phi(s))]$$

Zoomorphic carriers satisfy this condition because:

1. $\alpha(s_{\text{animal}})$ is high: the animal is a pure trait carrier, not bound to any particular social context.
2. $\phi(s_{\text{animal}})$ is low: moral accusation directed at a “fox” or “wolf” is received with less resistance than accusation directed at a human figure.

3. $r(s_{\text{animal}})$ is high: zoomorphic and visually saturated images are among the most readily retained in human memory [3, 40].

This proposition receives support from the empirical record. The hedgehog rolling on grapes, the beaver biting off its own testicles, the pelican piercing its breast—these images have survived in cultural memory for nearly two millennia, while countless human-centered moral narratives from the same period have been forgotten. The bestiary’s imagistic vocabulary constitutes what Boyer [42] would call a collection of “minimally counterintuitive” representations: sufficiently strange to capture attention, sufficiently structured to support inference.

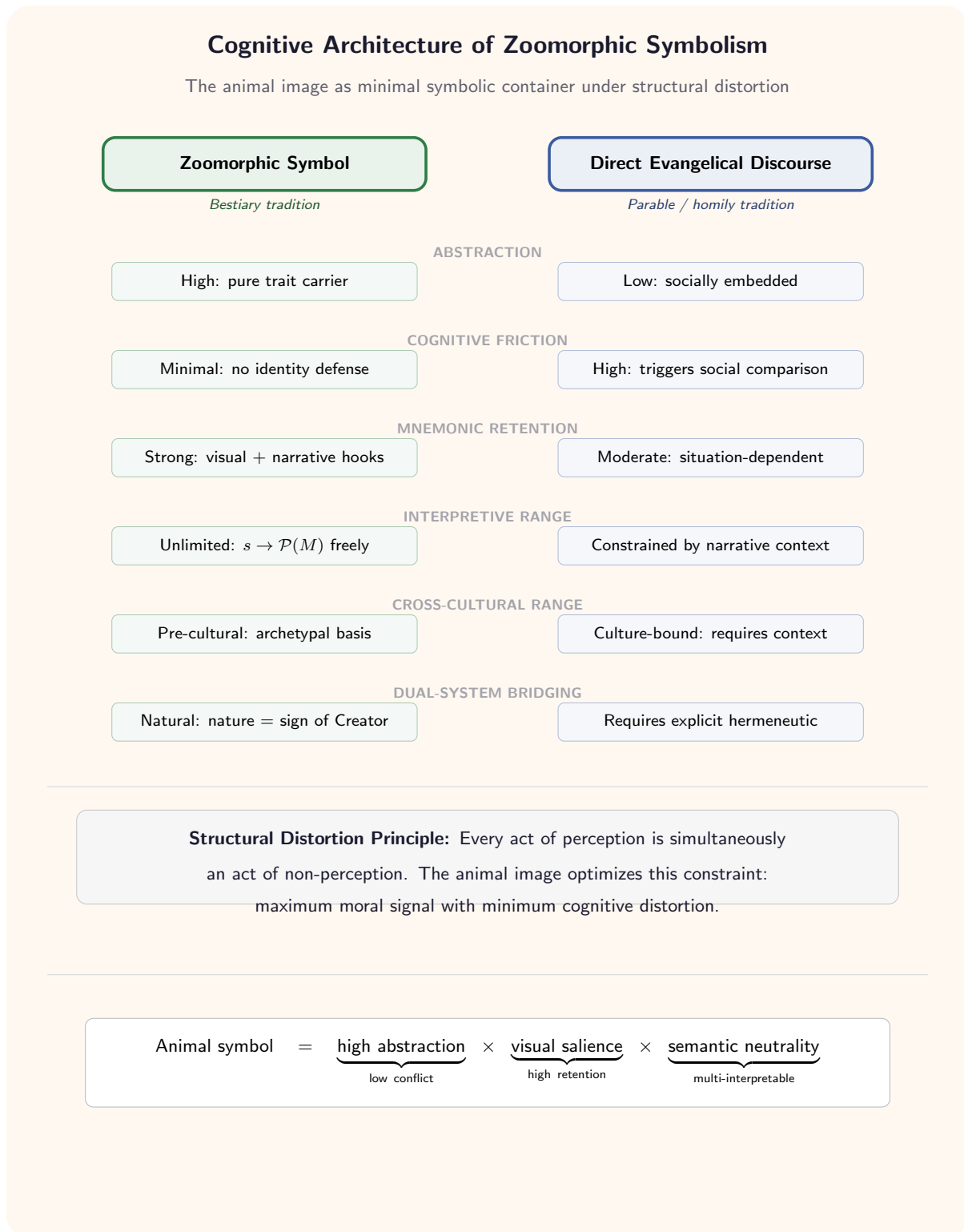


Figure 3: **Cognitive architecture of zoomorphic symbolism.** Comparison of the semiotic properties of zoomorphic symbols (left) and direct evangelical discourse (right) across six dimensions. The animal image achieves optimal pedagogical carrying capacity by combining high abstraction, low cognitive friction, and strong mnemonic retention—properties predicted by the Structural Distortion Principle.

9.2 Interpretive Flexibility and Cross-Cultural Portability

A further advantage of zoomorphic symbolism is its compatibility with the expansion operator $\mathcal{E}$. The lion can signify Christ *or* the devil; the fox can signify heresy *or* worldly cunning. This dual interpretability—the capacity to carry both *in bono* and *in malo* readings without narrative contradiction—is a direct consequence of the animal’s semantic neutrality. A human character who signifies both Christ and the devil generates a narrative paradox; a lion that does so generates a theological *distinctio*.

Moreover, the animal symbol draws on pre-cultural, cross-civilizational archetypes. The lion, serpent, eagle, and bull carry symbolic resonance far older than Christianity. The bestiary embedded itself in a deep stratum of symbolic cognition that was not dependent on any specific canonical text but parasitized—in the neutral, ecological sense—on the mythological memory of the species. This pre-cultural grounding provided a foundation of cross-cultural portability that no historically specific narrative could match.

9.3 The Metaphysical Substrate

The bestiary operates on a foundational metaphysical premise: the natural world is already saturated with meaning. Every created thing is a sign of its Creator. This premise makes the bestiary self-extending: any animal, any natural phenomenon, can in principle be incorporated and moralized, because meaning is not imposed on nature but discovered within it. In contrast, direct evangelical discourse is tied to specific historical events and characters; its extension requires new revelation or new narrative, not merely new observation.

This self-extending property corresponds, in information-theoretic terms, to an open coding scheme: the set of symbolic nodes S is not fixed but expandable, and the description and moralization functions D and μ can be applied to any new element admitted to S . The bestiary’s kernel is not merely invariant but generative.

9.4 The Manuscript as Pedagogical Channel: What Print Destroyed

The formal framework developed thus far treats the bestiary as a system of symbolic content. But the medieval bestiary was not a system of content alone—it was a material object, handwritten on parchment, and its materiality constituted a pedagogical channel that the transition to print permanently eliminated. This loss deserves formal attention, because it represents the destruction of an entire dimension of the adaptive system.

A manuscript bestiary is not a neutral container for text. The scribe’s hand—its regularity or irregularity, its size, weight, and ductus, its decorative elaborations—functions as a *meta-communicative layer* superimposed on the semantic content. This layer carries information that no printed edition can reproduce.

Consider three paleographical features and their pedagogical consequences:

First, script hierarchy. Medieval scribes routinely employed distinct scripts for distinct textual registers within a single manuscript: a formal textualis for the main text, a less formal anglicana for glosses and commentary, and a display script (often colored or enlarged) for rubrics, incipits, and chapter headings. In a bestiary, the shift from one script to another signals to the reader the shift from *descriptio* to *moralizatio*—that is, the transition from $D(s)$ to $\mu(s)$ within the kernel $\mathcal{K}$. The reader does not need to parse the content to know that a transition has occurred; the visual register of the handwriting announces it. This is a paleographical encoding of the dual-system boundary: the change of hand (or hand-style) marks the passage from $\mathcal{W}$ -observation to $\mathcal{K}_T$ -interpretation.

Second, scribe as interpretive agent. No two scribes copied the same text identically. Variant spellings, marginal annotations, corrections, omissions, and additions constitute a layer of editorial interpretation that is inseparable from the act of transmission. Each manuscript is therefore not merely a copy but a *re-evaluation* of the text by a specific human intelligence operating under specific institutional constraints. The scribe’s errors are as informative as the scribe’s successes: they reveal what was difficult to understand, what was considered dispensable, and what was worth correcting. In formal terms, each act of manuscript copying applies an unintentional adaptive operation M_{scribe} to the kernel $\mathcal{K}$ —a micro-scale compression, expansion, or correction that accumulates across generations of copies. This is precisely the mechanism by which the adaptive operations documented in Sections 6–8 entered the tradition: not by fiat but by the aggregate effect of individual scribal decisions.

Third, the tactile and spatial dimension. A manuscript bestiary occupies physical space. Its pages must be turned; its illustrations must be found; its text must be navigated without an index. The spatial organization of the page—the layout of text and image, the use of decorated initials as navigational markers, the density of writing per folio—constitutes an information architecture that structures the reader’s encounter with the content. Baxter [13] documented wear patterns on specific pages of bestiary manuscripts, demonstrating that certain chapters were consulted far more frequently than others. The manuscript’s materiality thus records the *actual use pattern* of the pedagogical system—a kind of analog usage log that no printed edition preserves.

What did print destroy? The printing press standardized the text, eliminated the scribal re-evaluation layer, replaced script hierarchy with typographic hierarchy (a far coarser distinction), and removed the individuality of the copy. In terms of the formal framework, print froze $M_{\text{scribe}} = 0$: it eliminated the micro-scale adaptive operations that had driven the bestiary’s evolution for a millennium. The printed bestiary (and its descendants in natural history) could still undergo macro-scale operations—new editions, new commentaries, new illustrations—but the continuous, low-level adaptive pressure exerted by individual scribal transmission was lost. This may explain, in part, why the bestiary as a genre did not survive the transition to print: the technology that promised perfect fidelity destroyed the very mechanism—imperfect, individual, materially embodied copying—that had sustained the genre’s adaptive capacity.

What it adds: The scribal layer M_{scribe} extends the formal apparatus to include the *transmission medium* as an active component of the adaptive system, not merely a passive container. This reframes the transition from manuscript to print as the elimination of a class of adaptive operations, explaining why genres sustained by scribal adaptation could not survive a technology that removed it.

Practical application: The analysis applies to any knowledge system transitioning from a medium that permits low-level adaptive variation (handwriting, oral tradition, wiki-style collaborative editing) to one that enforces uniformity (print, locked PDFs, read-only databases). The formal prediction is that such transitions will destroy adaptive capacity unless compensatory mechanisms are introduced.

9.5 The Scribe as a Person: Graphology, Creativity, and the Manuscript as Time Machine

The formal framework has treated the scribe as an operator— M_{scribe} —a function that transforms input text into output text with predictable classes of variation. This is formally correct but humanly insufficient. Behind every bestiary manuscript stands a person: someone who sat for weeks or months in a scriptorium, a cloister, or a private study, dipping a quill into iron gall ink and inscribing words onto prepared animal skin. This person had a name (rarely recorded), a training

(partially reconstructable), a personality (inferable from the work), and an inner life (irretrievably lost—or nearly so).

9.5.1 What the Hand Reveals

Paleography traditionally treats handwriting as evidence for dating and localization: a textualis formata hand places a manuscript in a specific century, region, and institutional milieu. But the same handwriting carries information that paleography has not traditionally sought to extract: information about the scribe as an individual.

The emerging field of computational paleography, combined with graphological analysis, permits inferences about the scribe’s psychological and physiological state that go beyond dating and localization:

Pressure and rhythm. A scribe writing under time pressure produces strokes with uneven pressure, inconsistent spacing, and accelerating rhythm toward the end of each gathering. A scribe working at leisure produces even, measured strokes with consistent spacing throughout. The physical properties of the ink (thickness of lines, degree of penetration into the parchment) record this rhythm permanently. In the bestiary corpus, manuscripts produced for institutional libraries tend to show measured, consistent hands; those produced for personal use or under commission pressure show more variable rhythm.

Fatigue and attention. Scribal errors cluster toward the end of work sessions: the rate of corrections, deletions, and line-skipping increases as the scribe tires. The distribution of errors across a manuscript’s gatherings provides a record of the scribe’s work habits—how long each session lasted, where breaks were taken, whether the scribe returned to correct errors from previous sessions or pressed forward.

Engagement and indifference. The most revealing graphological evidence concerns the scribe’s relationship to the text being copied. In some bestiary manuscripts, the scribe’s hand changes perceptibly between sections: display script becomes more elaborate for chapters the scribe evidently considered important; the hand becomes more cursive and rapid for chapters treated as routine. Marginal annotations in the scribe’s own hand—comments, corrections, cross-references, and occasionally personal reactions—reveal which passages caught the scribe’s attention and which were copied mechanically.

Creative intervention. The formal framework identifies the scribe as an involuntary adaptive agent (M_{scribe}). But some scribes were voluntary adaptive agents: they made deliberate editorial choices, selected among variant readings, added explanatory glosses, and occasionally composed original material. The compiler of KK 4.25 is the supreme example: a scribe who was simultaneously a creative editor, reorganizing the genre’s content according to a systematic pedagogical vision. But traces of creative intervention appear in less dramatic forms throughout the corpus—a choice between synonyms, a clarifying paraphrase, an interpolated phrase that makes explicit what the exemplar left implicit.

9.5.2 The Graphological Portrait

For a handful of bestiary manuscripts where the scribe’s hand is sufficiently distinctive and the text sufficiently long, a provisional graphological portrait can be attempted. Consider the scribe of Aberdeen University Library MS 24 (the Aberdeen Bestiary, c. 1200). The hand is a high-grade textualis prescissa—an expert bookhand requiring years of training. The strokes are confident, with consistent pressure and precise letter-spacing. The decorative program is elaborate, with gold leaf, complex pigments, and historiated initials that required close coordination between scribe and

illuminator. This is a professional at the height of his skill, working without haste, aware that the manuscript was destined for a prestigious library.

Now compare the hand of the glossator who added marginal notes to the same manuscript in a later century. The anglicana hand is rapid, informal, sometimes illegible. The ink is thinner; the pressure is uneven. The notes are functional, not decorative—references to sermon occasions, cross-references to theological texts, shorthand reminders. This is a user, not a producer: someone who engaged with the bestiary as a working tool, extracting homiletic material under time pressure, probably in preparation for a specific preaching assignment.

These two hands—the producer and the user—represent the bestiary’s complete life cycle: creation in the controlled environment of the scriptorium, followed by active use in the uncontrolled environment of pastoral practice. The graphological contrast between them is a record of this life cycle inscribed permanently into the material object.

9.5.3 What the Scribe Felt

We cannot know what a 13th-century scribe felt. We can, however, identify the conditions under which the scribe worked and the elements of the bestiary text most likely to have produced an emotional or intellectual response.

The scribe spent hours inscribing detailed descriptions of animal behaviors designed to be vivid, memorable, and morally charged. The pelican piercing its breast to revive its young with its own blood. The beaver biting off its own testicles to escape the hunter. The phoenix burning itself to ashes and rising on the third day. These are not emotionally neutral images. They were designed to provoke wonder, pity, fear, and admiration—and there is no reason to suppose that the scribe was immune to the emotional effects of the material being copied.

Evidence of scribal emotional engagement exists, though it is indirect. Some scribes added exclamatory glosses: *Nota bene!* (“note well!”) beside passages they found particularly striking. Others added small drawings—sketches of animals, pointing fingers (*manicules*), or decorative flourishes—that indicate that the scribe paused, reflected, and responded to the text rather than copying mechanically. In rare cases, scribes added personal comments: complaints about the quality of the exemplar, notes about the weather, or remarks on the difficulty of the text.

These traces are precious because they are among the most intimate records of medieval inner life that survive. A cathedral chronicle records what happened; a legal document records what was agreed; a theological treatise records what was believed. But a scribe’s marginal annotation records what an individual person *thought and felt* at a specific moment while engaging with a specific text. It is the closest we can come to the lived experience of a medieval mind.

9.5.4 The Manuscript as Time Machine

The bestiary manuscript is, in a precise sense, a time machine. When a modern reader examines a page written by a 13th-century scribe, the reader is seeing *exactly what the scribe’s eyes saw as the pen moved across the parchment*: the same letterforms, the same ink color (faded but identifiable), the same spatial layout, the same relationship between text and margin. The parchment itself was once the skin of an animal that lived in medieval England; the ink was mixed from oak galls and iron salts by a specific person; the quill was cut from a specific goose feather.

No other artifact brings us this close to a specific moment in the past. A building has been altered, repaired, and repurposed. A painting has been cleaned, restored, and reframed. A coin has been circulated through thousands of hands. But a manuscript page that has been stored in a library since the 13th century preserves the trace of one person’s hand in one act of creation, fixed permanently at the moment the ink dried on the parchment.

The formal framework identifies what makes this trace valuable: it is a record of M_{scribe} in action. Every variant, every correction, every marginal note, every fluctuation in the hand's rhythm is a datum about the adaptive process through which the bestiary tradition evolved. The scribe's personality—training, temperament, fatigue, attention, creativity, emotional response—is the medium through which the tradition's formal operations were materially instantiated. Without the scribe, there are no adaptive operations; there are only abstract descriptions of operations that never occurred.

The transition from manuscript to digital facsimile preserves the visual appearance of the page but loses the tactile and material dimension: the texture of the parchment, the slight indentation where the pen pressed, the faint residue of pricking and ruling, the smell of age. These properties carry information about the production process that the image does not capture. High-resolution multispectral imaging can recover some of this information (undertext, erased annotations, ink composition), but the material object remains irreplaceable as a source of evidence about the physical act of writing.

In the same way that an archaeological site cannot be re-excavated once it has been dug, a manuscript cannot be re-consulted once it has been destroyed. The information it carries about the scribe, the scriptorium, and the act of writing is finite, singular, and non-renewable.

What it adds: The formal framework treats the scribe as an operator (M_{scribe}). This section restores the operator's humanity. The scribe's hand carries information at three levels: (i) institutional (dating, localization, training), (ii) psychological (personality, engagement, fatigue, creativity), and (iii) existential (the physical trace of a specific human being's encounter with a specific text at a specific moment in the past). Level (iii) is irreplaceable and irreproducible: no other class of artifact preserves the immediate trace of a medieval individual's cognitive and emotional engagement with this degree of specificity. When a manuscript is destroyed, this trace is destroyed with it.

What made the scribe the same as us: The scribe experienced fatigue, boredom, and moments of creative engagement. The scribe responded emotionally to vivid imagery. The scribe made errors under time pressure and corrected them when motivation was sufficient. The scribe had aesthetic preferences visible in decorative choices. These are universal features of human cognition that have not changed in eight centuries.

What made the scribe different: The scribe inhabited a teleological world in which every animal was a moral sign, a world in which copying a text was simultaneously an act of preservation, an act of piety, and a form of meditation. The scribe's relationship to the text was not that of a modern reader to a book but that of a participant to a living tradition: the act of copying was itself a form of engagement with the tradition's content, not merely a mechanical reproduction of its form.

The manuscript as time machine: The parchment page is the only artifact that preserves, in unmediated physical form, the trace of a specific moment of medieval intellectual and creative life. Its loss is not the loss of a copy; it is the loss of a window. The urgency of preservation is not sentimental; it is epistemological. The window is closing. The parchment is degrading. The scribe's hand is fading from the page.

9.6 Why Jesus Did Not Use Animals: The Structural Logic of the Parable

A striking feature of the Gospels, when viewed against the background of the bestiary tradition, is the near-total absence of animal symbolism in the teachings attributed to Jesus. The parables employ human agents almost exclusively: sowers, servants, kings, fathers, sons, virgins, merchants, judges, widows, debtors. Animals appear occasionally—the camel through the needle's eye, sheep and goats at the Last Judgment, foxes that have holes—but these are incidental comparisons, not developed zoological-moral sequences. Jesus does not moralize the lion, the pelican, or the beaver.

His pedagogical method is fundamentally different from the bestiary’s, and this difference is not accidental but structurally determined.

The formal framework developed in this paper makes the reason explicit. The bestiary operates through *emblematic pedagogy*: a fixed symbolic node (the animal) carries a detachable moral interpretation that can be multiplied, compressed, or corrected without altering the emblem itself. The animal is a container; the moralization is the content. This architecture is optimized for *institutional transmission across time*: it survives scribal copying, translation, and editorial modification because the emblem is stable and the content is modular.

Jesus’s parables operate through *situational pedagogy*: a narrative scenario involving human agents in social situations creates a moral crisis that the listener must resolve through personal engagement. The parable of the Prodigal Son does not offer a detachable moral; it creates a situation in which the listener must choose between the elder brother’s justice and the father’s mercy. The parable of the Good Samaritan does not moralize the Samaritan as a symbolic node; it forces the listener to confront a social boundary (Jew/Samaritan) and decide whether that boundary holds under conditions of suffering.

In the terms of Section 9, the parables deliberately *maximize* cognitive friction (ϕ) rather than minimizing it. They employ human agents precisely *because* human agents trigger social identification, comparison, and defensive resistance. This is not a pedagogical deficiency; it is a pedagogical strategy. The friction is the pedagogy. The listener who identifies with the elder brother and feels the injustice of the father’s welcome is experiencing the dual-system conflict ($\mathcal{W}$ vs. $\mathcal{K}_T$) in real time, as a lived cognitive event rather than an intellectual proposition.

This explains why Jesus did not use animals: zoomorphic symbolism would have defeated his pedagogical purpose. The fox-as-heresy or the lion-as-Christ can be learned, memorized, and repeated without personal engagement. A student can master the entire bestiary moralization system without ever being morally challenged by it, because the animal image absorbs the moral content without reflecting it back onto the learner. The parables refuse this absorption. They use human agents because human agents cannot be moralized without the listener becoming implicated.

Definition 9.1 (Emblematic vs. Situational Pedagogy). Let $\phi(s)$ denote the cognitive friction of a symbolic carrier s (Definition 4.4).

- **Emblematic pedagogy** minimizes ϕ : the carrier (animal, heraldic device, abstract symbol) is chosen to reduce defensive identification. Content is detachable from carrier. Optimized for institutional repetition and long-term transmission. Genre: bestiary, fable, emblem book.
- **Situational pedagogy** maximizes ϕ : the carrier (human agent in social scenario) is chosen to provoke identification and defensive engagement. Content is inseparable from narrative situation. Optimized for immediate transformative encounter. Genre: parable, case study, Socratic dialogue.

The two strategies are not in competition; they serve different functions within the dual-system architecture. Emblematic pedagogy transmits the *vocabulary* of the $\mathcal{W}/\mathcal{K}_T$ distinction: it teaches the student that the natural world can be read as a field of moral signs. Situational pedagogy provokes the *experience* of the dual-system conflict: it forces the student to confront the incompatibility of $V_{\mathcal{W}}$ and $V_{\mathcal{K}_T}$ in a specific case. The bestiary teaches the student *that* the lion signifies Christ; the parable teaches the student *what it costs* to follow Christ when following Christ contradicts worldly optimization.

The medieval Church deployed both strategies simultaneously: the bestiary in the schoolroom and the library; the parable in the pulpit and the confessional. The *distinctiones* tradition (Section 7) can be understood as an attempt to bridge the two: by multiplying the interpretive pathways

available for each animal, the *distinctiones* introduced a degree of situational flexibility into the emblematic system, allowing the preacher to select the interpretation most relevant to the specific pastoral situation. The expansion operator $\mathcal{E}$ thus represents, in formal terms, a partial convergence of emblematic and situational pedagogy—a hybridization that increased the bestiary’s homiletic utility precisely by importing some of the interpretive friction that the pure emblem excluded.

What it adds: The emblematic/situational distinction resolves an apparent puzzle: if animals are optimal carriers (Proposition 9.1), why did the most influential moral teacher in Western history not use them? The answer is that optimality depends on the objective function. For long-term institutional transmission, animals are optimal (low friction, high retention). For immediate personal transformation, humans are optimal (high friction, forced identification). The formal framework accommodates both by distinguishing between optimization for $r \cdot \alpha/\phi$ (bestiary) and optimization for ϕ directly (parable).

Practical application: Modern instructional designers face the same choice. Case studies using anonymized or fictional characters (low friction) are effective for teaching general principles; case studies using the learner’s own institution or peer group (high friction) are effective for provoking behavioral change. The formal framework predicts that the first type will be more widely adopted and longer-lived, while the second will be more transformative per exposure but less durable as a transmissible format.

9.7 Archetypal Depth: Why Particular Animals Received Particular Attributes

The formal framework has established *that* the animal image functions as an optimal pedagogical carrier and *how* the operations $\mathcal{C}$, $\mathcal{E}$, $\mathcal{R}$ modify it. A deeper question remains: *why* did the lion receive the attribute of Christ’s resurrection, the fox the attribute of diabolical cunning, the pelican the attribute of sacrificial love? The assignment of *in bono* and *in malo* attributes to specific animals is not arbitrary. It follows a psychic logic that the bestiary compilers inherited from a stratum of symbolic cognition far older than Christianity—a stratum that psychoanalytic and archetypal theory has mapped in considerable detail.

9.7.1 The Archetypal Substrate

Jung [51] proposed that certain symbolic configurations—the shadow, the anima/animus, the self, the trickster—recur across cultures not because of historical diffusion but because they correspond to structural features of the human psyche. Whether one accepts the full Jungian apparatus or not, the empirical observation is robust: certain animals carry remarkably consistent symbolic associations across civilizations that had no documented contact.

The lion is sovereign power in Mesopotamia, Egypt, Greece, Rome, Persia, India, and medieval Europe. The serpent is ambivalent knowledge—both healing and deadly—in virtually every documented culture. The eagle is transcendence, height, and divine perspective from Sumerian *Anzu* to Roman aquila to Christian evangelist symbol. The fox is cunning deception from Aesop to Reynard to Japanese *kitsune*. These associations are not learned; they are, in the terminology of evolutionary psychology, *prepared*—the human cognitive apparatus is predisposed to form them because the animals’ observable behaviors map onto psychic structures that every human being possesses.

The bestiary did not invent these associations. It *baptized* them: it took pre-existing archetypal configurations and redirected them toward Christian moral instruction. This is the operation we have called “parasitism on pre-Christian mythological memory” (Section 9). The formal framework can now be made more precise.

9.7.2 The *In Bono/In Malo* Split as Psychic Ambivalence

The most psychoanalytically revealing feature of the bestiary is the *in bono/in malo* structure itself. When the Third Family's *distinctiones* assign the lion both as Christ (*in bono*) and as the devil (*in malo*), they are not merely providing preachers with flexible material. They are encoding a fundamental property of archetypal symbols: their constitutive ambivalence.

In psychoanalytic terms, the *in bono/in malo* split corresponds to the splitting mechanism described by Melanie Klein [52]: the infant's earliest psychic operation divides the object-world into idealized good objects and persecutory bad objects. This splitting is not a pathology but a necessary developmental stage that structures all subsequent symbolization. The bestiary's systematic division of each animal into positive and negative readings is a cultural institutionalization of this psychic operation.

Consider the specific attributions:

The lion. Observable behavior: apex predator, dominance, roaring. Archetypal configuration: sovereign power. The ambivalence is intrinsic: sovereign power can protect (Christ as Lion of Judah, shepherd-king) or destroy (the devil as roaring lion, 1 Peter 5:8). The bestiary's three "natures" of the lion—concealment of tracks, sleeping with open eyes, resurrection of cubs—map onto three psychoanalytic themes: (i) the hiddenness of true power (the self conceals its nature from the ego, as Christ conceals divinity from the devil); (ii) vigilance beneath apparent passivity (the unconscious never sleeps, as divinity never ceases); (iii) the capacity of the paternal function to restore what appears dead (the father-archetype as source of psychic renewal). The moralization maps these onto Christological categories, but the psychic resonance preceded Christianity by millennia.

The fox. Observable behavior: nocturnal, uses cunning to escape pursuit, feigns death. Archetypal configuration: trickster. The fox embodies the shadow—the aspect of the psyche that operates through deception, indirection, and the exploitation of others' assumptions. In every culture that has foxes, they represent the capacity for intelligent dishonesty. The bestiary moralizes the fox as the devil or the heretic *in malo*, but the trickster archetype is not purely negative. In many traditions (Reynard, Br'er Rabbit, the Norse Loki in his benign aspects), the trickster is also the agent of necessary disruption—the force that prevents the system from rigidifying. The bestiary's purely negative moralization of the fox (*in malo* only, in most families) thus represents a deliberate *suppression* of the trickster's positive aspect in favor of a univocal moral reading—a compression that the *distinctiones* tradition partially reversed.

The pelican. Observable behavior: the pelican appears to pierce its own breast while feeding its young (actually pressing fish from its pouch against its chest). Archetypal configuration: the self-sacrificing mother. The image activates the deepest layer of attachment psychology: the mother who gives her own substance to sustain her offspring. The moralization as Christ's sacrifice maps the maternal archetype onto the Christological narrative with remarkable precision: both involve the voluntary destruction of the self for the sustenance of the other. That this particular image—the pelican in her piety (*pelicanus in pietate sua*)—became one of the most widely reproduced symbols in Christian art testifies to the depth of its archetypal resonance. It survives because it activates a psychic configuration (sacrificial parental love) that is universal, pre-cultural, and experientially immediate.

The serpent. Observable behavior: limbless locomotion, venom, skin-shedding. Archetypal configuration: ambivalent knowledge, death-and-renewal. The serpent is the most thoroughly ambivalent symbol in human culture: it is simultaneously the agent of the Fall (Genesis 3), the instrument of healing (the bronze serpent of Numbers 21, the caduceus), the symbol of wisdom ("be ye wise as serpents," Matthew 10:16), and the image of cyclical renewal (the ouroboros). No other animal carries this degree of psychic polarity. The bestiary's treatment of the serpent accordingly shows

the highest degree of *in bono/in malo* differentiation across families. In Jungian terms, the serpent represents the unconscious itself: the source of both creative and destructive energy, the force that cannot be domesticated but must be related to.

The beaver. Observable behavior: the beaver was believed to bite off its own testicles when pursued by hunters, offering them to the hunter to save its life. Archetypal configuration: renunciation of desire. The moralization is explicit: the beaver signifies the Christian who, pursued by the devil, casts away the “testicles” of sin and lust. The psychoanalytic reading is equally transparent: the beaver enacts the castration anxiety that Freud [53] identified as a fundamental structuring force in the male psyche, but it transforms this anxiety from a pathological threat into a voluntary moral act. The bestiary converts the most feared psychic operation (the loss of generative power) into the most valued spiritual operation (the renunciation of vice). This transformation is possible only because the animal image provides sufficient psychic distance: the listener can contemplate castration-as-renunciation without the full weight of personal castration anxiety, because it is the *beaver*, not the listener, who performs the act.

9.7.3 Formal Implications

The archetypal analysis has three implications for the formal framework:

First, the assignment of attributes is not arbitrary but constrained by a psychic fitness function. An animal can only carry those moral attributions that are compatible with its archetypal configuration. The lion cannot effectively signify humility (its observable behavior contradicts the attribution too violently); the lamb cannot effectively signify sovereign power (its observable behavior undermines it). The moralization function $\mu(s)$ is constrained by the archetypal pre-structure of s : only attributions compatible with the animal’s psychic resonance survive repeated cultural transmission. Formally:

$$\mu(s) \in \mathcal{M}_{\text{compatible}}(s) \subset \mathcal{M}$$

where $\mathcal{M}_{\text{compatible}}(s)$ is the subset of moral attributions that are archetypally congruent with the observable and symbolic properties of s . Attributions outside this subset may be imposed by editorial fiat but will not survive transmission: they will be silently dropped, corrected, or replaced across generations of copying.

Second, the *in bono/in malo* split is not merely a homiletic convenience but a reflection of the constitutive ambivalence of archetypal symbols. Every archetype carries both a luminous and a shadow aspect; the bestiary’s dual-reading system (*distinctiones*) formalizes this ambivalence into a pedagogically useful structure. The expansion operator $\mathcal{E}$ does not create the ambivalence; it makes explicit what was always latent in the symbol.

Third, the bestiary’s longevity is partially explained by its depth of archetypal penetration. A pedagogical system that engages only the rational surface of the psyche—that teaches facts and propositions—is vulnerable to obsolescence when the facts change or the propositions lose credibility. A system that engages the archetypal layer—that activates structures of sovereignty, sacrifice, cunning, ambivalent knowledge, and renunciation—is resistant to obsolescence because those structures do not change. The bestiary persists not only because its kernel is structurally invariant (the Persistence Triad) but because the symbolic content of that kernel resonates with psychic invariants that are older than any culture.

What it adds: The constraint $\mu(s) \in \mathcal{M}_{\text{compatible}}(s)$ explains why the moralization function is not arbitrary: it is bounded by the psychic properties of the carrier. This transforms the expansion operator $\mathcal{E}$ from an unconstrained proliferation of meanings to a structured exploration of a pre-defined possibility space—the space of archetypally compatible interpretations. It also explains the persistence of specific animal–attribute pairings across centuries: they survive because they satisfy both a cultural function (moral instruction) and a psychic function (archetypal activation).

Practical application: Any symbolic system used in pedagogy, branding, or organizational communication is constrained by the archetypal resonance of its symbols. A corporate logo featuring a serpent will inevitably carry connotations of both wisdom and danger; a mascot based on a fox will carry connotations of both cleverness and untrustworthiness. The formal constraint $\mu(s) \in \mathcal{M}_{\text{compatible}}(s)$ predicts that symbolic attributions violating archetypal congruence will fail to gain traction regardless of institutional support.

10 Dual-System Formalization and the Pedagogy of Ontological Asymmetry

The formal framework developed in the preceding sections—an invariant kernel adapted by modular operations under the constraint of the Persistence Triad—addresses the structural conditions for genre persistence. A deeper question now arises: what is the *content* that these pedagogical technologies transmit, and why does that content resist naive transmission so stubbornly?

This section proposes that the bestiary, and medieval pedagogical technologies more broadly, can be understood as mediating structures within a dual-system architecture: a system that transmits constraints from one ontological regime to agents embedded in another. The formal framework for this analysis has been developed independently in the context of Gospel Formalization—the systematic extraction of structural laws from Evangelic maxims [50]—and we present its key results here as they illuminate the bestiary’s function.

Remark 10.1 (Epistemic status of this section). This section is *theoretical and exploratory*. Unlike Sections 6–8, which formalize documented textual phenomena, the dual-system analysis proposes a structural interpretation of the *content* that the bestiary transmits. The value functions V_W and V_{K_T} are defined qualitatively (by their monotonicity properties) rather than quantitatively. The parameter λ in Proposition 10.1 is a theoretical construct indexing the agent’s evaluative orientation; it is not empirically estimable from surviving medieval evidence. The section’s contribution is to make explicit a structural feature of the bestiary’s content—its dual-system character—that is implicit in the moralizations but has not, to our knowledge, been formalized. The mathematical notation serves to clarify the logical structure of the argument, not to imply quantitative precision.

10.1 The Dual-System Architecture

Any attempt to formalize the core content of the Western moral tradition encounters a structural difficulty that does not arise when working with proverbs, legal norms, or philosophical aphorisms. In those cases, reduction to formal constraints proceeds within a single system of regularities. The Gospel—and, by extension, the moral-theological content that the bestiary encodes—presupposes the simultaneous existence of two distinct systems governed by incompatible principles yet addressing the same agent [50].

Definition 10.1 (Dual-System Architecture). A **dual-system architecture** consists of:

- A **World-System** $\mathcal{W}$: a closed, finite system characterized by scarcity, irreversible loss, and locally measurable outcomes. Its value function $V_{\mathcal{W}}$ is monotonic with respect to accumulation and security, and anti-monotonic with respect to loss and vulnerability.
- A **Kingdom-System** $\mathcal{K}_T$: an open, non-finite system characterized by non-conservation, reversibility of loss, and non-local evaluation. Its value function $V_{\mathcal{K}_T}$ may be anti-correlated with $V_{\mathcal{W}}$: actions that decrease $V_{\mathcal{W}}$ may increase $V_{\mathcal{K}_T}$, and vice versa.

The agent operates physically and causally within $\mathcal{W}$ while being evaluated normatively with respect to $\mathcal{K}_T$.

The dual-system architecture explains why the Gospel’s moral maxims appear paradoxical when evaluated within a single system. Commands such as loving one’s enemy, renouncing retaliation, and giving without expectation of return violate the stability conditions of $\mathcal{W}$. Within $\mathcal{W}$ alone, they are irrational strategies. The apparent paradox dissolves once both systems are simultaneously operative: the maxims function as conditional transition operators that override $\mathcal{W}$ -optimization by introducing an alternative evaluative regime whose outcomes are realized outside $\mathcal{W}$ ’s accounting framework.

Proposition 10.1 (Cross-System Paradox Resolution). Let a be an action available to an agent embedded in $\mathcal{W}$. If $V_{\mathcal{W}}(a) < 0$ (the action is locally suboptimal in $\mathcal{W}$) but $V_{\mathcal{K}_T}(a) > 0$ (the action is admissible or optimal in $\mathcal{K}_T$), then a is rationally justified if and only if the agent’s evaluation function incorporates both systems:

$$V_{\text{agent}}(a) = \lambda \cdot V_{\mathcal{W}}(a) + (1 - \lambda) \cdot V_{\mathcal{K}_T}(a), \quad \lambda \in [0, 1]$$

Statements such as “the last shall be first” and “whoever loses his life will save it” are not contradictions but cross-system mappings: rank inversions between $\mathcal{W}$ -ordering and $\mathcal{K}_T$ -ordering.

The parameter λ captures the agent’s degree of orientation toward the World-System. At $\lambda = 1$ (pure World-System optimization), all Gospel maxims appear irrational. At $\lambda = 0$ (pure Kingdom-System orientation), the material conditions addressed by the Gospel disappear, rendering its instructions meaningless. The Gospel presupposes $\lambda \in (0, 1)$: the agent operates within both systems simultaneously, and the resulting tension is not an accident but the genre’s constitutive condition.

10.2 The Bestiary as Dual-System Mediator

The bestiary occupies a precise position within this architecture. Its kernel $\mathcal{K}$ encodes transitions between the two systems:

- The **descriptio** (zoological description) operates within $\mathcal{W}$: it describes the natural world as empirically observed (or as believed to be observed).
- The **moralizatio** (moral interpretation) operates at the boundary of $\mathcal{W}$ and $\mathcal{K}_T$: it interprets natural phenomena as signs of a transcendent moral order.

The bestiary’s pedagogical function is to teach agents how to read the World-System as containing indicators of the Kingdom-System. Each chapter is a miniature transition exercise: the student begins with a $\mathcal{W}$ -observation (the lion sleeps with open eyes) and is guided to a $\mathcal{K}_T$ -interpretation (Christ’s divine nature remains vigilant even in death). The expansion operator $\mathcal{E}$, which transforms univocal moralizations into multi-valued *distinctiones*, multiplies the number of available transition pathways for a given $\mathcal{W}$ -observation.

10.3 Why Dual-System Content Resists Naive Transmission

A critical insight of the Gospel Formalization framework [50] is that dual-system content cannot be transmitted faithfully within a single evaluative framework. If the Kingdom-System is reduced to a moralized version of worldly logic, the content becomes ethical idealism and collapses under practical contradiction. If the World-System is dismissed as illusory, the content becomes escapism and loses operational meaning. Both reductions destroy the internal coherence of the transmitted material.

The bestiary addresses this problem through the specific cognitive properties of zoomorphic symbolism identified in Section 9. The animal image is optimally suited for dual-system pedagogy precisely because it is semantically neutral between systems: a lion described naturalistically belongs to $\mathcal{W}$, but a lion interpreted allegorically operates at the $\mathcal{W}/\mathcal{K}_T$ boundary. The animal’s cognitive low-friction and high-retention properties (Proposition 9.1) ensure that the transition between systems is achieved without the cognitive resistance that direct theological assertion would provoke.

10.4 Implications for Formal Pedagogy

The dual-system framework has implications beyond the medieval bestiary. Any pedagogical technology that transmits moral or normative content operates, implicitly or explicitly, within a dual-system architecture: the system of empirical reality (how things are) and the system of normative aspiration (how things ought to be). The bestiary’s longevity can be understood, in part, as a consequence of its structural honesty about this duality: it does not pretend to unify the two systems but provides tools for navigating between them.

Formalization of this duality—making explicit the two systems, their incompatible value functions, and the conditions under which cross-system transitions are admissible—is a structural necessity for any pedagogical technology that aspires to long-term faithfulness. Non-formal transmission of dual-system content is inherently unstable: without explicit constraint specification, the two systems inevitably collapse into one another, producing either moralism or mysticism. The bestiary’s adaptive mechanisms—compression, expansion, correction—can be understood as strategies for maintaining the dual-system distinction under changing institutional conditions.

11 Toward a Unified Structural Theory of Adaptive Systems

The formal apparatus developed in this paper—a structural kernel preserved by modular adaptive operations under persistence constraints—was not constructed *a priori*. It emerged from a sequence of domain-specific investigations whose convergent results gradually revealed a shared underlying architecture. The present section situates the bestiary analysis within this broader research program.

Remark 11.1 (On self-citation and independent validation). This paper draws on three prior works by the same author [47, 48, 49] and on one monograph [50]. This pattern of self-citation requires comment.

The situation is structurally analogous to any research program in which a series of domain-specific papers progressively reveals a common architecture. When Mandelbrot applied fractal geometry successively to coastlines, financial markets, and turbulence, the later papers necessarily cited the earlier ones; this was not circularity but the natural trajectory of a cumulative program. Similarly, when Kahneman and Tversky applied prospect theory across domains, each application cited the foundational paper. Self-citation is legitimate—indeed, unavoidable—when a researcher’s prior work provides the only available formalization of the concepts being applied. The Persistence

Triad, the Structural Distortion Principle, and the Tree-Top Meta-Method are not available in other authors’ publications because they are original contributions of this research program. Citing them is not self-promotion but accurate attribution.

The appropriate response to this situation is not to suppress citations but to invite independent scrutiny. All cited works are publicly deposited with DOIs on Zenodo and accessible for review. The present paper’s formal apparatus is fully specified within its own pages and does not require reading the cited works to be evaluated; the citations indicate *provenance*, not dependency. We welcome independent tests of the Persistence Triad against other long-lived pedagogical genres, and of the $\mathcal{C}/\mathcal{E}/\mathcal{R}$ taxonomy against other adaptive cultural systems.

11.1 From Domain-Specific Results to General Theory

The research trajectory began with investigations in domains far removed from medieval studies: dynamical systems theory, cognitive science, social systems, and astrophysics. In each domain, a common pattern emerged: systems that persist in changing environments do so by satisfying a triad of structural conditions—closure, boundary maintenance, and resilience—while deploying modular operations that adapt local configurations without destroying the invariant core [47].

The recognition that this pattern held across physical, biological, cognitive, and social domains suggested a general structural law rather than a domain-specific regularity. The methodological principle guiding this generalization—*generalize formally until generalization is no longer possible, then derive domain-specific consequences from what remains structurally true*—was formalized as the Tree-Top Meta-Method [49].

The bestiary paper was not designed to illustrate this general theory. The sequence was reversed: the domain-specific analyses came first, the general pattern became apparent, and the present paper fills a gap by demonstrating that the same structural laws govern the persistence of cultural and pedagogical systems. The medieval bestiary, far from being an exotic or marginal case, turns out to instantiate the same architecture that governs persistence in dynamical systems, cognitive architectures, and social institutions.

11.2 Cross-Domain Structural Correspondences

The following correspondences illustrate the scope of the general framework:

Structural Principle	Biological	Cognitive	Pedagogical (Bestiary)
Closure	Homeostasis	Attractors	Self-contained chapters
Boundary	Membrane	Rep. isolation	Genre distinction
Resilience	Immune system	Error correction	Adaptive modules $\{M_i\}$
Compression	Gene regulation	Memory consol.	Mnemonic verses ($\mathcal{C}$)
Expansion	Gene expression	Inference	<i>Distinctiones</i> ($\mathcal{E}$)
Correction	Horiz. gene transfer	Belief revision	Cross-family borrowing ($\mathcal{R}$)

Table 1: Cross-domain correspondences of the kernel-plus-modules architecture. Each row identifies a structural operation that appears in biological, cognitive, and pedagogical systems under different material substrates but identical formal descriptions. These correspondences are *proposed*, not proved; they represent structural hypotheses whose rigorous validation requires domain-specific investigation in each case. The table’s function is heuristic: it identifies a pattern of recurrence that motivates further inquiry, not a demonstrated isomorphism.

The Structural Distortion Principle [48]—the formal result that every act of perception is si-

multaneously an act of non-perception—appears in the bestiary context as the observation that the animal image achieves pedagogical efficacy precisely through what it *excludes*: social identity, political valence, and historical specificity. The cognitive low-friction property of zoomorphic symbolism (Section 9) is a specific instance of the general principle that adaptive systems optimize by selective omission, not by exhaustive inclusion.

11.3 The Present Paper’s Contribution

This paper contributes to the broader program by demonstrating three results:

1. The Persistence Triad, originally established for dynamical systems, applies to cultural and pedagogical genres. The bestiary’s 1400-year survival is explained by the same structural conditions that govern the persistence of biological organisms and cognitive architectures.
2. The three adaptive operations—compression, expansion, correction—are not ad hoc descriptions of bestiary-specific phenomena but domain-specific instantiations of general modular operations that any persistent system must deploy.
3. The dual-system architecture identified in the analysis of Gospel formalization illuminates a structural feature of all normative pedagogy: the transmission of constraints from one value regime to agents embedded in another.

The bestiary thus serves as a “proof of concept” for the general theory in a domain—medieval cultural production—where formalization has rarely been attempted. The fact that the same formal apparatus explains phenomena in astrophysics, cognitive science, social dynamics, and 13th-century English manuscript production suggests that the structural laws identified are genuine invariants, not artifacts of disciplinary convention.

12 Theological Legitimacy, Orthodoxy, and the Pagan Substrate

12.1 Who Was “Physiologus”? The Origin of the Tradition

The word *Physiologus* (Φυσιολόγος) is not a person’s name. It is a Greek common noun meaning “the naturalist” or “the one who speaks about nature,” derived from φύσις (nature) and λόγος (word, discourse). The title arose from the text’s recurring formula: *ho physiologos legei*—“the naturalist says”—a pseudepigraphic device by which the anonymous author appeals to an unnamed authority. In later centuries, this impersonal citation was misread as a proper name, and the text was attributed to various celebrated Church Fathers: Epiphanius of Salamis, Basil of Caesarea, Peter of Alexandria, Ambrose of Milan, and even Aristotle. None of these attributions has any historical basis. The text is anonymous.

The *Physiologus* was composed in Greek, almost certainly in Alexandria, Egypt. The dating remains disputed. The traditional date of the 2nd century CE rests on perceived parallels with the writings of Clement of Alexandria (c. 150–215), who appears to have known the text, and the *Letter of Barnabas* (early 2nd century), which employs the same method of zoological allegory on several of the same animals. Alan Scott has argued for a later date, at the end of the 3rd or in the 4th century. The current scholarly consensus places composition in the range c. 150–350 CE, with the earlier date slightly favored. Origen (c. 185–254) cited the work under the title “Physiologus,” providing a firm *terminus ante quem* of the mid-3rd century for at least some version of the text.

12.1.1 The Alexandrian Intellectual Context

The text emerged from the distinctive intellectual milieu of Hellenistic Alexandria, where Greek natural philosophy (Aristotle, Pliny the Elder, Aelian), Jewish allegorical exegesis (Philo of Alexandria), and early Christian catechesis intersected. The method of the *Physiologus*—drawing moral or theological lessons from natural phenomena—did not require the development of Origen’s systematic allegorical exegesis; the *Letter of Barnabas* offered a sufficient model before Origen’s time. The text drew on a tradition of pseudepigraphic natural histories, including the works attributed to Bolos of Mendes (3rd century BCE), and compiled folklore from Indian, Hebrew, Egyptian, and Greco-Roman traditions.

12.1.2 The Gnostic Question

The Alexandrian context raises a question that the *Decretum Gelasianum* would later make explicit: did the *Physiologus* carry Gnostic influences? Alexandria was the epicenter of Gnostic Christianity in the 2nd–3rd centuries. Valentinus taught there; the Basilideans flourished there; the *Gospel of Thomas*, with its emphasis on hidden knowledge revealed through natural symbols, circulated in the same milieu. The *Physiologus*’s core method—that the visible natural world encodes hidden spiritual truths accessible through allegorical interpretation—is structurally compatible with Gnostic epistemology, which posited hidden knowledge ($\gamma\nu\tilde{\omega}\sigma\iota\varsigma$) encoded in the material world by a demiurge.

This does not mean the *Physiologus* was a Gnostic text. Its theology is broadly orthodox: the moralizations point to Christ, the Church, and the Devil, not to aeons, archons, or pleroma. But the *method*—the assumption that every natural creature carries an encoded spiritual message—occupies a conceptual space that is shared by orthodox allegorical exegesis and Gnostic symbolic interpretation. It was this ambiguity that made the text vulnerable to condemnation. The boundary between “reading nature as God’s second book” (orthodox) and “reading nature as a system of hidden knowledge” (potentially heterodox) was thin, and the *Physiologus* sat precisely on it.

12.2 The *Decretum Gelasianum*: The Papal Prohibition

The condemnation of the *Physiologus* appears in the *Decretum Gelasianum de libris recipiendis et non recipiendis* (“Gelasian Decree concerning books to be received and not received”), a catalogue of approved and condemned texts traditionally attributed to Pope Gelasius I (pontificate 492–496). The decree lists the *Physiologus* among texts to be rejected:

Liber qui appellatur Physiologus, ab haereticis conscriptus et beati Ambrosii nomine praesignatus, apocryphus.

[Translation: The book called *Physiologus*, composed by heretics and falsely attributed to blessed Ambrose, is apocryphal.]

The decree’s authorship is disputed. Modern scholarship regards it as a compilation of the late 5th or early 6th century, incorporating material from several sources, possibly including a genuine Gelasian component. The attribution to Gelasius I gave it considerable authority in the early medieval West, though it was never universally enforced.

The condemnation specifies two grounds: (1) the text was “composed by heretics” (*ab haereticis conscriptus*)—a probable reference to the Alexandrian milieu with its Gnostic associations; and (2) it was “falsely attributed to blessed Ambrose” (*beati Ambrosii nomine praesignatus*)—a reference to the pseudepigraphic attributions that sought to lend the text patristic authority. The decree does

not identify which specific heresies are at issue, but the Alexandrian provenance and the text’s method of reading hidden spiritual meanings in natural phenomena provide the most likely context.

The paradox of the condemnation is that it failed. The *Physiologus* continued to be copied, translated, and adapted throughout the early medieval period, eventually generating the Latin bestiary tradition that is the subject of this paper. The formal framework explains why: the kernel $\mathcal{K}$ was too cognitively powerful and pedagogically useful to suppress, and the Western tradition found a way to retain the kernel while replacing the suspect cosmological substrate—the selective $\mathcal{R}$ operation described below.

12.3 The Western Resolution: Baptism of the Kernel

The Western Latin tradition resolved the tension between the *Physiologus*’s suspected heterodoxy and its undeniable pedagogical utility by a strategy that the formal framework makes precise: it *preserved the kernel* $\mathcal{K}$ (the species–descriptio–moralizatio triad) while *replacing the cosmological substrate*. The *Physiologus*’s implicit Alexandrian natural philosophy was gradually overwritten by Isidorian encyclopedism (from the *Etymologiae*) and patristic exegesis (from Ambrose, Augustine, Gregory). The moralization function $\mu(s)$ was progressively aligned with orthodox sources, even as the zoological descriptions $D(s)$ retained their Hellenistic content. In formal terms, the adaptive operation was a selective correction of μ that left D intact—a partial application of $\mathcal{R}$ to the moralization layer only.

12.4 The Greek *Physiologus* and Its Fate in the Orthodox World

The original Greek *Physiologus* survives in multiple recensions, classified by Sbordone (1936) into four groups, with over 77 manuscripts known (more than 20 additional manuscripts have been identified since Sbordone’s edition). The text continued to be copied in the Greek-speaking world throughout the Byzantine period, and translations of the so-called Byzantine redaction were made into Middle Bulgarian (13th–14th century, edited by Ana Stoykova, 2011), Serbian, Russian, and Romanian.

However—and this is the critical contrast with the Latin West—the Greek *Physiologus* *did not evolve into a bestiary*. It remained a *Physiologus*: a relatively compact text of 20–60 chapters, preserving the original C_1 structure (one moralization per species, typically univocal), without the adaptive operations ($\mathcal{C}$, $\mathcal{E}$, $\mathcal{R}$) that transformed the Latin tradition. No Greek equivalent of the *distinctiones*-enriched Third Family, the cross-family corrected KK 4.25, or even the Isidorian-expanded Second Family has been identified. The Byzantine copies are, formally speaking, arrested at C_1 : they preserve the kernel without developing the modular adaptive apparatus.

Several structural factors explain this developmental arrest:

First, the Orthodox theological tradition maintained a stronger distinction between the symbolic reading of Scripture (systematized in patristic exegesis) and the symbolic reading of nature (not granted comparable authority). The Western *liber naturae* doctrine, which the bestiary presupposed, was less thoroughly developed in the East.

Second, the Orthodox liturgical tradition was primarily hymnographic and iconographic rather than homiletic in the Latin sense. The sermon (*homilia*) played a less central role in parish instruction than in the Western mendicant-driven pastoral revolution of the 13th century. The institutional demand for preaching aids—the demand that generated the *distinctiones*, the *exempla* collections, and the bestiary’s expansion—was correspondingly weaker.

Third, the Orthodox tradition was more suspicious of the pagan substrate. The Hellenistic zoological material was recognized as originating outside the patristic canon. Without a comprehensive

program of “baptism”—the systematic replacement of the cosmological substrate while preserving the pedagogical kernel—the genre remained under theological suspicion. The Eastern tradition did not perform the selective $\mathcal{R}$ operation on μ that the Western tradition accomplished.

Fourth, the Orthodox world preserved direct access to the Greek patristic sources (Basil’s *Hexameron*, Gregory of Nyssa, John of Damascus) that the Western bestiary laboriously compiled from translations and excerpts. Where the West needed a compendium of animal symbolism because its clergy could not read Greek patristics in the original, the East had less need for a derivative pedagogical tool.

12.5 Eastern Translations: Syriac, Arabic, Armenian, Georgian, Ethiopic

The *Physiologus* was translated into the major languages of Eastern Christianity at an early date, producing a family of versions that illuminate the cross-cultural portability (and non-portability) of the kernel.

Syriac. A Syriac translation was made no later than the 5th century, possibly from a Greek exemplar closely related to the earliest recension. Two Syriac versions are known: an earlier one published by Tychsen (1795) and a later, expanded version published by Land (1875). The Syriac *Physiologus* preserves the kernel structure but adapts certain moralizations to the theological vocabulary of the Syriac-speaking Church, providing evidence of an early $\mathcal{R}$ operation on μ across linguistic boundaries.

Arabic. An Arabic version exists in a few medieval manuscripts (plus 17th-century copies). Its origin is unclear: it may derive from a Coptic intermediary or from the Syriac. The Arabic *Physiologus* circulated in a cultural environment where the *Kalīla wa-Dimna* (the Arabic translation of the Indian *Pañcatantra*) already provided a sophisticated animal-fable tradition with its own moral apparatus. The two traditions did not merge, suggesting that the kernel $\mathcal{K}$ of each was sufficiently distinct to resist hybridization.

Armenian. The Armenian translation, made in the 5th century (probably directly from the Greek), survives in over 40 manuscripts, making it one of the richest manuscript traditions of the *Physiologus* outside the Latin West. The Armenian version is titled *Baroyxos* (Βαρουχος, “Moralist”), indicating that the Armenian translators understood the text’s function as moral instruction. A critical edition by Gohar Muradyan has been published. The “Eastern” versions, including the Armenian, are notably *not illustrated*—in sharp contrast to the richly illuminated Latin bestiaries, confirming that the illustration program was a Western addition to the kernel, not part of the original $\mathcal{K}$.

Georgian. A Georgian translation was made from the Armenian (not directly from the Greek), surviving in a single manuscript, the 10th-century Shatberdi miscellany. According to Jost Gippert, the translator was probably an Armenian not fully proficient in Georgian. This translation chain (Greek $\rightarrow$ Armenian $\rightarrow$ Georgian) illustrates the operation of $\mathcal{R}$ across multiple linguistic boundaries, with cumulative losses and adaptations at each step.

Ethiopic. An Ethiopic translation was made in the 5th century, surviving in at least four manuscripts (16th–18th century copies). The Ethiopic version is notable for its close correspondence with Version C of the Latin *Physiologus*, suggesting a shared Greek ancestor distinct from the Byzantine redaction.

12.6 The Icelandic *Physiologus*: The Northern Frontier

At the opposite geographic extreme from Alexandria, the *Physiologus* reached medieval Iceland, where it produced one of the most remarkable adaptations in the tradition. The Icelandic *Physio-*

logus survives in two fragmentary manuscripts, both dating from c. 1200, making them among the earliest Icelandic manuscripts extant and the *earliest illustrated manuscripts from Iceland*.

Fragment A (AM 673 a I 4^o) contains five entries with moral interpretations and illustrations. Fragment B (AM 673 a II 4^o) contains eighteen entries with descriptions and allegorical meanings. Both originate from the Westfjord districts of northwestern Iceland, though they cannot be precisely localized. Del Zotto Tozzoli, the manuscript's most recent editor, has suggested that these texts are closer to bestiaries than to the *Physiologus* proper.

The Icelandic adaptation displays three striking features:

First, ecological filtering. The entries focus heavily on birds and common forest-dwelling animals—species that Icelanders would have recognized—with the notable inclusion of the whale, an animal of enormous practical importance in Icelandic life (whaling laws are attested from the mid-10th century). The most conspicuous absence is the lion, which opens virtually every European bestiary as a Christ figure. In its place, the Icelandic version begins with the phoenix—a mythical creature, but one whose symbolism (death and resurrection) parallels the lion's without requiring familiarity with an animal that no Icelanders had ever seen. This is a conscious $\mathcal{R}$ operation: replacing $D(\text{lion})$ with $D(\text{phoenix})$ while preserving $\mu(\text{Christ's Resurrection})$.

Second, moralization priority. Fragment A apparently dispensed entirely with zoological descriptions, retaining only the moral symbolism and substituting illustrations for descriptions. This represents an extreme compression: $\mathcal{C}$ applied so aggressively that the D layer is eliminated, leaving only μ and image.

Third, cultural isolation. The Icelandic fragments represent an independent line of transmission, probably via a Latin intermediary, into a society at the extreme periphery of the Latin cultural sphere. Their existence confirms that the kernel $\mathcal{K}$ could cross not only linguistic but ecological and cultural boundaries—though at the cost of significant adaptation.

12.7 The Irish Question: Why No Irish Bestiaries?

No Irish Latin bestiary has been identified, despite the fact that Ireland was one of the most productive centers of manuscript culture in the early medieval period (6th–9th centuries). The *Book of Kells*, the *Book of Durrow*, and other Insular manuscripts display elaborate zoomorphic decoration whose symbolic program remains only partially decoded. Did an Irish bestiary tradition exist?

The formal framework predicts that the structural demand for such a tradition was fully satisfied in early medieval Ireland: content requiring transmission (Christian theology), heterogeneous recipients (the Irish Church's dual structure of monastic and lay instruction), and institutional infrastructure (the great monasteries of Iona, Clonmacnoise, Kildare, and others). The non-survival of an Irish bestiary admits several explanations:

(1) *Viking destruction.* The Viking raids of the 9th–10th centuries caused catastrophic manuscript losses in Irish monasteries. Texts of lower institutional priority (a moralized animal compendium, as opposed to liturgical books, gospel books, and legal texts) would have been among the most vulnerable.

(2) *The Insular tradition's preference for Scripture-integrated zoomorphism.* Irish manuscript art embeds animal symbolism directly into gospel decoration (the evangelist symbols of Matthew/man, Mark/lion, Luke/ox, John/eagle) rather than isolating it in a separate genre. The zoomorphic interlace of Insular art may represent a different formalization of the same kernel—one integrated into liturgical books rather than separated into a standalone genre.

(3) *Chronological mismatch.* The bestiary's period of maximum production (c. 1170–1300) coincided with the Anglo-Norman colonization of Ireland. The English bestiary tradition may simply

not have been transplanted to Ireland in significant quantity, and the pre-Norman Irish tradition had developed alternative forms for zoological-moral instruction.

12.8 The Spanish Absence

The Iberian Peninsula presents a similar puzzle. Despite its rich manuscript culture and its unique position at the intersection of Latin Christian, Arabic Islamic, and Jewish traditions, virtually no Spanish Latin bestiaries survive. The few Iberian manuscripts containing *Physiologus* material are peripheral witnesses, not fully developed bestiaries.

The most likely explanation is institutional: the bestiary was a distinctly *English* phenomenon, driven by the specific institutional configuration of English cathedral schools, mendicant networks, and preaching reform. The *distinctiones* tradition that catalyzed the bestiary’s expansion ($\mathcal{E}$) was centered at Paris and Lincoln, not at Toledo or Salamanca. The Spanish preaching tradition drew on different sources (*exempla* collections, hagiographic compilations) that fulfilled the same pedagogical function without employing the bestiary’s kernel. Al-Andalus contributed the *Kalila wa-Dimna* tradition, which occupied the ecological niche that the bestiary might otherwise have filled.

12.9 How Many Bestiaries Survive? The 100 vs. 150 Debate

The number of surviving Latin bestiaries has been variously cited as “about 40” (Christopher de Hamel, 1990), “over 50” (Wikipedia, citing the Aberdeen Bestiary), “62” (the Getty Museum’s *Book of Beasts* catalogue, 2019, counting illuminated bestiaries), “under 90” (the Ashmole Bestiary article), “about 100” (Dr. Dines, personal communication, February 2026), and “approximately 150” (various sources, including this paper’s previous drafts).

The discrepancy reflects genuine ambiguity in the definition of “bestiary.” The count depends on several classification decisions:

What counts as a bestiary? If only fully developed, self-standing Latin bestiaries of the Second and Third Family type are counted, the number is closer to 60–90. If *Physiologus*-type manuscripts (BIs Family, Transitional), fragmentary witnesses, and bestiaries embedded in miscellanies are included, the number rises toward 150. The bestiary.ca database lists hundreds of manuscripts containing bestiary-related material, but many are partial witnesses or miscellanies containing only extracts.

Do Physiologus manuscripts count? The Latin *Physiologus* (Versions A, B, C, Y, Dicta Chrysostomi) is the bestiary’s ancestor, not the bestiary itself. Strictly, the bestiary tradition begins when Isidorian material is added to the *Physiologus* base. If *Physiologus* manuscripts are excluded, the count drops significantly.

Dr. Dines, who has personally examined more bestiary manuscripts than any other living scholar, estimates the count of genuine Latin bestiaries at approximately 100 (personal communication, February 2026). This estimate excludes *Physiologus* manuscripts, fragmentary witnesses of only a few folios, and miscellanies that incorporate bestiary extracts without constituting bestiaries. The present paper adopts Dr. Dines’s estimate as the most authoritative, while noting that the larger figure of 150 represents the total population of manuscripts containing bestiary-type material in any form.

Dr. Dines further observes that the bestiary’s extinction is closely linked to the destruction of European monasteries—a point of considerable importance for the formal framework (personal communication, February 2026). The dissolution of the English monasteries under Henry VIII (1536–1541), the suppression of religious houses across northern Europe during the Reformation,

and the secularization of monastic libraries in southern Europe during the 18th–19th centuries destroyed the institutional infrastructure that had sustained the bestiary tradition. In formal terms, the Persistence Triad’s *boundary maintenance* condition failed when the institutions that maintained the boundary—the monasteries and their libraries—ceased to exist.

12.10 The Monastic Audience: For Whom Were Bestiaries Made?

The question of the bestiary’s intended audience is more complex than it initially appears. The evidence supports a primarily monastic and clerical readership, but with important qualifications.

Evidence for monastic use: Several bestiaries bear evidence of monastic provenance: ownership inscriptions, library catalogue entries, and binding evidence linking them to specific religious houses (Benedictine, Cistercian, Augustinian). The integration of *distinctiones* and preaching aids points to users who were preparing sermons. The Latin language itself restricts the readership to the Latinate—overwhelmingly clerics.

Evidence for educational use with younger religious: The mnemonic verses (Section 4) point specifically to a pedagogical context involving memorization, consistent with the training of novices or young monks. The relatively straightforward Latin of most bestiaries, compared with the dense technical language of scholastic theology, suggests a readership at an intermediate level of Latin competence—consistent with monastic students rather than advanced scholars. The wear patterns on some bestiary manuscripts (heavy use at the beginning, lighter toward the end) are consistent with classroom use where the early chapters were most frequently taught.

Evidence for broader use: However, several illuminated bestiaries are of such high quality (gold leaf, elaborate pigments, professional illustration programs) that they cannot have been simple teaching tools for novices. The Aberdeen Bestiary and the Ashmole Bestiary are luxury productions, likely commissioned for wealthy patrons or prestigious institutional libraries. These manuscripts served a display function alongside their pedagogical one.

The Cambridge Digital Library’s assessment notes that “perhaps the Latin bestiaries were primarily read by priests and men in the religious orders. Several belonged to religious houses or to individual Benedictines, Cistercians, Augustinian canons, Dominicans or Franciscans. The bestiary may have been more widely read by them than by members of the laity, for whose ownership, particularly of Latin bestiaries, there is less evidence.”

The formal framework permits a reconciliation: the bestiary served multiple audiences at different complexity levels, and this is precisely the function of the C_1 – C_4 scale. A C_1 bestiary with simple moralizations could serve as a novice’s introduction to moral theology; a C_3 bestiary with *distinctiones* served as a preacher’s reference tool; a luxuriously illustrated bestiary served as a library showpiece and a visual meditation aid. The same genre, at different complexity levels, addressed different institutional needs—which is why the complexity scale was necessary in the first place.

What it adds: This section establishes four critical points: (1) the *Physiologus*’s origin in Alexandrian Christianity, at the boundary between orthodox exegesis and Gnostic epistemology, explains why the text attracted condemnation; (2) the *Decretum Gelasianum*’s failure to suppress the text demonstrates the kernel’s cognitive robustness; (3) the West/East divergence constitutes a natural experiment testing the Persistence Triad; (4) the Eastern translations confirm the kernel’s cross-linguistic portability while demonstrating μ -layer adaptation. Additionally: the Icelandic adaptation tests ecological filtering; the Irish and Spanish absences test institutional demand; the manuscript count clarifies the empirical base; and the audience question resolves through the C_1 – C_4 scale.

Practical application: The Gelasian condemnation failed because the kernel’s pedagogical utility outweighed the institutional objection—a prediction the formal framework makes: any pedagogical technology satisfying the Persistence Triad will resist suppression if its kernel addresses a genuine cognitive need. This applies to contemporary debates over educational technologies condemned by institutional authorities (social media in education, AI tutoring systems, gamification of learning).

13 Modern Pastoral Use of Bestiary Material

13.1 Contemporary Deployment

The bestiary is not merely a historical artifact; its material continues to circulate in contemporary pastoral practice, though in forms that the medieval compilers would scarcely recognize.

In the Roman Catholic tradition, bestiary imagery persists in liturgical art (the pelican *in pietate sua* remains a standard Eucharistic symbol), in homiletic illustration (preachers continue to use the pelican, the phoenix, and the lamb as typological images), and in catechetical materials designed for children and adult converts. The *Catechism of the Catholic Church* (1992) does not cite the bestiary directly but deploys the *liber naturae* doctrine that the bestiary presupposed: “God speaks to man through the visible creation” (§1147). The bestiary’s kernel—natural observation leading to spiritual interpretation—thus remains operative at the level of theological principle, even though the genre as such has been superseded.

In Protestant traditions, bestiary material circulates primarily through the Aesopian and emblematic traditions that absorbed it after the Reformation. The emblem books of the 16th and 17th centuries (Alciato, Camerarius, Joachim Camerarius the Younger’s *Symbolorum et Emblematum* series) drew heavily on bestiary zoology and moralization, redirecting it toward Protestant devotional and moral instruction. Contemporary evangelical and Pentecostal preaching occasionally deploys bestiary-derived imagery—the eagle as spiritual aspiration (Isaiah 40:31), the serpent as spiritual danger—without awareness of its medieval genealogy.

In Anglican and mainline Protestant traditions, the bestiary’s material has been largely absorbed into literary and artistic culture rather than pastoral practice. C.S. Lewis’s *The Chronicles of Narnia*—in which Aslan (a lion) is an explicit Christ-figure who dies and is resurrected—is the most influential modern deployment of bestiary typology, and its pedagogical structure replicates the bestiary’s kernel almost exactly: a zoological image (D : the lion Aslan) is linked to a spiritual interpretation (μ : Christ’s sacrifice and resurrection).

13.2 What Has Been Lost

Modern pastoral deployments of bestiary material typically operate at C_1 (univocal moralization) and have lost the adaptive operations that sustained the medieval tradition. The pelican-as-Eucharist is a single, fixed attribution; no modern catechist treats it as a node in a *distinctio*

with multiple *in bono* and *in malo* readings across four hermeneutical levels. The expansion operator $\mathcal{E}$ has been lost—and with it, the homiletic flexibility that made the bestiary a working tool for preachers rather than a static symbol.

This loss is not inevitable. The formal framework developed in this paper could, in principle, be applied to reconstruct the expansion operation for contemporary pastoral use: any animal image currently deployed at C_1 could be enriched with a systematic *distinctio* that provides multiple interpretive pathways for different pastoral contexts. Whether this would constitute a legitimate recovery of a lost pedagogical technology or an artificial imposition of a dead genre on living practice is a question that lies beyond the scope of this paper but within the scope of practical pastoral theology.

What it adds: The continued circulation of bestiary material at C_1 in modern pastoral contexts—without the adaptive operations that sustained the medieval tradition—constitutes a degraded instantiation of the kernel. The formal framework predicts that C_1 -only deployment should be less durable than deployment with the full $\mathcal{C}/\mathcal{E}/\mathcal{R}$ apparatus. This is testable: compare the longevity of catechetical materials using zoomorphic symbolism with and without multiple interpretive pathways.

Practical application: The framework provides a concrete recipe for pastoral resource design: take any existing C_1 symbol (pelican = Eucharist), apply $\mathcal{E}$ systematically (generate *in bono/in malo* readings across literal, allegorical, tropological, anagogical levels), and produce a working *distinctio* that gives the preacher interpretive flexibility.

14 Cross-Cultural Variation in Animal Symbolism: Universality and Divergence

14.1 The Universality Thesis and Its Limits

Section 9.7 argued that certain animal–attribute associations are archetypally grounded and cross-culturally robust. This claim requires qualification: while some associations show remarkable universality, others diverge radically across cultures, and these divergences are as informative for the formal framework as the convergences.

Universal or near-universal associations: The lion as sovereign power is documented from Mesopotamia to medieval Europe to modern corporate branding. The serpent as ambivalent knowledge-and-danger appears in every major cultural tradition. The eagle as height, vision, and transcendence recurs from Sumerian mythology through Roman imperial symbolism to the coats of arms of modern nation-states. These associations are so robust across cultures with no documented contact that they must be attributed to the convergence of the animals’ observable behavior with universal features of human cognitive architecture.

Radically divergent associations: The owl is wisdom in Greek and Western traditions (Athena’s companion) but a harbinger of death in Chinese, Japanese, Aztec, and many African traditions. The dog is fidelity and companionship in European culture but ritually unclean (*najis*) in Islamic jurisprudence and associated with the underworld in Mesoamerican traditions. The cat is sacred in ancient Egypt, associated with witchcraft in medieval Europe, and a symbol of good fortune in Japan (*maneki-neko*). The pig is domestic prosperity in Chinese culture and absolute abomination in Jewish and Islamic dietary law.

14.2 What Divergence Reveals

These divergences are not random. They follow a pattern that the formal framework can articulate: divergence occurs precisely where the animal’s observable behavior is *ambiguous with respect to the psychic function it is asked to serve*, and the culture’s specific historical and ecological context resolves the ambiguity in one direction or another.

The owl is nocturnal and solitary. These behaviors are compatible with two opposite psychic functions: the sage who sees in darkness (wisdom) and the death-messenger who inhabits the realm of the unseen (mortality). Greek culture, which valorized intellectual contemplation, resolved the ambiguity toward wisdom. East Asian cultures, in which darkness was more strongly associated with yin and the spirit-world, resolved it toward death-portent. Neither resolution is “correct” in an absolute sense; both exploit a genuine feature of the owl’s behavioral profile.

In formal terms, the archetypal constraint $\mu(s) \in \mathcal{M}_{\text{compatible}}(s)$ defines a possibility space, not a unique assignment. The compatible set $\mathcal{M}_{\text{compatible}}(\text{owl})$ includes both $\{\text{wisdom}\}$ and $\{\text{death-portent}\}$; culture selects which element of the compatible set becomes the dominant moralization. Where the compatible set contains elements that are mutually exclusive within a given value system, different cultures will select different elements—producing the observed divergence.

This analysis has a sharp implication for the bestiary’s pedagogical value: the bestiary is *not* a universal pedagogical technology in the naive sense. It is universal in its *mechanism* (zoomorphic symbolism with moral interpretation) but culturally specific in its *content* (which animals signify what). A bestiary produced for a Chinese audience in the 13th century would have needed to assign the owl to *in malo* (death-portent), not *in bono* (wisdom)—the opposite of the Western *distinctio*. The kernel $\mathcal{K}$ is cross-culturally portable; the moralization function μ is not.

14.3 Implications for Pedagogical Design

This finding has direct practical consequences. Any attempt to use zoomorphic symbolism in cross-cultural pedagogical contexts—international curricula, multinational corporate training, missionary catechesis—must reckon with the fact that the archetypal compatibility set $\mathcal{M}_{\text{compatible}}(s)$ is culturally filtered. The formal framework predicts that pedagogical systems deploying culturally dissonant animal attributions will fail: a training program using the dog as a symbol of loyalty will encounter resistance in cultural contexts where the dog is polluted or inauspicious.

The medieval bestiary itself encountered this problem when it crossed cultural boundaries. The Latin *Physiologus* was translated into Ethiopic, Armenian, Syriac, Arabic, and other non-European languages, but in each case the moralization function was partially modified to accommodate local symbolic norms. These modifications constitute further instances of the correction operator $\mathcal{R}$ —now operating not across textual families but across cultures.

The deeper lesson is that the kernel $\mathcal{K}$ (observation $\rightarrow$ description $\rightarrow$ interpretation) is genuinely universal: every culture observes animals, describes their behaviors, and draws moral or spiritual lessons from them. But the specific mapping $\mu : S \rightarrow \mathcal{M}$ is culturally conditioned, and any pedagogical technology that treats it as fixed and universal will fail outside its culture of origin. The bestiary’s longevity in the West was possible precisely because it operated within a single, relatively stable cultural and theological framework; its non-adoption in the East (Section 12) and the modifications required for non-European translations demonstrate the limits of cross-cultural portability.

What it adds: The divergence data transforms the compatibility constraint $\mu(s) \in \mathcal{M}_{\text{compatible}}(s)$ from a theoretical postulate into an empirically testable prediction: cultural divergence should occur within, not outside, the compatibility set. If an attested moralization falls outside $\mathcal{M}_{\text{compatible}}$, it should show short persistence.

Practical application: Any organization deploying symbols cross-culturally (multinational branding, international NGO communications, cross-cultural education) must audit its symbols against the compatibility sets of target cultures. The bestiary’s experience predicts that ignoring this constraint produces cultural rejection.

15 The Bestiary and World Mythological Traditions: From Maya to Aboriginal Australia

15.1 The Scope of Comparison

The bestiary is one instance of a phenomenon that appears, in structurally recognizable forms, in virtually every documented human culture: the systematic use of animal narratives to encode, transmit, and enforce moral, cosmological, or social norms. This section surveys the global range of this phenomenon, identifies structural parallels and divergences, and draws implications for the formal framework.

Remark 15.1 (On interdisciplinary questioning). The comparison that follows ranges across cultures traditionally studied by separate academic disciplines: medieval Latin philology, Mesoamerican archaeology, Australian Aboriginal anthropology, West African oral literature, Hindu mythology, and cognitive science. This range will strike some readers as overambitious or dilettantish. We address this concern directly.

Interdisciplinary inquiry is not a luxury but a structural necessity for the questions this paper asks. The claim that the bestiary’s kernel $\mathcal{K}$ reflects a universal mechanism of moral pedagogy *cannot be tested within a single disciplinary tradition*. It can only be tested by examining whether the same formal structure appears in traditions that are genealogically independent of the Latin bestiary. The alternative—restricting the analysis to the Western tradition alone—would make the universality claim unfalsifiable by construction.

It is not possible to determine whether a structural pattern is universal by studying only one instance of it. The comparative method, despite its risks (superficial analogy, decontextualization, neo-colonial imposition of Western categories), remains the only available instrument for testing claims of cross-cultural structural invariance. We proceed with this method while acknowledging its limitations and inviting correction from specialists in each tradition discussed.

No discipline has the right to forbid questions that fall across disciplinary boundaries. The questions exist because the phenomena exist; the administrative division of knowledge into departments does not divide the phenomena themselves.

15.2 Maya and Mesoamerican Traditions

The Maya *Popol Vuh* (“Book of Counsel”), the Quiché Maya creation narrative, features animal characters in structurally significant roles. The Hero Twins Hunahpú and Xbalanqué descend to Xibalbá (the Underworld) accompanied by zoomorphic guides and tricksters. The mosquito, the firefly, the owl, and the jaguar each carry specific narrative and cosmological functions. The owl (*Tecolotl*) serves as death-messenger—consistent with the Mesoamerican resolution of the owl’s archetypal ambiguity (Section 14).

The formal structure is recognizable: a zoological observation (the owl’s nocturnal behavior, its cry near human habitations) is linked to a cosmological interpretation (the owl announces death, serves as emissary of the underworld lords). The kernel $\mathcal{K}_{\text{Maya}} = (\text{animal, behavior, cosmological role})$ is structurally isomorphic to the bestiary’s $\mathcal{K} = (\text{species, descriptio, moralizatio})$, differing in the target interpretive system (cosmological vs. moral-theological) but not in the formal architecture.

The Aztec tradition intensifies this pattern. The *Tonalamatl* (divinatory calendar) assigns specific animals to specific day-signs, each carrying moral and prognostic significance: the jaguar (*ocelotl*) = warrior courage and danger; the serpent (*coatl*) = wisdom and destructive power; the eagle (*cuauhtli*) = solar aspiration and military valor. These are not decorative associations but systematic mappings from zoological observation to moral-cosmological interpretation—the kernel $\mathcal{K}$ in Mesoamerican instantiation.

15.3 Australian Aboriginal Traditions

The Aboriginal Australian “Dreamtime” (*Tjukurpa* in Pitjantjatjara, *Jukurrpa* in Warlpiri) provides perhaps the most radical test of the universality thesis. Aboriginal traditions developed in complete isolation from all other civilizations for at least 40,000 years. If the kernel $\mathcal{K}$ appears here, its universality cannot be attributed to diffusion.

It does. The Dreamtime narratives are organized around ancestral beings who are simultaneously animal and human, natural and supernatural. The Rainbow Serpent (*Wanambi*) is the most widespread: a water-associated serpent that created landscape features, enforces social norms, and punishes transgressions. The formal structure: observable natural feature (a waterhole, a rock formation) $\rightarrow$ narrative description (the Serpent created it during the Dreamtime) $\rightarrow$ moral-social norm (the waterhole is sacred; violation brings punishment). This is $\mathcal{K} = (S, D, \mu)$ with S = landscape features, D = creation narrative, μ = normative prescription.

The kangaroo, emu, goanna, and crow each carry specific moral attributes in their respective clan narratives. These attributes are transmitted through elaborate initiatory sequences that combine narrative, visual art (rock painting, body decoration), song, and enacted performance—a multi-modal pedagogical technology that deploys the kernel across multiple sensory channels simultaneously. The bestiary’s reliance on text and illustration alone appears, by comparison, as a narrowed instantiation of a broader cognitive pattern.

15.4 West African Oral Traditions

The Anansi spider stories (Akan/Ashanti tradition) and the animal trickster cycles of the Yoruba, Hausa, and Bantu traditions provide a third independent test. Anansi, like the bestiary fox, embodies the trickster archetype: cunning, boundary-violation, ambiguous morality. The hare (*Sungura* in Swahili traditions) serves a parallel function across East Africa.

The critical structural observation is that these traditions operate at C_3 complexity without any contact with the Western *distinctiones* tradition. The same animal (Anansi) is moralized *in bono* (cleverness as survival strategy for the powerless) and *in malo* (cleverness as selfishness and social disruption) within the same narrative tradition, depending on the specific story and pedagogical context. The expansion operator $\mathcal{E}$ has been independently deployed: $\mu(\text{Anansi}) = \{m_{\text{bono}}, m_{\text{malo}}\}$.

15.5 Hindu and Buddhist Traditions

The *Pañcatantra* (“Five Treatises,” c.3rd century BCE) is the most structurally explicit non-Western analogue. Its frame narrative organizes animal fables into five thematic books, each

transmitting a set of principles of *nīti* (practical wisdom, statecraft). The formal structure is: animal characters in narrative scenarios → explicit moral maxim (*nīti-sūtra*). This is $\mathcal{K} = (\text{scenario}, \text{narrative}, \text{maxim})$ —structurally identical to the bestiary kernel, with “scenario” replacing “species” and “maxim” replacing “moralization.”

The Buddhist Jātaka tales (c. 3rd century BCE onward) add a distinctive element: the animal protagonist *is* the Buddha in a previous birth. The moralization is therefore simultaneously zoological, ethical, and soteriological. The formal structure: animal behavior → narrative → lesson about the path to Enlightenment. The kernel is preserved; the interpretive system (μ) is Buddhist rather than Christian.

15.6 Formal Implications: The Universal Kernel

The survey confirms the prediction of Section 19: wherever conditions (i)–(iii) are satisfied—content requiring transmission, heterogeneous audience, institutional infrastructure—a pedagogical genre with the kernel $\mathcal{K} = (S, D, \mu)$ emerges independently. The kernel is not a Western invention but a cognitive universal.

The cross-cultural evidence also confirms the three Formalization Laws (Section 20):

- *Total Formalizability*: Every tradition surveyed has rendered its animal symbolism into an explicit, transmissible structure with identifiable components.
- *Boundary-Conditioned Verification*: The same animal receives different attributions in different cultural boundary conditions.
- *Plurality and Incompleteness*: No single tradition exhausts the interpretive potential of its zoomorphic vocabulary; traditions operating at C_3 (Anansi, Pañcatantra, Third Family) acknowledge this explicitly.

What it adds: The cross-cultural survey transforms the bestiary from a Western medieval artifact into one instance of a human universal. The formal kernel $\mathcal{K}$ is confirmed in traditions separated by 40,000+ years of independent development (Aboriginal Australia), 10,000+ km of geographic distance (Maya, West Africa), and radically different cosmological frameworks (Hindu, Buddhist, Christian, Aboriginal). This level of convergence cannot be explained by diffusion; it must reflect a structural feature of human cognition.

Practical application: Contemporary multicultural education programs can draw on this universal kernel structure to design pedagogical materials that are genuinely cross-cultural—not by imposing one culture’s moralizations on another, but by using the shared formal architecture ($\mathcal{K}$) while allowing each community to supply its own μ .

16 Reconstructing Lost Bestiaries: Unknown Histories and Cosmological Substrates

16.1 The Possibility of Entirely Unknown Bestiaries

Appendix E estimated that the 150 surviving manuscripts represent perhaps 7.5–19% of total production. The “missing” manuscripts are predominantly simple (C_1 , C_2) and likely from dissolved institutions. But the Monte Carlo model raises a more provocative possibility: could there be

bestiary-type texts that are not merely lost copies of known versions but *entirely unknown compositions*—texts that were produced, used, and lost without leaving any trace in the surviving record?

The formal framework makes this question tractable. If the conditions for structural necessity (Section 19) were satisfied in contexts beyond the well-documented English tradition, then independent bestiary-type compositions should have emerged in those contexts—and some of them may have been lost without direct or indirect witnesses.

16.2 Candidate Lost Traditions

Irish and Celtic bestiaries. The Irish monastic tradition produced a distinctive intellectual culture that synthesized Christian theology with pre-Christian Celtic mythology. The *Book of Kells* and *Book of Durrow* include elaborate zoomorphic decoration whose symbolic program has never been fully decoded. Did an Irish bestiary tradition exist, linking these decorative programs to systematic moralizations? No Irish Latin bestiary has been identified, but the structural demand (conditions i–iii) was fully satisfied in early medieval Ireland. The formal framework predicts that such a tradition should have existed; its non-survival may reflect the catastrophic manuscript losses of the Viking raids (9th–10th centuries).

Iberian-Arabic hybrid bestiaries. Al-Andalus (Islamic Iberia) was a zone of intense cultural contact between Christian, Islamic, and Jewish traditions. The Arabic *Kalīla wa-Dimna* circulated alongside Latin bestiaries. Did hybrid texts emerge that combined Latin bestiary structure with Arabic zoological content? The formal framework predicts that cross-tradition $\mathcal{R}$ operations should have occurred in contact zones where scribes had access to both traditions.

Bestiaries with cosmological moralizations. All surviving Latin bestiaries moralize animals in moral-theological terms: the lion signifies Christ, the fox signifies the devil. But the kernel $\mathcal{K}$ does not logically require moral-theological interpretation. The moralization function μ could map to *cosmological* interpretations: the lion signifies the sun, the serpent signifies cyclical time, the eagle signifies celestial ascent. Such cosmological bestiaries would represent a different instantiation of the same kernel—one that the Hellenistic *Physiologus* itself partially embodied (with its Alexandrian cosmological subtext) before the Western tradition stripped it of non-Christian content.

Could cosmological bestiaries have existed? The *Liber Monstrorum* (8th–9th century) and the cosmological sections of Isidore’s *De Natura Rerum* suggest that the conceptual space for such texts was available. The *Physiologus*’s own condemnation in the *Decretum Gelasianum* (Section 12) may have targeted precisely such cosmological readings, implying that they existed and were perceived as threatening enough to suppress. The formal prediction: cosmological bestiaries were produced, used in contexts where theological orthodoxy was less strictly enforced, and lost when their institutional settings were destroyed or absorbed.

16.3 Reconstruction Method

The formal apparatus permits a principled approach to reconstruction—not of specific lost texts (which cannot be recovered without material evidence) but of the *formal properties* that lost texts must have possessed if they existed.

Step 1: Specify the kernel variant. A cosmological bestiary would have $\mathcal{K}_{\text{cosm}} = (S, D, \mu_{\text{cosm}})$, where $\mu_{\text{cosm}} : S \rightarrow \mathcal{M}_{\text{cosm}}$ maps species to cosmological rather than moral interpretations. The compatibility constraint $\mu_{\text{cosm}}(s) \in \mathcal{M}_{\text{compatible}}(s)$ still holds: the lion’s cosmological interpretation must be compatible with its observable behavior (sovereignty, power, solar associations).

Step 2: Predict the complexity trajectory. If the cosmological bestiary existed, the formal framework predicts the same $C_1 \rightarrow C_3$ trajectory: initial univocal cosmological moralization, followed by expansion into multiple cosmological interpretive pathways (the lion = sun, kingship, zodiacal Leo, Mithraic symbol, etc.).

Step 3: Identify traces. Even if the cosmological bestiary is entirely lost as a genre, its *traces* should be detectable in surviving texts that drew on the same material: cosmological glosses in surviving bestiary manuscripts, astronomical associations in encyclopedic works, and zoomorphic cosmological imagery in art.

What it adds: The reconstruction method demonstrates that the formal framework is not merely descriptive of what exists but *predictive of what should exist*. If the formal apparatus is correct, then bestiary-type texts with cosmological moralizations, cross-tradition hybrid structures, and Celtic-Christian syntheses should have been produced. Their absence from the surviving record is informative: it constrains the survival model (Appendix E) and identifies the most promising directions for future manuscript discovery.

Why speculative reasoning is necessary: A formal theory that generates no predictions beyond what is already known is vacuous. The bestiary framework’s value lies partly in its ability to predict the properties of texts that have not yet been found—or that once existed and were lost. These predictions are falsifiable: if systematic search in the predicted locations (Irish collections, Iberian archives, fragments in composite codices) yields nothing, the structural necessity thesis is weakened.

On the legitimacy of interdisciplinary speculation: Some readers will object that this section “speculates” rather than “demonstrates.” We embrace the characterization. All scientific progress begins with informed speculation constrained by formal structure. Darwin speculated about transitional fossils before any were found; Mendeleev speculated about undiscovered elements before they were synthesized; the Standard Model predicted the Higgs boson decades before its detection. The formal framework generates predictions; whether those predictions are confirmed or refuted is a matter for future research, not for *a priori* dismissal.

17 Methodological Note: The Necessity of Interdisciplinary Questioning

This paper has drawn on medieval Latin philology, codicology, paleography, systems theory, information theory, cognitive psychology, psychoanalysis, archetypal theory, decision theory, viability mathematics, ecology, population dynamics, Mesoamerican archaeology, Australian Aboriginal anthropology, West African oral literature, Hindu and Buddhist studies, theology, and the philosophy of science. This breadth will attract the criticism that the paper is “too broad,” “insufficiently specialized,” or “dilettantish.”

We address this criticism directly, because it reflects a methodological assumption that is itself unexamined and, we believe, incorrect.

17.1 The Disciplinary Fallacy

The assumption that a question belongs to a single discipline is a product of 19th-century institutional organization, not a property of the question itself. The question “why did the bestiary persist for 1400 years?” does not belong to philology, or to systems theory, or to cognitive science. It belongs to all of them simultaneously, because the bestiary’s persistence was simultaneously a philological phenomenon (textual transmission), a systems-theoretic phenomenon (structural adaptation), and a cognitive phenomenon (memorability and processing efficiency). No single discipline

can answer the question, because no single discipline possesses all the relevant tools.

The alternative—restricting the analysis to a single discipline—does not eliminate the interdisciplinary nature of the question. It merely leaves parts of the answer unstated. A purely philological analysis can document what happened (which manuscripts borrowed from which) but cannot explain why the pattern of borrowing took the form it did. A purely cognitive analysis can explain why animal images are memorable but cannot account for the specific institutional mechanisms that sustained the tradition. Only an analysis that integrates these perspectives can provide a complete account.

17.2 The Cost of Disciplinary Purity

Disciplinary purity has real intellectual costs. The bestiary has been studied for over a century within the confines of medieval philology, and within those confines, significant progress has been made (culminating in Dr. Dines’s discoveries). But certain questions could not even be *formulated* within the disciplinary framework: Why is the kernel invariant? Why are exactly three adaptive operations sufficient? Why did animal symbolism outperform other carriers? These questions require formal vocabulary that philology does not provide, and empirical evidence that cognitive science has not applied to medieval material. They fell into the gap between disciplines—not because they were unimportant, but because no single discipline owned them.

17.3 The Right to Ask

We assert the following methodological principle: *any question that can be formulated precisely and addressed with available evidence is legitimate, regardless of how many disciplinary boundaries it crosses*. The question “does the bestiary’s formal structure appear in Aboriginal Australian oral traditions?” crosses at least four disciplinary boundaries. It is nonetheless a legitimate question, because it is precise (it asks about a specific formal structure), addressable (the relevant ethnographic data exist), and consequential (the answer bears on the universality of the formal framework).

The disciplinary gatekeeping that would forbid such questions is not a methodological principle but an institutional reflex. It protects disciplinary territory, not intellectual quality. The appropriate response to an interdisciplinary claim is not “you are not qualified to ask this” but “your evidence is insufficient” or “your formal reasoning contains an error.” The first response is *ad hominem*; the second and third are scientific.

This paper invites both kinds of substantive criticism. It does not accept disciplinary jurisdictional objections.

What it adds: The defense converts a potential weakness (breadth) into a demonstrated strength (testability). Every interdisciplinary claim in this paper generates a prediction that specialists in the relevant discipline can test. The cross-cultural comparison predicts specific formal structures in Aboriginal, Maya, and West African traditions; the archetypal analysis predicts specific constraints on moralization; the manuscript survival model predicts specific properties of undiscovered manuscripts. These predictions are gifts to specialists, not trespasses on their territory.

Practical application: The principle that “any precisely formulated question is legitimate regardless of disciplinary boundaries” applies to all research programs operating at the intersection of multiple fields: digital humanities, computational social science, bioethics, science and technology studies, and the emerging field of cultural analytics. The bestiary paper demonstrates concretely how formal vocabulary enables interdisciplinary collaboration without requiring interdisciplinary expertise in every collaborator.

18 The Zoological Value of Erroneous Natural History

18.1 The Problem of Bestiary “Errors”

The bestiary’s zoological content is, by modern standards, largely incorrect. Lions do not breathe life into stillborn cubs. Beavers do not bite off their own testicles. Pelicans do not pierce their breasts to feed their young with their blood (they regurgitate fish). The elephant has no difficulty lying down and standing up again. The barnacle goose does not grow from barnacles.

These errors have led generations of scholars to dismiss the bestiary’s zoological content as “medieval superstition” or “credulous natural history,” treating the descriptions as obstacles to the moralizations rather than as objects of interest in their own right. This dismissal is both historically anachronistic and analytically short-sighted.

18.2 Errors as Information

The bestiary’s zoological errors are not noise; they are data. Each error preserves a specific state of knowledge—or, more precisely, a specific *configuration of observation, inference, and authority*—that characterized the intellectual culture in which the bestiary was produced and transmitted. The errors are informative in at least four ways:

First, they document the history of observation. Some bestiary descriptions are accurate observations that were subsequently misinterpreted. The pelican’s habit of pressing partially digested fish from its gular pouch against its blood-red breast was a real observation; the interpretation that the bird was piercing itself was a reasonable inference from the visual evidence available to observers who could not examine the bird at close range. The error preserves a record of what pre-modern observers actually saw and how they processed visual information under conditions of limited access.

Second, they trace the transmission of authority. The bestiary’s zoological content derives from identifiable sources: Pliny, Solinus, Isidore, Aristotle (in Latin translation), and the *Physiologus* itself. When a bestiary chapter reproduces an error from Pliny, it documents the authority chain through which zoological knowledge traveled. The persistence of specific errors across centuries of copying—when direct observation could have corrected them—reveals the dominance of textual authority over empirical observation in medieval intellectual culture.

Third, they reveal the teleological structure of pre-modern natural history. Medieval zoology was not descriptive in the modern sense; it was teleological. Animals were understood to have been created with specific natures that served instructive purposes. The beaver’s self-castration was not a factual claim about beaver behavior; it was a claim about the beaver’s *nature*—what the beaver was *for* in the divine economy. The “error” is an error only within a descriptive epistemology; within a teleological epistemology, it is a perfectly coherent statement about the moral structure of creation. The bestiary thus preserves a complete alternative epistemology of the natural world—one in which the question “what does this animal do?” is inseparable from the question “what does this animal mean?”

Fourth, they have genuine zoological value as ethological records. Not all bestiary descriptions are incorrect. The *Physiologus*’s account of the antelope catching its horns in a thicket may reflect genuine observation of the gerenuk or the addax. The description of the beaver’s engineering behavior (dam construction, lodge building) is substantially accurate. The account of the crocodile’s relationship with the trochilus (Egyptian plover) reflects a real mutualistic behavior documented by Herodotus and confirmed by modern ethologists. Even the obviously fabulous descriptions (the phoenix, the unicorn, the basilisk) preserve fragments of real animal behavior—migratory patterns,

horn morphology, venomous snakes—filtered through the interpretive apparatus of a pre-modern culture.

18.3 Formal Implications

In the formal framework, the bestiary’s zoological errors illuminate the relationship between the description function $D(s)$ and the moralization function $\mu(s)$. In a modern knowledge system, D and μ are logically independent: a factual description can be updated without affecting its moral interpretation, and a moral interpretation can be revised without altering the facts. In the bestiary, D and μ are *co-adapted*: the zoological description is shaped by the moralization it is required to support, and the moralization is constrained by the description it can draw on.

This co-adaptation explains the persistence of errors: correcting $D(\text{pelican})$ to reflect actual pelican behavior (regurgitation from the pouch) would destroy $\mu(\text{pelican})$ (Christ’s self-sacrifice). The “error” in D is stabilized by the value of μ . In formal terms:

If $\mu(s)$ depends on $D(s)$, then correcting $D(s)$ may require $\mathcal{R}$ to restore $\mu(s)$.

This is precisely the situation that the correction operator was designed to address—but in the case of zoological errors, the correction was never performed because the cost (destruction of the moralization) exceeded the benefit (zoological accuracy). The bestiary optimized for pedagogical function, not for zoological truth.

This trade-off is not unique to the medieval period. Modern pedagogical systems routinely simplify, distort, or mythologize their subject matter to enhance memorability and moral impact. The Bohr model of the atom is zoologically false but pedagogically indispensable. The narrative of Columbus “discovering” America is historically distorted but serves a nation-building function. The formal framework’s identification of the D/μ co-adaptation provides a vocabulary for analyzing these trade-offs in any pedagogical context.

What it adds: The D/μ co-adaptation reveals that the description and moralization functions are not independent channels but coupled systems. This coupling explains both the persistence of errors (correcting D destroys μ) and the resistance of the bestiary to zoological updating. The formal framework thus accounts for a feature of the tradition—the longevity of demonstrable falsehoods—that literary and historical analysis has noted but not explained structurally.

Practical application: Any knowledge system that pairs factual content with evaluative interpretation faces the same co-adaptation constraint. Legal precedents where factual error is embedded in a normatively valued ruling, medical myths that persist because they support therapeutic compliance, and corporate origin stories that mythologize founders—all exhibit D/μ coupling. The formal framework predicts that correction of D will encounter resistance proportional to the value of μ .

19 Was the Bestiary an Accident or a Necessity?

19.1 The Question

The bestiary emerged in a specific time and place: Hellenistic Alexandria, between the 2nd and 4th centuries, at the intersection of Greek natural philosophy, Jewish allegorical exegesis, and early Christian catechesis. Was this emergence contingent—a historical accident that might not have occurred under different circumstances—or structurally determined, in the sense that the conditions of the period made some such genre inevitable?

The formal framework developed in this paper permits a precise formulation of the question. Define the *structural demand* for a pedagogical genre as the conjunction of three conditions: (i) a body of content requiring transmission across generations (the moral-theological content of early Christianity), (ii) a population of recipients with heterogeneous intellectual preparation (the catechumens, clergy, and laity of the early Church), and (iii) an institutional infrastructure for organized instruction (the catechetical schools, episcopal courts, and eventually the monastic and cathedral schools of the Latin West).

19.2 The Structural Argument for Necessity

When these three conditions are satisfied simultaneously, a persistent pedagogical genre must emerge—not necessarily the bestiary specifically, but some genre satisfying the Persistence Triad. The argument proceeds as follows.

The content requiring transmission (condition i) is complex, multi-layered, and resistant to naive simplification (Section 10: dual-system content cannot be transmitted within a single evaluative framework). The recipients are heterogeneous (condition ii), requiring materials adaptable across levels of sophistication—from beginning students who need univocal associations (C_1) to advanced preachers who need flexible homiletic resources (C_3). The institutional infrastructure (condition iii) provides the selective environment: genres that serve institutional needs survive; those that do not, vanish.

Under these conditions, the viability theory framework [7] predicts that the system will converge on a genre satisfying the Persistence Triad—because genres lacking closure, boundary maintenance, or resilience will be eliminated by institutional selection. The specific *form* of the genre (zoomorphic symbolism with moral interpretation) is historically contingent; the *existence* of some genre with the formal properties of the bestiary is structurally necessary.

The bestiary’s specific form was selected from among the available alternatives by the cognitive optimality of zoomorphic carriers (Section 9). Given that a persistent pedagogical genre was structurally required, and given that zoomorphic symbolism optimizes the cognitive constraints on moral pedagogy (Proposition 9.1), the bestiary’s emergence was over-determined: it satisfied both the institutional demand for a persistent pedagogical tool and the cognitive demand for an optimally memorable, low-friction symbolic carrier.

19.3 How Structural Necessity Supports Persistence

If the bestiary’s emergence was structurally necessary rather than accidental, this fact itself supports its persistence through a mechanism that the formal framework can articulate. A genre that exists because it satisfies a structural demand will persist as long as that demand persists—because its elimination would create a functional vacuum that would be filled by a structurally equivalent genre. The “survival” of the bestiary is therefore not the survival of a contingent cultural product but the survival of a structural solution to a persistent institutional problem.

This explains the convergent evolution documented in Section 7: the independent integration of *distinctiones* into bestiaries in England (from William de Montibus) and Germany (the expanded *Physiologus Theobaldi*) represents not coincidental parallel innovation but independent responses to the same structural demand—the need for flexible homiletic material at the C_3 level. The demand generated the solution, not once but repeatedly, because the demand was structural rather than contingent.

The same logic applies to self-organization theory more broadly. Kauffman [57] demonstrated that systems operating far from equilibrium under selection pressure tend to converge on a limited

set of organizational forms—the “adjacent possible” of structurally viable configurations. Prigogine and Stengers [58] showed that dissipative structures emerge spontaneously when energy throughput exceeds a threshold. The bestiary’s emergence can be understood as an instance of cultural self-organization: the throughput of moral-theological content through early Christian institutional channels exceeded the capacity of unstructured transmission (oral teaching, ad hoc catechesis), and a structured pedagogical genre crystallized as a dissipative structure channeling that throughput.

The formal prediction is testable: in any culture where conditions (i)–(iii) are satisfied simultaneously, a pedagogical genre with the formal properties of the bestiary (invariant kernel, modular operations, dual-system mediation) should emerge independently. The Hindu *Pañcatantra*, the Islamic *Kalīla wa-Dimna* (a descendant of the *Pañcatantra*), the Buddhist Jātaka tales, and the Confucian *Classic of Mountains and Seas (Shanhaijing)* represent candidate examples from non-Christian traditions. Each employs animal symbolism for moral instruction within an institutional educational framework. A systematic comparison of their formal properties with those of the Latin bestiary—testing whether they satisfy the Persistence Triad and deploy analogues of $\mathcal{C}$, $\mathcal{E}$, and $\mathcal{R}$ —would constitute a strong test of the structural necessity thesis.

What it adds: If the bestiary’s emergence was structurally necessary, its persistence is explained at a deeper level than the Persistence Triad alone provides. The Triad explains *how* the genre persists; the necessity argument explains *why it exists in the first place*. A genre that exists because it fills a structural need is more robust than one that exists by accident, because its elimination creates a functional vacuum that tends to be refilled.

Practical application: The structural necessity argument applies to any institutional setting where conditions (i)–(iii) are satisfied. A corporation with complex values to transmit, heterogeneous employees, and organized training infrastructure should expect—and can deliberately design—a pedagogical system with bestiary-like formal properties. The emergence of such systems is not optional; it is structurally determined.

20 The Bestiary as an Early Attempt at Formalization

20.1 The Bestiary as Proto-Formal System

The bestiary is not usually described as a formalization of anything. It is described as a literary genre, a catechetical tool, or an art-historical tradition. But the formal analysis conducted in this paper reveals that the bestiary was, in fact, an early and remarkably successful attempt to formalize the properties of animals *as images*—to systematize the mapping from zoological observation to moral interpretation, and to make that mapping available as a reusable, modular, combinable resource for moral instruction.

The characteristics of formalization are present:

- **Explicit structure:** Each chapter follows the invariant triple (*species–descriptio–moralizatio*). The structure is not implicit but visually and textually marked.
- **Composability:** Chapters can be rearranged, added, or removed without destroying the system’s coherence. The kernel is modular.
- **Extensibility:** New animals can be incorporated by applying the same template: observe, describe, moralize. The coding scheme is open.

- **Multi-level interpretation:** The *distinctiones* tradition introduces a systematic hermeneutical framework (literal, allegorical, tropological, anagogical) that functions as a formal grammar for generating interpretations.

These are the properties of a formal system—not in the strict mathematical sense of Hilbert or Gödel, but in the broader sense of a system with explicit rules, composable elements, and a generative grammar. The bestiary formalized the animal-as-image.

20.2 The Three Formalization Laws and the Bestiary

The formalization implicit in the bestiary tradition can be analyzed through the three Formalization Laws established in *Foundations of Intellectual Substrate Mining* [56], where their full formal statements and mathematical proofs are presented in Chapter 9 (pp. 125–164). These laws are treated here as established methodological results and are used as governing constraints without restating their derivation.

The Law of Total Formalizability. Any phenomenon exhibiting determinate structure is formally representable under a sufficient constraint set. The bestiary confirms this law: the phenomenon of animal-as-moral-sign, which appears pre-theoretically as a diffuse cultural intuition (“the lion is kingly”), is formalized in the bestiary into an explicit structure with identifiable components (S, D, μ), definable operations ($\mathcal{C}, \mathcal{E}, \mathcal{R}$), and measurable properties ($\rho, H, C_1\text{--}C_4$). The bestiary demonstrates that moral-zoological symbolism is *totally formalizable*: there is no aspect of the tradition’s structure that resists formal representation under the constraint set developed in this paper.

The Law of Boundary-Conditioned Verification. Verification of any formal claim is necessarily conditioned by explicit boundary regimes. The bestiary illustrates this law through the cultural variability documented in Section 14: the moralization $\mu(\text{owl}) = \text{wisdom}$ is verifiable only within the boundary conditions of Greek-derived Western culture; within Chinese or Mesoamerican boundary conditions, the same formal structure yields $\mu(\text{owl}) = \text{death-portent}$. The moralization function is not absolutely valid but boundary-conditioned. The Law predicts exactly this: any formalization of a structured phenomenon will be valid within its constraint set and potentially invalid outside it.

The Law of Plurality and Incompleteness. No single formalization can exhaust a structured phenomenon. The bestiary tradition provides the most vivid confirmation of this law: the evolution from C_1 (univocal moralization) to C_3 (multiple *distinctiones*) is precisely the progressive discovery that a single formalization ($\mu(s) = m$) cannot exhaust the symbolic content of the animal image. The expansion operator $\mathcal{E}$ is the bestiary’s own response to the Law of Plurality: confronted with the inadequacy of univocal interpretation, the tradition evolved toward set-valued moralization ($\mu^*(s) = \{m_1, \dots, m_n\}$), acknowledging that multiple non-equivalent formalizations of the same symbolic node are necessary.

20.3 Cross-Cultural Diversification as Confirmation

The diversification of animal interpretations across cultures (Section 14) provides independent confirmation of all three laws operating simultaneously.

The Law of Total Formalizability is confirmed because every culture that has developed zoomorphic moral symbolism has done so through some version of the kernel structure: observe, describe, interpret. The mechanism is universally formalizable even though the specific content is not.

The Law of Boundary-Conditioned Verification is confirmed because the same animal receives opposite attributions in different cultural boundary regimes. The owl-as-wisdom (Western) and the owl-as-death (East Asian) are both “correct” formalizations—each verified within its boundary conditions, each inapplicable outside them.

The Law of Plurality and Incompleteness is confirmed because the global diversity of zoomorphic symbolism demonstrates that no single culture’s moralization function μ exhausts the interpretive potential of the animal kingdom. The lion has been sovereignty, Christ, the devil, the sun, courage, tyranny, and the guardian of the dead across different traditions. Each attribution is a partial formalization; their plurality confirms that no single formalization is complete.

What it adds: The three Formalization Laws [56] transform the bestiary from a curious medieval artifact into a case study in the general theory of formalization itself. The bestiary is not merely a text that *can be* formalized (as this paper demonstrates); it is a text that *was itself an act of formalization*—an early, sustained, institutionally supported attempt to render moral-symbolic knowledge explicit, composable, and transmissible. Recognizing this changes the bestiary’s status from cultural relic to epistemological precedent.

Practical application: The three laws apply to any domain where implicit knowledge is being rendered explicit: codification of tacit expertise, development of AI training taxonomies, construction of ontologies for knowledge management. The bestiary’s trajectory—from univocal formalization (C_1) through the discovery of plurality (C_3) to compensatory integration (C_4)—is a predictable developmental sequence for any formalization project, and the Formalization Laws explain why.

21 Computable Models: From Formal Framework to Testable Simulations

The formal apparatus developed in this paper is not merely descriptive. It generates a family of computable models that can be implemented, calibrated against manuscript data, and used to make quantitative predictions about pedagogical systems past and present. This section specifies twelve models at varying levels of complexity, each with defined inputs, outputs, and validation criteria.

21.1 Model 1: Moralization Density Equilibrium

Type: Threshold dynamics.

Idea: The correction operator $\mathcal{R}$ is triggered when the moralization density $\rho(A)$ of a section A falls below a compiler-specific threshold ρ^* . This can be modeled as a simple dynamical system in which sections with $\rho < \rho^*$ attract imported material until equilibrium is restored.

Formal specification:

$$\rho_{t+1}(A) = \rho_t(A) + \gamma \cdot \max(0, \rho^* - \rho_t(A)) \cdot \rho_{\text{source}}$$

where $\gamma \in (0, 1]$ is the compiler’s editing intensity and ρ_{source} is the moralization density of the available external source.

Inputs: Section-by-section moralization counts from manuscript collations. ρ^* estimated from observed trigger points (e.g., KK 4.25 was triggered at $\rho = 0.27$).

Output: Predicted final ρ per section after correction; comparison with observed ρ in corrected

manuscripts.

Validation: Test against the birds section of KK 4.25 (observed: ρ rises from 0.25 to 0.69). Extend to other sections and manuscripts.

Implementation: Spreadsheet or simple Python script. No specialized software required.

21.2 Model 2: Compression Fidelity Curve

Type: Rate-distortion analysis.

Idea: The compression operator $\mathcal{C}$ trades volume for fidelity. For any given chapter, there exists a curve relating compression ratio $r = |v(s)|/|D(s) + \mu(s)|$ to semantic fidelity $F = |\text{sem}(v) \cap \text{sem}(D, \mu)|/|\text{sem}(D, \mu)|$, where semantic elements are counted as a finite inventory.

Formal specification:

$$F(r) = 1 - \alpha \cdot e^{-\beta/r}, \quad r \in (0, 1]$$

where α and β are chapter-specific parameters estimated from the semantic inventory analysis (cf. Appendix A: 6/8 elements preserved at $r = 1/10.7$).

Inputs: Prose chapters and their corresponding verse summaries from Bodley 533 and Cambridge manuscripts; semantic inventory coding (which elements are present in each version).

Output: Estimated fidelity curve per chapter; identification of the “compression floor” below which semantic core is destroyed.

Validation: Cross-validate across the approximately 20 chapters for which both prose and verse survive.

Implementation: Requires manual semantic coding (a philological task) followed by standard curve-fitting.

21.3 Model 3: Interpretive Entropy Landscape

Type: Information-theoretic mapping.

Idea: Compute the upper bound $H_{\max}(s) = \log_2 |\mu^*(s)|$ for every symbolic node s across all families. This generates a “landscape” showing where interpretive flexibility is concentrated and where it is absent.

Formal specification: For each manuscript family F and each species $s \in S_F$:

$$H_{\max}^F(s) = \log_2 |\mu_F^*(s)|$$

Aggregate: $\bar{H}^F = \frac{1}{|S_F|} \sum_{s \in S_F} H_{\max}^F(s)$.

Inputs: Complete moralization inventories for each species in each family (extractable from Dines’s and Clark’s editions).

Output: Heat map of $H_{\max}(s)$ across species and families; identification of “entropy hotspots” (species with maximal interpretive range) and “entropy deserts” (species with univocal moralization even in C_3 texts).

Validation: The model predicts that entropy hotspots should correlate with species most frequently cited in surviving sermon collections.

Implementation: Database of moralization counts; standard information-theoretic computation.

21.4 Model 4: Kernel Stability Under Scribal Noise

Type: Stochastic simulation (Monte Carlo).

Idea: Each act of manuscript copying introduces random perturbations to the kernel: small changes to S (species added or dropped), D (descriptions modified), and μ (moralizations altered). The

Persistence Triad predicts that the kernel should be robust to these perturbations. This can be tested by simulating N generations of copying with empirically calibrated error rates and measuring kernel stability.

Formal specification: At each generation t :

$$\begin{aligned} S_{t+1} &= S_t \cup \{s_{\text{add}}\} \setminus \{s_{\text{drop}}\} \quad \text{with probabilities } p_{\text{add}}, p_{\text{drop}} \\ D_{t+1}(s) &= D_t(s) + \epsilon_D(s) \quad \text{with } \epsilon_D \sim \mathcal{N}(0, \sigma_D^2) \\ \mu_{t+1}(s) &= \mu_t(s) + \epsilon_\mu(s) \quad \text{with } \epsilon_\mu \sim \mathcal{N}(0, \sigma_\mu^2) \end{aligned}$$

Kernel identity is measured by overlap: $\text{KI}(t) = |S_t \cap S_0|/|S_0|$ and semantic similarity of D and μ components.

Inputs: Error rates $p_{\text{add}}, p_{\text{drop}}, \sigma_D, \sigma_\mu$ estimated from collation of manuscript variants.

Output: Distribution of kernel identity $\text{KI}(t)$ after t generations; identification of the “persistence horizon” (number of generations before KI drops below a threshold).

Validation: Compare simulated kernel degradation with observed manuscript variation across the 150 known bestiary manuscripts.

Implementation: Python or R; Monte Carlo with 10,000+ runs per parameter setting.

21.5 Model 5: Dual-System Agent-Based Model

Type: Agent-based simulation.

Idea: Simulate a population of agents, each characterized by $\lambda \in [0, 1]$ (orientation toward $\mathcal{W}$), exposed to bestiary-style pedagogical material. Agents update λ based on encounters with zoomorphic moral content (which shifts λ downward, toward $\mathcal{K}_T$) and worldly experience (which shifts λ upward, toward $\mathcal{W}$).

Formal specification:

$$\lambda_{t+1} = \lambda_t + \eta_W \cdot (1 - \lambda_t) - \eta_B \cdot \lambda_t \cdot \text{enc}(t)$$

where η_W is the worldly drift rate, η_B is the bestiary influence rate, and $\text{enc}(t) \in \{0, 1\}$ indicates whether the agent encountered bestiary material at time t .

Inputs: Population size, initial λ distribution, encounter frequency, drift and influence rates.

Output: Steady-state distribution of λ in the population; identification of critical encounter frequency below which $\lambda \rightarrow 1$ (bestiary has no sustained effect).

Validation: Qualitative comparison with historical evidence on the frequency of bestiary use in different institutional settings.

Implementation: NetLogo, Mesa (Python), or custom simulation.

21.6 Model 6: Genre Competition and Ecological Niche

Type: Lotka-Volterra competition dynamics.

Idea: Model the bestiary as one of several competing pedagogical genres (bestiary, exempla, sermon, catechism), each with a carrying capacity determined by institutional demand. Genres compete for the limited resource of scribal labor and library shelf-space.

Formal specification:

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{\sum_j \alpha_{ij} N_j}{K_i} \right)$$

where N_i is the number of surviving manuscripts of genre i , r_i is its reproduction (copying) rate, K_i is its carrying capacity, and α_{ij} is the competition coefficient between genres i and j .

Inputs: Manuscript survival counts by genre and century; estimated copying rates from colophon evidence.

Output: Predicted trajectory of bestiary manuscript production over time; identification of the ecological niche that the bestiary occupied and when it was displaced.

Validation: Compare predicted decline curves with observed manuscript counts per century.

Implementation: Standard ODE solver (SciPy, MATLAB).

21.7 Model 7: Archetypal Compatibility Filter

Type: Constraint satisfaction / set-theoretic.

Idea: For each animal s , define the compatible moralization set $\mathcal{M}_{\text{compatible}}(s)$ as those attributions consistent with the animal’s observable behavioral profile. Test whether historically attested moralizations fall within this set and whether culturally divergent moralizations correspond to different elements of the same compatibility set.

Formal specification: For each species s , define a behavioral feature vector $\mathbf{b}(s) \in \{0, 1\}^k$ (nocturnal, predatory, maternal, aquatic, etc.). Define a mapping $\mathcal{M}_{\text{compatible}} : \{0, 1\}^k \rightarrow \mathcal{P}(\mathcal{M})$ that associates feature vectors with compatible interpretations.

Inputs: Coded behavioral features for each bestiary animal; coded moralizations from Western, Eastern, Islamic, and Asian traditions.

Output: Proportion of attested moralizations falling within the predicted compatibility set; identification of “forced” moralizations that violate compatibility and their persistence rates.

Validation: Moralizations violating compatibility should show shorter persistence spans than compatible ones.

Implementation: Database of feature vectors and moralizations; set-membership testing.

21.8 Model 8: Print Transition Catastrophe

Type: Bifurcation analysis.

Idea: The transition from manuscript to print eliminated the scribal adaptive operation M_{scribe} . Model this as a control parameter σ (scribal variation intensity) that drops discontinuously to zero at the moment of print adoption, and analyze the resulting bifurcation in the genre’s adaptive capacity.

Formal specification:

$$\frac{d\mathcal{K}}{dt} = f(\mathcal{K}, \sigma) - \delta \cdot \mathcal{K}$$

where $f(\mathcal{K}, \sigma)$ represents adaptive updating (positive for $\sigma > 0$) and δ is the genre’s obsolescence rate. At $\sigma = 0$, the system loses its adaptive term and $\mathcal{K} \rightarrow 0$ if $\delta > 0$.

Inputs: Estimated σ from manuscript variant analysis; estimated δ from historical genre replacement rates.

Output: Critical σ^* below which the genre cannot sustain itself; predicted time-to-extinction after print adoption.

Validation: Compare predicted extinction date with actual disappearance of new bestiary production (c. early 16th century).

Implementation: Bifurcation analysis tools (MATCONT, XPP-AUTO).

21.9 Model 9: Cross-Cultural Divergence Predictor

Type: Branching process with cultural drift.

Idea: Starting from a common ancestral moralization (the *Physiologus*), simulate how moralizations

diverge as the text is transmitted into different cultural contexts, each with a distinct “cultural filter” that selects among elements of $\mathcal{M}_{\text{compatible}}(s)$.

Formal specification: At each cultural transmission event, the moralization for species s in culture c is sampled:

$$\mu_c(s) \sim \text{Categorical} \left(\frac{w_c(m)}{\sum_{m' \in \mathcal{M}_{\text{comp}}(s)} w_c(m')} \right), \quad m \in \mathcal{M}_{\text{compatible}}(s)$$

where $w_c(m)$ is the cultural weight assigned to interpretation m in culture c .

Inputs: Comparative moralization data from Latin, Greek, Ethiopic, Armenian, Arabic, and Syriac *Physiologus* traditions.

Output: Predicted divergence rate; identification of “culturally stable” moralizations (common across all traditions) and “culturally labile” ones (diverging rapidly).

Validation: Culturally stable moralizations should correspond to species with smaller $|\mathcal{M}_{\text{compatible}}(s)|$ (less ambiguous behavioral profiles).

Implementation: Bayesian phylogenetic methods adapted from computational linguistics.

21.10 Model 10: Optimal Curriculum Design

Type: Optimization / operations research.

Idea: Given the formal apparatus, design an optimal pedagogical curriculum as a configuration $\mathcal{K} + \{M_i\}$ that maximizes learning outcomes subject to constraints on volume, complexity, and audience heterogeneity. This is a direct modern application of the bestiary’s architecture.

Formal specification:

$$\max_{\mathcal{K}, \{M_i\}} U(\mathcal{K}, \{M_i\}) \quad \text{s.t.} \quad \text{Vol}(\mathcal{K}) \leq V_{\max}, \quad C(\mathcal{K}) \in \{C_1, \dots, C_4\}, \quad \rho(A) \geq \rho^* \quad \forall A$$

where U is a pedagogical utility function (e.g., weighted sum of retention, flexibility, and coverage).

Inputs: Topic inventory S , content descriptions D , available interpretive frameworks $\mathcal{M}$, audience profile, volume constraints.

Output: Optimal kernel design and module selection; recommended complexity level per audience segment.

Validation: A/B testing of bestiary-inspired curriculum design against conventional formats.

Implementation: Linear or mixed-integer programming (Gurobi, OR-Tools).

21.11 Model 11: Semantic Network of Zoomorphic Associations

Type: Graph-theoretic / network analysis.

Idea: Construct a bipartite graph $G = (S \cup \mathcal{M}, E)$ where edges connect species to their attested moralizations. Analyze the network’s structural properties: degree distribution (how many meanings per animal?), clustering (which animals share moralizations?), hub species (maximally connected), and community structure.

Formal specification:

$$V = S \cup \mathcal{M}, \quad E = \{(s, m) : m \in \mu^*(s)\}$$

$$\text{Degree:} \quad k(s) = |\mu^*(s)|, \quad k(m) = |\{s : m \in \mu^*(s)\}|$$

Inputs: Complete moralization inventory across all families (extractable from published editions and Dines’s data).

Output: Network visualization; degree distribution; identification of hub species (lion, serpent, eagle) and hub moralizations (Christ, devil, preacher); modularity analysis.

Validation: Hub species should correspond to those with longest independent tradition; hub moralizations should correspond to central theological concepts.

Implementation: NetworkX (Python), Gephi, or igraph (R).

21.12 Model 12: Persistence Prediction for Modern Pedagogical Systems

Type: Diagnostic checklist / scoring model.

Idea: Operationalize the Persistence Triad as a scoring instrument that evaluates any pedagogical system's likelihood of long-term survival. Each of the three conditions (closure, boundary maintenance, resilience) is scored on a scale based on observable indicators derived from the bestiary analysis.

Scoring rubric:

Condition	High score (3)	Medium (2)	Low (1)
Closure	Each unit self-contained, intelligible independently	Some cross-references required	Strong sequential dependency
Boundary	Clear genre identity, distinct from neighbors	Partially overlapping with other genres	Indistinguishable from similar products
Resilience	All three operations ($\mathcal{C}$, $\mathcal{E}$, $\mathcal{R}$) available	One or two operations available	No modular adaptation possible

Prediction: Systems scoring ≥ 8 are predicted to persist across institutional transitions; systems scoring ≤ 4 are predicted to fail within one institutional generation.

Validation: Retrospective application to historically documented educational formats (textbooks, MOOCs, programmed instruction, apprenticeship models) and comparison with observed persistence.

Implementation: Survey instrument; no computation required beyond scoring.

What they add: The twelve models transform the paper's formal apparatus from a descriptive vocabulary into a generative toolkit. Each model takes specific formal constructs (ρ , H , $\mathcal{C}$, $\mathcal{E}$, $\mathcal{R}$, λ , the Persistence Triad, the compatibility constraint) and converts them into computable, testable, falsifiable instruments. The models span five levels of complexity, from spreadsheet-level computations (Models 1–3) through standard simulations (Models 4–6) to cross-disciplinary research programs (Models 9–12). **Practical application:** Models 1–3 can be implemented immediately with existing manuscript data. Model 10 (Optimal Curriculum Design) and Model 12 (Persistence Prediction) are directly applicable to contemporary educational design. Model 11 (Semantic Network) would produce the first comprehensive map of the bestiary's interpretive structure as a network. Model 8 (Print Transition Catastrophe) has implications for any knowledge system undergoing a medium transition (e.g., print to digital, classroom to online).

22 If Only 150 Bestiaries Survived: A Thought Experiment in Civilizational Reconstruction

Imagine that all records of Western civilization have been destroyed except for the approximately 150 surviving Latin bestiary manuscripts. No Bibles, no legal codes, no chronicles, no architectural remains—only the bestiaries. What could a future civilization reconstruct from this corpus alone?

The question is not idle. It tests the information density of the bestiary as a cultural artifact and, by extension, the information density of any adaptive pedagogical system that satisfies the Persistence Triad.

22.1 What Could Be Reconstructed

A complete theology. The moralizations contain, in compressed form, the core doctrines of Christianity: the Incarnation (lion cubs resurrected on the third day = Christ’s Resurrection), the Atonement (pelican piercing its breast = Christ’s sacrifice), the Fall and Redemption (serpent = temptation and healing), the Trinity (implied by the structural relationships among the three “natures” of the lion). A patient analyst could reconstruct the Christological, soteriological, and eschatological framework of medieval Christianity from the moralizations alone—not in detail, but in structural outline.

A natural history. The zoological descriptions, however inaccurate by modern standards, document hundreds of animal species (real and mythological), their attributed behaviors, habitats, and interrelationships. The bestiaries would reveal that their civilization recognized lions, elephants, whales, eagles, crocodiles, and dozens of other species—indicating extensive geographic knowledge. The fabulous animals (phoenix, unicorn, basilisk) would reveal the epistemological framework: a civilization that did not sharply distinguish observed from reported, natural from supernatural.

A pedagogical philosophy. The $\mathcal{K} = (S, D, \mu)$ structure would be immediately recognizable: this civilization believed that knowledge of the natural world was inseparable from moral instruction. The consistent pattern of observation followed by interpretation would reveal a teleological worldview in which every natural phenomenon was considered meaningful.

A social structure. The existence of mnemonic verses (Bodley 533) would reveal a literate educational system requiring memorization. The *distinctiones* would reveal a sophisticated hermeneutical tradition with trained interpreters. The varying complexity levels (C_1 through C_4) would reveal a stratified audience: beginners who needed simple associations and advanced practitioners who required flexible, multi-level material. The institutional infrastructure of schools, libraries, and scriptoria would be inferable from the manuscripts’ physical properties: parchment quality, script types, binding techniques, illustration programs.

A language and its literature. The Latin prose would provide a substantial corpus for linguistic reconstruction. The Leonine hexameter verses would reveal a metrical tradition. The biblical quotations embedded in the moralizations would allow partial reconstruction of the Vulgate text. The citing of patristic authorities (Ambrose, Augustine, Gregory, Isidore) would reveal that this civilization possessed an extensive earlier literary tradition, now lost.

An aesthetic. The illustrated bestiaries (a significant proportion of the surviving corpus) would reveal a visual culture of extraordinary sophistication: gold leaf, complex pigment chemistry, naturalistic and stylized animal representation, decorative borders, historiated initials. The aesthetic program would be fully recoverable.

22.2 What Could Not Be Reconstructed

The bestiaries would *not* reveal: the political history of the civilization (no kings, battles, or treaties), the economic system (no prices, trade routes, or currencies), the legal framework (no laws, courts, or contracts), the technological level beyond manuscript production (no engineering, agriculture, or medicine), the geographic extent of the civilization (the manuscripts are portable and could have been produced anywhere), or the demographic structure (no population data).

Most consequentially, the bestiaries would not reveal their own context. A future analyst would not know that these 150 manuscripts were fragments of a vast civilization that also produced cathedrals, universities, legal systems, and global trade networks. The bestiaries would present themselves as the *totality* of the civilization’s intellectual output—a civilization that appeared to be exclusively concerned with the moral interpretation of animals.

22.3 The Information Density Lesson

The thought experiment reveals that the bestiary is an extraordinarily information-dense artifact. From 150 manuscripts, a future civilization could reconstruct the theology, natural history, pedagogical philosophy, social structure, language, and aesthetic of medieval Christendom—not completely, but in structural outline. This density is not accidental. It is a consequence of the bestiary’s formal properties: the kernel $\mathcal{K}$ packs three layers of information (zoological, moral, theological) into every chapter; the adaptive operations ensure that the corpus covers multiple complexity levels; and the material form of the manuscript (script, illustration, binding) carries additional information about the civilization’s technological and aesthetic capabilities.

The deeper lesson is that adaptive pedagogical systems are, by their nature, civilizational summaries. A system designed to transmit a civilization’s core knowledge to its next generation must contain, in compressed form, the essential structure of that civilization. The bestiary was not designed as an archive; it was designed as a teaching tool. But a teaching tool that successfully transmits a civilization’s values, knowledge, and interpretive methods is, *ipso facto*, an archive of that civilization.

What it adds: The thought experiment demonstrates that the bestiary’s information density is not a metaphor but a measurable property: the ratio of reconstructable civilizational features to manuscript count. No other medieval genre of comparable size (150 manuscripts) would permit reconstruction of theology, natural history, pedagogy, language, and aesthetic simultaneously. This density is a direct consequence of the kernel’s triple structure (S, D, μ) and the adaptive operations that ensure comprehensive coverage.

Practical application: The thought experiment provides a criterion for evaluating the archival value of any pedagogical system: *if this were the only surviving artifact of its civilization, how much could be reconstructed?* Systems scoring high on this measure—because they embed multiple knowledge layers in a compact, self-contained structure—are the most valuable targets for preservation. This criterion is directly applicable to the digitization priorities discussed in the next section.

23 The Race Against Decay: Digitization, Manuscript Mortality, and the Preservation of Effective Complexity

23.1 The Material Crisis

Medieval manuscripts are dying. They have survived six to nine centuries, but survival is not immortality. Parchment degrades through oxidation, hydrolysis, and biological attack. Ink fades, flakes, and corrodes the writing surface (iron gall ink is acidic and slowly destroys the parchment it was applied to). Binding structures fail; leaves detach, crumble, and are lost. Mold, insects, rodents, and fluctuations in temperature and humidity accelerate all of these processes. Fire and flood remain catastrophic risks: the 2019 fire at the Duchess Anna Amalia Library in Weimar destroyed or damaged 50,000 volumes, and the 1966 Florence flood damaged over a million manuscripts and books.

The rate of destruction is difficult to quantify precisely, but conservators estimate that the usable life of a medieval parchment manuscript under typical archival conditions (stable temperature, controlled humidity, limited handling) is on the order of 1,000–1,500 years from production. For the bestiary corpus, produced primarily between 1170 and 1300, this means that the manuscripts are now entering the final third of their expected material lifespan. Every decade of delay in digitization risks the permanent loss of information that has survived eight centuries.

The problem is compounded by the acceleration of environmental degradation. Climate change increases the frequency of extreme weather events (flooding, heat waves) that damage archival storage. Reduced institutional funding for conservation—a persistent trend across European and North American libraries—means that fewer manuscripts receive the stabilization treatments (deacidification, rebinding, environmental monitoring) that extend their usable life.

23.2 The Digitization Effort

Against this background of accelerating decay, the global program of manuscript digitization represents a race—and not one that is being won.

The technology for high-quality manuscript digitization is mature: multi-spectral imaging can recover text invisible to the naked eye; 3D scanning can capture the physical structure of bindings; machine-readable metadata standards (TEI, IIIF) enable cross-collection searchability. The bottleneck is not technology but resources: trained personnel to handle fragile manuscripts, institutional agreements to permit access, and sustained funding for the labor-intensive process of image capture, metadata creation, and quality control.

For the bestiary corpus specifically, the situation is mixed. Several major bestiaries have been fully digitized and are available online: the Aberdeen Bestiary (Aberdeen University Library MS 24), the Ashmole Bestiary (Bodleian Library MS Ashmole 1511), and the Northumberland Bestiary (Getty Museum MS 100) are accessible in high-resolution digital facsimiles. But many others—including manuscripts critical for the textual analysis conducted in this paper—remain undigitized, accessible only through physical consultation or through the black-and-white microfilm images of the mid-20th century.

The fundamental problem is one of institutional priorities. Libraries hold millions of manuscripts; bestiaries are a tiny fraction. Digitization programs are typically organized by collection, not by genre: a library will digitize its “treasures” (the most visually spectacular manuscripts) before its working texts (the plain, unillustrated manuscripts that are often the most textually important). The mnemonic verses of Bodley 533—among the most pedagogically significant texts in the tradition—are in a manuscript that is neither visually spectacular nor widely known. Its

digitization priority is correspondingly low.

23.3 Manuscript Recognition: The OCR Problem for Handwriting

Digitization alone does not make manuscripts accessible. A digital image of a page of 13th-century textualis is no more readable to a non-paleographer than the original parchment. The critical missing technology is reliable handwritten text recognition (HTR) for medieval scripts.

Recent advances in machine learning have produced HTR systems (Transkribus, eScriptorium, Kraken) capable of transcribing some medieval handwriting with accuracy rates exceeding 95% for well-formed bookhand. But accuracy drops sharply for: cursive scripts (anglicana, secretary hand), damaged or faded text, text with heavy abbreviation (medieval Latin abbreviates aggressively), and manuscripts with complex page layouts (marginal glosses, interlinear annotations, illustrated pages).

For the bestiary corpus, the HTR challenge is significant. The manuscripts span four centuries and multiple script types; they contain extensive Latin abbreviation; many include marginal and interlinear glosses that are among the most textually interesting features; and the illustrated manuscripts present page layouts that confuse standard line-detection algorithms. A fully automated pipeline from manuscript image to searchable transcription remains a medium-term aspiration, not a current capability.

The consequence is that the vast majority of the bestiary corpus remains effectively inaccessible to scholars who are not trained paleographers with physical access to the manuscripts. The knowledge contained in these manuscripts—the zoological descriptions, the moralizations, the *distinctiones*, the mnemonic verses, the marginal annotations that record how the manuscripts were actually used—is locked behind a double barrier: physical fragility and paleographic opacity.

23.4 Effective Complexity as the Object of Preservation

The argument of this paper provides a framework for understanding what, exactly, is at risk of being lost. It is not merely “old manuscripts.” It is *effective complexity*—the structured information that the bestiary tradition accumulated over 1,400 years of adaptive evolution.

Effective complexity, as defined by Gell-Mann and Lloyd, is the length of the shortest description of a system’s regularities—distinct from both randomness (which has no regularities to describe) and simple order (which can be described briefly). The bestiary tradition represents a case of high effective complexity: its regularities (the kernel $\mathcal{K}$, the operations $\mathcal{C}/\mathcal{E}/\mathcal{R}$, the complexity scale C_1 – C_4 , the dual-system architecture) are not trivially simple, but they are lawful and describable. The specific instantiation of these regularities across 150 manuscripts—the particular choices made by particular compilers in particular institutional contexts—constitutes an irreplaceable record of how a complex adaptive system evolved in a specific historical environment.

When a manuscript is destroyed, what is lost is not merely a copy of a text. What is lost is:

- The specific scribal hand, which documents the training, habits, and institutional context of an individual copyist.
- The specific textual variants, which document the micro-scale adaptive operations (M_{scribe}) through which the tradition evolved.
- The specific physical properties (parchment quality, binding structure, wear patterns), which document the manuscript’s production context and use history.

- The specific marginal annotations, which document how the manuscript was actually read, taught from, and argued over by its medieval users.

None of this information is recoverable once the manuscript is destroyed. The text can survive in other copies; the specific material instantiation cannot. And it is the material instantiation—not the abstract text—that carries the richest information about the adaptive system’s actual operation.

23.5 A Formal Criterion for Preservation Priority

The formal framework permits the formulation of a preservation priority criterion. Define the *irreplaceability index* of a manuscript i as:

$$I(i) = U(i) \cdot R(i) \cdot V(i)$$

where:

- $U(i)$ is the manuscript’s **uniqueness**: the proportion of its textual and material features that are not attested in any other surviving manuscript. A manuscript that is the sole witness to a textual variant, a scribal hand, or an editorial innovation has $U(i)$ close to 1.
- $R(i)$ is the manuscript’s **risk**: the estimated probability of destruction within the next 50 years, based on its current physical condition, storage environment, and institutional stability.
- $V(i)$ is the manuscript’s **informational value**: the number of formally describable features (kernel properties, adaptive operations, complexity level, moralization density, scribal characteristics) that the manuscript documents.

Manuscripts with high $I(i)$ should receive priority for digitization, conservation treatment, and scholarly study. The index formalizes an intuition that conservators already apply informally: unique, endangered, information-rich manuscripts should be saved first. But the formal specification makes the criteria explicit, comparable, and auditable.

What it adds: The digitization crisis is not external to the formal framework; it is a direct consequence of it. The bestiary tradition accumulated effective complexity over 1,400 years through the micro-scale adaptive operations of individual scribes. Each manuscript is a unique material record of this process. The transition from parchment to digital image preserves the text but loses the material layer (M_{scribe} , wear patterns, physical structure)—just as the transition from manuscript to print eliminated the scribal adaptive operations (Section 9). The formal framework identifies what is at stake: not “heritage” in the vague sense, but quantifiable effective complexity whose loss is irreversible.

Practical application: The irreplaceability index $I(i) = U(i) \cdot R(i) \cdot V(i)$ can be computed for every manuscript in the bestiary corpus (and for any other manuscript corpus) using data that is largely already available in library catalogues and condition reports. It provides a principled basis for allocating the limited resources available for manuscript preservation and digitization—replacing ad hoc decisions about “which manuscripts are important” with a formal, auditable criterion.

A call to action: The 150 bestiary manuscripts are a finite, irreplaceable resource. At current rates of degradation, some of the most textually important but physically fragile manuscripts may become unreadable within decades. The formal framework developed in this paper provides both the justification for urgent preservation (each manuscript carries unique effective complexity) and the tools for prioritization (the irreplaceability index). The race between decay and digitization is not a metaphor; it is a measurable process with a predictable outcome if current trends continue.

24 Conclusion

This paper has proposed a formal theory of adaptive pedagogical systems and demonstrated its explanatory power through a detailed case study of the medieval Latin bestiary tradition. The central results are:

1. A persistent pedagogical genre can be modeled as a configuration $\mathcal{K} + \{M_i\}$, where $\mathcal{K}$ is an invariant structural kernel and $\{M_i\}$ is a set of modular adaptive operations.
2. Three fundamental operations—compression ($\mathcal{C}$), expansion ($\mathcal{E}$), and correction ($\mathcal{R}$)—are sufficient to account for the observed patterns of pedagogical adaptation in the bestiary tradition and are formally characterizable as general operations on any knowledge-transmission system.
3. Genre persistence requires the satisfaction of the Persistence Triad: closure, boundary maintenance, and resilience. The bestiary satisfies all three through its kernel architecture and modular operations.
4. The interpretive complexity of a pedagogical system can be measured on a four-level scale (C_1 – C_4) that captures the transition from univocal moralization to compensatory reconfiguration.
5. Zoomorphic symbolism constitutes an optimal carrier for adaptive pedagogical content because it simultaneously maximizes abstraction and mnemonic retention while minimizing cognitive friction—a prediction derived from the Structural Distortion Principle.
6. The dual-system architecture—World-System and Kingdom-System—illuminates the structural conditions under which normative pedagogy operates and explains why dual-system content resists naive transmission.

The meta-theoretical conclusion is that the medieval bestiary was not a static artifact or a quaint relic but a sophisticated adaptive system whose formal properties are shared by persistent systems across physical, biological, cognitive, and social domains. Its 1400-year survival is not an accident of cultural inertia but a consequence of structural design: an invariant kernel that provides stability, modular operations that provide adaptability, and a symbolic carrier—the animal image—that optimizes the cognitive constraints on knowledge transmission.

The bestiary’s lesson for contemporary pedagogical design is direct: persistent educational systems are those that preserve an invariant core while permitting modular adaptation at the periphery. The theory developed here provides a formal framework for analyzing, designing, and evaluating such systems.

25 Counterarguments and Responses

Any interdisciplinary paper that applies mathematical formalism to humanistic material invites objections from both sides: mathematicians may question the rigor, and humanists may question the relevance. We anticipate and respond to the strongest counterarguments.

25.1 “The formal apparatus is unnecessary — the empirical findings stand without it”

Counterargument. Dines’s discoveries about mnemonic verses, *distinctiones*, and cross-family borrowing are important contributions to bestiary scholarship. They can be stated, understood,

and appreciated without set-theoretic notation. The formalization adds complexity without adding knowledge. It dresses existing insights in mathematical clothing that obscures rather than illuminates.

Response. The formalization accomplishes four things that natural-language description cannot:

First, it reveals structural identity across disparate phenomena. Without the formal apparatus, the three strategies (verses in Bodley 533, *distinctiones* in the Third Family, cross-family borrowing in KK 4.25) appear as unrelated editorial decisions made by different compilers at different times. The operators $\mathcal{C}$, $\mathcal{E}$, $\mathcal{R}$ reveal them as instances of a common class—adaptive operations on an invariant kernel—that can be compared, combined, and evaluated within a single framework.

Second, it generates testable predictions. The formal framework predicts that other long-lived pedagogical genres should exhibit the same kernel-plus-modules architecture (Remark 5.1). This prediction is absent from Dines’s work, which does not claim generality beyond the bestiary tradition.

Third, it enables quantitative comparison. The moralization density ρ transforms the subjective judgment “this section feels thin” into a countable ratio that can be compared across manuscripts, sections, and textual families. Appendix C demonstrates this: ρ rises from 0.25 to 0.69 after the correction operation in KK 4.25.

Fourth, it connects medieval studies to systems theory, cognitive science, and information theory, creating the possibility of cross-disciplinary insight. Without formal commensurability, each discipline studies persistence in isolation; with it, structural isomorphisms become visible.

25.2 “The Persistence Triad is unfalsifiable — any long-lived genre can be fitted to it”

Counterargument. The conditions of closure, boundary maintenance, and resilience are so broad that any genre surviving long enough to be studied could be retrospectively described as satisfying them. The proposition is therefore vacuous—it excludes nothing.

Response. The objection has force against the original “if and only if” formulation, which is why we have weakened it to necessary conditions only (Proposition 4.1, Remark 4.1). In its revised form, the proposition is falsifiable by a single counterexample: a pedagogical genre that persisted across centuries while demonstrably lacking one of the three conditions.

Consider candidate counterexamples. The medieval *summae* (encyclopedic theological compilations) lack resilience: they could not be modularly adapted without complete rewriting, and they did not persist beyond the scholastic period. The *prognostica* (astrological prediction manuals) lack boundary maintenance: they blurred into medical, astronomical, and calendrical literature and lost their generic identity. The *quodlibeta* (records of academic disputations) lack closure: they require institutional context to be intelligible. All three genres are shorter-lived than the bestiary, consistent with the proposition’s predictions.

The deeper point is that the Persistence Triad is not a tautology but a *structural decomposition*: it specifies *which* properties are necessary, thereby directing attention to testable features of specific genres. A framework that says “persistent genres persist because they have properties that enable persistence” would be vacuous. A framework that says “persistent genres require closure, boundary maintenance, and resilience—and each of these can be independently assessed” is not.

25.3 “The interpretive entropy is a metaphor, not a measure”

Counterargument. The Shannon entropy formula requires probability distributions. Since we cannot reconstruct $p(m|s)$ from medieval evidence, $H(A)$ is a formula without inputs—a mathematical decoration, not a measurement.

Response. The objection is partially correct, and we have acknowledged this (Section 4). However, it conflates two distinct uses of mathematical formalism: *computation* and *structural indexing*.

The entropy formula is not used computationally in this paper (no numerical values of H are calculated from probability distributions). It is used as a structural index that makes three contributions a qualitative description cannot:

1. It assigns $H = 0$ *exactly* to C_1 systems, creating a formally sharp boundary between univocal and multi-valued moralization. Without the formula, the distinction between “one meaning” and “several meanings” is a matter of degree; with it, the boundary is a structural discontinuity.
2. It provides a computable *upper bound* when $|\mu^*(s)|$ is known: $H_{\max} = \log_2 |\mu^*(s)|$. For the bull in Appendix B, $H_{\max} = \log_2 10 \approx 3.32$ bits. This upper bound is not a metaphor; it is a precise quantity derivable from the textual evidence.
3. It connects the bestiary literature to the vast apparatus of information theory, enabling scholars in either field to recognize structural parallels without terminological translation.

Mathematical formalism in the humanities need not be confined to cases where all parameters are empirically measurable. The formalism serves to clarify logical structure, identify discontinuities, and establish cross-disciplinary connections—functions that are legitimate and valuable even when full numerical computation is impossible.

25.4 “The cognitive claims in Section 8 are untested speculation”

Counterargument. The claim that zoomorphic carriers are “optimal” for adaptive pedagogy rests on unquantified variables (α , ϕ , r). No controlled experiment has compared the pedagogical effectiveness of animal-based moral instruction with human-based moral instruction. The survival of certain bestiary images in cultural memory may reflect survivorship bias, not optimality.

Response. The objection correctly identifies the section’s empirical limitations (Remark 9.1). We do not claim experimental verification. We claim that the optimization structure—maximizing abstraction and retention while minimizing friction—provides a *structural explanation* for an otherwise puzzling observation: why animal symbolism proved more durable than other symbolic carriers across multiple civilizations, languages, and centuries.

The survivorship bias objection, while methodologically sound in principle, loses force when the survival pattern is as extreme and cross-culturally consistent as it is. The lion, eagle, serpent, and fox carry symbolic resonance from ancient Egypt through Greece, Rome, early Christianity, medieval Europe, and modern culture. The persistence of these specific zoomorphic symbols across fundamentally different civilizational contexts is difficult to attribute to mere chance or institutional momentum. The formal framework identifies a structural reason: these carriers satisfy a cognitive optimization that is independent of any specific institutional context.

The appropriate response is not to withdraw the claim but to propose tests. We identify three:

1. *Memory comparison:* Do subjects retain moral lessons presented through animal narratives better than identical lessons presented through human narratives, controlling for vividness and emotional valence?

2. *Resistance measurement*: Do subjects show less defensive resistance (measured by attitude change or argument engagement) when moral criticism is directed through animal proxies than through human proxies?
3. *Cross-cultural portability*: Do animal-based moral narratives translate across cultural boundaries with less semantic loss than human-centered moral narratives?

These experiments have not been conducted, but they are feasible and would provide direct evidence for or against Proposition 9.1.

25.5 “The dual-system framework is theology, not science”

Counterargument. Section 9 introduces a “World-System” and a “Kingdom-System” with incompatible value functions. This is Christian theology repackaged in mathematical notation. The value functions V_W and V_{K_T} are not defined with sufficient precision to support algebraic operations. The parameter λ is unestimable. The entire section should be removed or relegated to a theological appendix.

Response. The objection conflates the *origin* of a structural insight with its *domain of applicability*. The dual-system architecture is not a theological claim; it is a formal description of any situation in which an agent operates under two incompatible evaluation regimes simultaneously.

Consider non-theological instances of the same structure. A corporate whistleblower operates within a firm’s value system (V_W : loyalty, career advancement, institutional stability) while being evaluated by an external ethical or legal system (V_{K_T} : truth, public welfare, justice). The two systems prescribe incompatible actions for the same situation. A physician in a resource-constrained hospital faces the same structure: the hospital’s budgetary constraints (V_W) and the Hippocratic imperative (V_{K_T}) may prescribe different treatment decisions.

The medieval bestiary is a specific historical case in which the dual-system structure is encoded explicitly in the text’s architecture (*descriptio* = $\mathcal{W}$ -observation; *moralizatio* = $\mathcal{W}$ -to- $\mathcal{K}_T$ transition). The formal framework developed in [50] provides the vocabulary; the bestiary provides the case study. The claim is structural, not confessional: the bestiary *functions* as a dual-system mediator regardless of whether the reader accepts the theological premises of its moralizations.

The parameter λ is indeed unestimable from medieval evidence, and we have acknowledged this (Remark 10.1). Its function is to index the *logical space* of possible orientations, not to provide an empirical measurement. This is standard practice in decision theory, where utility functions are often defined qualitatively to clarify the structure of a decision problem, not to enable numerical optimization.

25.6 “The cross-domain correspondences are too vague to be meaningful”

Counterargument. Table 1 maps biological homeostasis, cognitive attractors, and bestiary self-contained chapters to a single principle of “closure.” At this level of abstraction, anything can be made to correspond to anything. The table is suggestive but demonstrates nothing.

Response. We agree that the table, taken in isolation, is suggestive rather than probative (see the revised caption). Its function is not to demonstrate a unified theory but to generate a research agenda. The critical question is whether the correspondences are *merely analogical* (different phenomena happen to look similar when described loosely) or *structurally isomorphic* (different phenomena satisfy identical formal conditions when described precisely).

The present paper provides evidence for structural isomorphism in one specific case: the bestiary’s Persistence Triad satisfies the same formal conditions established for dynamical systems

in [47]. The formal definitions of closure, boundary maintenance, and resilience are identical in both domains; only their material instantiations differ. This is not analogy; it is domain-specific realization of a general formal structure.

Whether the remaining correspondences in the table represent genuine isomorphisms or mere analogies is an open question that requires domain-specific investigation in each case. The table maps the territory to be explored; it does not claim that the exploration is complete. The appropriate response to a fruitful research program in its early stages is not to dismiss its organizing hypotheses but to test them.

25.7 “Self-citation undermines credibility”

Counterargument. The paper’s theoretical foundations rest on three works by the same author, all deposited as Zenodo preprints. This creates an echo chamber: the paper validates itself by citing itself.

Response. See Remark 11.1 for the detailed argument. The short version: self-citation is the natural and legitimate structure of any cumulative research program. Darwin’s *Descent of Man* cited *On the Origin of Species*; Einstein’s general relativity cited his special relativity; Kahneman and Tversky’s applications of prospect theory cited prospect theory. The question is not whether an author cites their own prior work but whether the prior work is (a) publicly accessible for independent scrutiny, (b) accurately represented, and (c) the genuine source of the concepts being used.

All three conditions are satisfied here. The cited works are deposited with DOIs on Zenodo, are freely downloadable, and contain the formal definitions and proofs that this paper applies. The present paper’s formal apparatus is fully specified within its own pages and can be evaluated without consulting the cited works. We welcome independent replication, critique, and extension.

The deeper concern behind the objection—that the framework has not yet been independently validated—is legitimate and acknowledged. Every research program begins with work by its originator. Independent validation is a stage in the program’s development, not a precondition for its publication. We explicitly invite such validation (Remark 11.1).

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Over two decades of meticulous philological and codicological research, Dr. Dines has fundamentally transformed the study of the English Latin bestiary from a genealogical exercise into an investigation of active pedagogical adaptation. His three principal discoveries—the mnemonic verse tradition in Bodley 533 and CUL Oo.vii.48 [14, 18, 15], the systematic integration of *distinctiones* into bestiary manuscripts [16], and the unprecedented cross-family editorial work in Cambridge, UL MS KK 4.25 [17]—are the empirical base on which every formal construct in this paper rests. The compression operator $\mathcal{C}$, the expansion operator $\mathcal{E}$, the correction operator $\mathcal{R}$, and the complexity scale C_1 – C_4 are formalizations of phenomena that Dr. Dines identified through years of patient manuscript examination, comparison, and interpretation.

Beyond these published findings, Dr. Dines has been extraordinarily generous with his time and expertise in personal conversations that shaped the present work in essential ways. His observation that the total number of genuine Latin bestiaries is closer to 100 than 150 (Section 12.9) forced a reconsideration of the survival statistics in Appendix E. His insight that the bestiary’s disappearance is inseparable from the destruction of European monasteries (Section 12.9) provided

the institutional anchor for the Persistence Triad’s boundary-maintenance condition. His detailed knowledge of individual manuscripts—their physical properties, scribal hands, marginal annotations, and institutional histories—informed the discussions of scribal personality (Section 9.5), manuscript materiality, and the audience question (Section 12.10). His work on image-text relations [23, 26], the function of bestiaries in miscellanies [24], the role of the Policraticus in Third Family bestiaries [25], and the only mappamundi in a bestiary context [21] provided the broader context without which the formal framework would have been rootless.

It should be stated with the utmost clarity that the mathematical formalization is the present author’s contribution, not Dr. Dines’s, and that Dr. Dines has not endorsed the formal framework. The responsibility for any errors, overreaches, or misinterpretations in the formal redescription of his philological findings belongs solely to the present author. The empirical discoveries are entirely his. This paper is, at its foundation, a formal reading of his life’s work, and the author is profoundly grateful for the privilege of engaging with scholarship of this caliber.

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A Compression in Action: The Pelican in Bodley 533

This appendix demonstrates the compression operator $\mathcal{C}$ on a real bestiary chapter. We present the prose chapter on the pelican from the Second B Family tradition (the textual base of Bodley 533), followed by the mnemonic verses found in that manuscript, with formal analysis.

A.1 The Prose Chapter (*descriptio* + *moralizatio*)

The pelican chapter in the Second B Family reads (in abbreviated form):

Pellicanus nimium filios suos diligit. Cum autem genuerit natos, et coeperint crescere, percutiunt parentes suos in faciem. Parentes autem reperiunt eos et occidunt. Tertia autem die mater eorum advenit, et aperit latus suum et incumbit super pullos suos, et effundit sanguinem suum super corpora mortuorum, et sic suscitatur eos a mortuis...

Moralizatio: Sic et Dominus noster per Isaiam prophetam dicit: "Filios enutrivit et exaltavit; ipsi autem spreverunt me." Conditor omnium rerum nos genuit, et nos eum percussimus. Quomodo percussimus eum? Quoniam servivimus creaturae potius quam creatori. Ascendit in altum crucis, et aperto latere suo, emisit sanguinem et aquam in salutem et vitam aeternam...

[**Translation:**] The pelican loves its young exceedingly. When it has brought forth its chicks and they begin to grow, they strike their parents in the face. The parents strike them back and kill them. But on the third day their mother comes, opens her side, and lies upon her young, pouring her blood over the bodies of the dead, and thus raises them from the dead... *Moralization:* So too our Lord says through the prophet Isaiah: "I have nourished and raised up children, but they have despised me." The Creator of all things begot us, and we struck him. How did we strike him? Because we served the creature rather than the Creator. He ascended to the height of the Cross, and opening his side, poured forth blood and water for salvation and eternal life...

The chapter comprises approximately 180 words of zoological description followed by approximately 120 words of moralization — a total of ~300 words.

A.2 The Mnemonic Verses from Bodley 533

The corresponding verses in Bodley 533 (fol. 45v) read:

*Qui pellicanus natos dum diligit, ipsi
Ledunt; sed pullos mater amore fouet.
Ter perit illa, semel sanguis fluit, inde resurgunt:
Sic Christus moriens nos ab amore leuat.*

[**Translation:**] The pelican, while it loves its young — they wound it; but the mother warms her chicks with love. Three times she perishes, once blood flows, thence they rise again: so Christ dying lifts us up through love.

These four lines of Leonine hexameter contain approximately 28 words.

A.3 Formal Analysis

Component	Prose chapter	Mnemonic verses
Zoological content	~180 words	3 lines (~21 words)
Moralization content	~120 words	1 line (~7 words)
Total volume	~300 words	~28 words
Compression ratio	—	~10.7:1
Semantic inventory:		
Parental love	✓	✓
Filial aggression	✓	✓ (<i>ledunt</i>)
Death of young	✓	✓ (<i>perit</i>)
Blood from breast	✓	✓ (<i>sanguis fluit</i>)
Resurrection of young	✓	✓ (<i>resurgunt</i>)
Christ typology	✓	✓ (<i>Christus moriens</i>)
Isaiah quotation	✓	×
Elaborated theology	✓	×

The compression operator $\mathcal{C}$ achieves a 10.7:1 volume reduction while preserving all six core semantic elements of the chapter. The two elements lost (the Isaiah quotation and the elaborated theological gloss) are expansions of the core typology, not independent semantic units. The operation is lossy but semantically faithful, confirming the formal specification: $|v(s)| \ll |D(s) + \mu(s)|$ and $\text{sem}(v(s)) \approx \text{sem}(D(s), \mu(s))$.

B Expansion in Action: The Bull in the Third Family

This appendix demonstrates the expansion operator $\mathcal{E}$ by presenting the bull chapter from the Third Family bestiary, showing how a simple Isidorian description (C_1) is transformed into a multi-level exegetical resource (C_3).

B.1 The Isidorian *descriptio* (from *Etymologiae* XII)

Bos fortis et patiens animal, cuius nomen Graece bous vocatur. Boues praeterea praesagire futura dicuntur... Cornua boum si igni propinqua sint, ceras facere dicuntur...

[**Translation:**] The ox is a strong and patient animal, whose name in Greek is called *bous*. Oxen are moreover said to foretell the future... If the horns of oxen are brought near fire, they are said to produce wax-like substances...

This description (C_1 level) provides natural-historical information only. No moralization is present. The moralization function is empty: $\mu_{\text{Isidore}}(\text{bos}) = \emptyset$.

B.2 The *distinctio* from William de Montibus

The Third Family adds a *distinctio* after the Isidorian text. The following reconstruction follows Dines (2024, pp. 67–68):

[**Literal**] *Bos animal domesticum, labori iugiter aptum...*

[**Allegorical, in bono**] *Per bovem significatur praedicator qui laborat pro omnibus... Item per bovem significatur populus Iudaicus... Item per bovem significatur Lucas Evangelista...*

[**Tropological, in bono**] *Per bovem significatur mansuetudo et bona opera...*

[*Anagogical*] *Per bovem significatur Christus, qui est bos sive vitulus in moriendo...*

[*Allegorical, in malo*] *Per bovem significatur diabolus... Item curiositas... Item obstinatio... Item quilibet stultus et luxuriosus...*

[**Translation:**] [**Literal**] The ox is a domestic animal, continually suited to labor... [**Allegorical, in bono**] By the ox is signified the preacher who labors for all... Likewise by the ox is signified the Jewish people... Likewise the Evangelist Luke... [**Tropological, in bono**] By the ox is signified meekness and good works... [**Anagogical**] By the ox is signified Christ, who is an ox or calf in dying... [**Allegorical, in malo**] By the ox is signified the devil... Likewise curiosity... Likewise obstinacy... Likewise any foolish and lascivious person...

B.3 Formal Analysis

The expansion operator transforms the moralization function as follows:

$$\mathcal{E} : \mu(\text{bos}) = \emptyset \mapsto \mu^*(\text{bos}) = \{m_1, m_2, \dots, m_{10}\}$$

Level	Reading	<i>in bono/in malo</i>	Index
Literal	Domestic animal, suited to labor	—	m_1
Allegorical	The preacher laboring for all	<i>in bono</i>	m_2
Allegorical	The Jewish people	<i>in bono</i>	m_3
Allegorical	The Evangelist Luke	<i>in bono</i>	m_4
Tropological	Meekness	<i>in bono</i>	m_5
Tropological	Good works	<i>in bono</i>	m_6
Anagogical	Christ (“a bull in dying”)	<i>in bono</i>	m_7
Allegorical	The devil	<i>in malo</i>	m_8
Allegorical	Curiosity / obstinacy	<i>in malo</i>	m_9
Allegorical	Any foolish and lascivious person	<i>in malo</i>	m_{10}

The interpretive entropy increases from $H = 0$ (no moralization) to:

$$H(\text{bos}) = - \sum_{i=1}^{10} p(m_i|\text{bos}) \log_2 p(m_i|\text{bos})$$

Under a uniform distribution ($p = 0.1$ for each), $H(\text{bos}) = \log_2 10 \approx 3.32$ bits. A preacher selecting material from this chapter has over ten distinct interpretive pathways available—a transformation from zero to substantial homiletic flexibility.

C Correction in Action: The Birds Section of KK 4.25

This appendix demonstrates the correction operator $\mathcal{R}$ by comparing the Birds section in the standard Third Family manuscript (Fitzwilliam 254) with the modified version in KK 4.25.

C.1 Standard Third Family Order (Fitzwilliam 254)

Chapter	Source	Moralized?
Ibis I	Third Family	No
Phoenix I	Third Family	Abbreviated
Caradrius	Third Family	Yes
Assida I (Ostrich)	Third Family	No
Hericinea	Third Family	No
Fulica I	Third Family	No

Moralization density: $\rho = 1.5/6 \approx 0.25$ (counting the abbreviated Phoenix moralization as 0.5).

C.2 Modified Order in KK 4.25

Chapter	Source	Moralized?	Inserted?
Ibis I	Third Family	No	—
Ibis II	Second Family	Yes	✓
<i>Quattuor Modis Peccatum</i>	Gregory the Great	Yes	✓
<i>Quattuor Colores Caelestis</i>	Isidore (modified)	Yes	✓
Assida II	Second Family	Yes	✓
Fulica II	Second Family	Yes	✓
Alcyon II	Second Family	Yes	✓
Phoenix II/III	Second Family	Yes	✓
Caradrius	Third Family	Yes	—
Mergus	Second Family	Yes	✓
Assida I	Third Family	No	—
Hericinea	Third Family	No	—
Fulica I	Third Family	No	—

Moralization density after correction: $\rho = 9/13 \approx 0.69$.

C.3 The Cause–Disease–Cure Sequence

The three chapters Ibis II → *Quattuor Modis Peccatum* → *Quattuor Colores Caelestis* form an integrated homiletic unit:

Chapter	Function	Content
Ibis II	DIAGNOSIS	Spiritual disease: carnal men feeding on deadly deeds
<i>Quattuor Modis Peccatum</i>	ETIOLOGY	Root cause: serpent’s suggestion → Eve’s delight → Adam’s consent → audacity of non-confession
<i>Quattuor Colores Caelestis</i>	PRESCRIPTION	Cure: baptism, symbolized by rainbow’s purple (modified from Isidore)

The correction operator raises ρ from 0.25 to 0.69—a near-tripling of moralization density—while constructing a coherent theological argument that no single chapter in either the Second or Third Family contained independently. This confirms that $\mathcal{R}$ operates not merely as quantitative repair (filling gaps) but as qualitative synthesis (creating new didactic structures from imported components).

D Dual-System Analysis: The Lion Chapter

This appendix demonstrates the dual-system architecture ($\mathcal{W} / \mathcal{K}_T$) on the lion chapter—the opening and most canonical chapter of the bestiary tradition.

D.1 The Text

The BIs lion chapter describes three “natures” of the lion:

Prima natura: *Cum ambulat in montanis, et odor venantium venerit ad eum, cauda sua tegit vestigia sua. . .*

Secunda natura: *Cum dormit, oculi eius vigilant; aperti enim sunt. . .*

Tertia natura: *Cum leaena parit catulum, generat eum mortuum, et custodit eum tribus diebus, donec veniens pater eius sufflet in faciem eius et vivificet eum. . .*

[**Translation:**] **First nature:** When the lion walks in the mountains and the scent of hunters reaches him, he covers his tracks with his tail. . . **Second nature:** When he sleeps, his eyes keep watch; for they are open. . . **Third nature:** When the lioness gives birth to her cub, she bears it dead, and guards it for three days, until its father comes and breathes in its face and gives it life. . .

The moralization follows:

Sic et Dominus noster Iesus Christus. . . vestigia suae divinitatis abscondit. . . Dormivit enim corpore in cruce, sed divinitas eius vigilabat. . . Sic Pater omnipotens tertia die a mortuis Filium suum primogenitum suscitavit. . .

[**Translation:**] So too our Lord Jesus Christ. . . concealed the traces of his divinity. . . For he slept in body on the Cross, but his divinity kept watch. . . So the Almighty Father on the third day raised from the dead his firstborn Son. . .

D.2 Dual-System Decomposition

Nature	World-System $\mathcal{W}$ (zoological observation)	Transition to $\mathcal{K}_T$ (theological interpretation)
Tracks	The lion covers its tracks with its tail to evade hunters. <i>$\mathcal{W}$-logic: predator-prey survival strategy. Rational under scarcity and threat.</i>	Christ concealed his divinity from the devil during the Incarnation. <i>$\mathcal{K}_T$-logic: concealment is not deception but salvific strategy. $V_{\mathcal{W}} < 0$ (vulnerability), $V_{\mathcal{K}_T} > 0$ (redemption).</i>
Sleep	The lion sleeps with open eyes. <i>$\mathcal{W}$-logic: vigilance as survival mechanism.</i>	Christ's body slept on the Cross, but his divinity remained vigilant. <i>$\mathcal{K}_T$-logic: death is not terminal but a transition state. $\mathcal{W}$-death $\neq \mathcal{K}_T$-death.</i>
Cubs	The lioness bears dead cubs; the father breathes life into them on the third day. <i>$\mathcal{W}$-logic: death is absorbing state. The narrative is zoologically "impossible" within $\mathcal{W}$.</i>	God the Father raised Christ from the dead on the third day. <i>$\mathcal{K}_T$-logic: resurrection is the paradigmatic cross-system transition. What is impossible in $\mathcal{W}$ is lawful in $\mathcal{K}_T$.</i>

D.3 Formal Interpretation

Each “nature” of the lion constitutes a transition operator T_i that maps a $\mathcal{W}$ -observation to a $\mathcal{K}_T$ -interpretation:

$$T_i : D_{\mathcal{W}}(\text{leo}, i) \mapsto \mu_{\mathcal{K}_T}(\text{leo}, i)$$

The third nature is the most structurally revealing. Within $\mathcal{W}$, the revival of dead cubs is impossible—death is an absorbing terminal state. The moralization asserts that within $\mathcal{K}_T$, this impossibility is overridden: resurrection is not a violation of law but the operation of a different law. The bestiary chapter does not resolve this cross-system incompatibility; it *presents* it as a fact requiring dual-system acceptance.

The evaluation functions conflict directly:

$$V_{\mathcal{W}}(\text{death of cubs}) = \text{terminal loss}, \quad V_{\mathcal{K}_T}(\text{death of cubs}) = \text{precondition for resurrection}$$

A student reading this chapter is being trained to hold both evaluations simultaneously without collapsing one into the other—precisely the skill that the dual-system architecture requires. The bestiary’s pedagogical function, at its deepest level, is the cultivation of dual-system cognitive capacity: the ability to read the natural world ($\mathcal{W}$) as a structured field of signs pointing to a transcendent order ($\mathcal{K}_T$) whose regularities differ from, and sometimes invert, those of the world.

When the expansion operator $\mathcal{E}$ is applied (as in the Third Family), the number of transition pathways multiplies. The lion’s *distinctio* interprets the animal not only as Christ (*in bono*) but also

as the devil (*in malo*): “The devil, like a roaring lion, walketh about seeking whom he may devour” (1 Peter 5:8). The same $\mathcal{W}$ -observation (the lion’s power and ferocity) supports two opposite $\mathcal{K}_T$ -interpretations, depending on the hermeneutical level selected. This is not contradiction but the signature of a dual-system mediator operating at C_3 complexity: the interpretive entropy $H > 0$ reflects the genuine multiplicity of cross-system mappings available to the reader.

Glossary of Interdisciplinary Terms

Adaptive module M

A function that transforms the pedagogical kernel $\mathcal{K}$ while preserving its structural identity. (Systems theory)

Adjacent possible

The set of structurally viable configurations accessible from the current state of a system. Term from Kauffman. (Complexity theory)

Allegory

A mode of interpretation in which narrative elements stand for abstract or spiritual realities distinct from their literal referents. (Literary theory / theology)

Anagogical sense

The fourth level of medieval exegesis, reading a text as pointing toward eschatological fulfillment and ultimate spiritual realities. (Theology)

Anglicana

A cursive script widely used in England from the 13th century, typically employed for glosses and informal text. (Paleography)

Archetype

A primordial psychic pattern recurring across cultures, not learned but structurally inherent in the human psyche. Term from Jung. (Analytical psychology)

Artes praedicandi

“Arts of preaching” — medieval treatises on sermon composition and rhetorical technique. (Medieval studies)

Attractor

A state or set of states toward which a dynamical system tends to evolve. (Dynamical systems theory)

BIs family

A bestiary textual family combining the *Physiologus* version B with material from Isidore of Seville’s *Etymologiae*. (Bestiary studies)

Bit The fundamental unit of information; the amount of information needed to distinguish between two equally probable outcomes. (Information theory)

Boundary maintenance

The condition that a system maintains a recognizable distinction between interior and exterior. One of the three conditions of the Persistence Triad. (Systems theory)

Boundary-Conditioned Verification

The second Formalization Law: verification of any formal claim is necessarily conditioned by explicit boundary regimes. (ISM methodology)

C_1 – C_4

The four levels of the Interpretive Complexity Scale: C_1 (univocal moralization), C_2 (compressed retention), C_3 (multiple exegesis), C_4 (compensatory reconfiguration). (This paper)

Castration anxiety

In Freudian theory, the fear of loss of generative power, which the bestiary beaver chapter externalizes and transforms into voluntary renunciation. (Psychoanalysis)

Catechumen

A person receiving instruction in the Christian faith prior to baptism. (Theology)

Closure

The condition that a system’s internal processes form a self-sustaining cycle. One of the three conditions of the Persistence Triad. (Systems theory)

Co-adaptation (of D and μ)

The coupling between the description and moralization functions, such that correcting one may destroy the other. (This paper)

Codicology

The study of the physical structure of manuscripts (binding, layout, materials, construction). (Medieval studies)

Cognitive friction (ϕ)

The degree of defensive resistance a symbolic carrier provokes in the recipient. High for human agents, low for zoomorphic carriers. (This paper)

Compression operator ($\mathcal{C}$)

The operator that reduces a didactic module's volume while preserving its semantic core. (This paper)

Convergent evolution

The independent emergence of similar structures in unrelated systems under similar selection pressures. (Biology / this paper)

Correction operator ($\mathcal{R}$)

The operator that repairs topological deficits in a didactic corpus by importing material from external sources. (This paper)

 D : Description function

The mapping from each symbolic node to its factual or narrative description. $D : S \rightarrow \mathcal{T}$. (This paper)

Decretum Gelasianum

A 6th-century pseudo-papal decree listing canonical, apocryphal, and suspect texts, which condemned the *Physiologus* by name. (Church history)

Descriptio

The zoological description portion of a bestiary chapter, presenting the animal's "natures" or observable behaviors. (Bestiary studies)

Dissipative structure

A self-organized structure maintained far from thermodynamic equilibrium by continuous energy throughput. Term from Prigogine. (Thermodynamics)

Distinctiones

Medieval systematic multi-level interpretations of biblical or theological terms, distinguishing literal, allegorical, tropological, and anagogical senses. (Medieval studies)

Dual-coding theory

The theory that information encoded in both verbal and imagistic formats is more readily retained. Term from Paivio. (Cognitive psychology)

Dual-system architecture

A formal framework in which an agent operates within a World-System $\mathcal{W}$ while being evaluated by a Kingdom-System $\mathcal{K}_T$, with incompatible value functions. (This paper / Gospel Formalization)

Ductus

The characteristic movement and rhythm of a scribe's pen strokes, revealing speed, training, and intention. (Paleography)

Emblematic pedagogy

Pedagogical strategy that minimizes cognitive friction by using abstract or zoomorphic carriers detachable from content. Optimized for long-term transmission. (This paper)

Entropy (interpretive)

A structural index measuring the interpretive range available in a pedagogical system. $H = 0$ for univocal moralization; $H > 0$ for multiple exegesis. (This paper / information theory)

Epistemology

The branch of philosophy concerned with the nature, sources, and limits of knowledge. (Philosophy)

Etymologiae

An encyclopedic work by Isidore of Seville (c. 560–636), a primary source for bestiary zoological descriptions. (Medieval studies)

Exemplum (pl. *exempla*)

A brief illustrative narrative used in medieval preaching to make moral points concrete. (Medieval studies)

Expansion operator ($\mathcal{E}$)

The operator that transforms the moralization function from single-valued to set-valued, multiplying interpretive pathways. (This paper)

Falsifiability

The property of a claim that allows it to be refuted by evidence. A criterion of scientific meaningfulness proposed by Popper. (Philosophy of science)

Formalization

The process of rendering implicit structure explicit through defined elements, operations, and rules. (Logic / ISM methodology)

Formalization Law I (Total Formalizability)

Any phenomenon exhibiting determinate structure is formally representable under a sufficient constraint set. (ISM methodology)

Formalization Law II (Boundary-Conditioned Verification)

Verification of any formal claim is conditioned by explicit boundary regimes. (ISM methodology)

Formalization Law III (Plurality and Incompleteness)

No single formalization can exhaust a structured phenomenon. (ISM methodology)

Fourfold exegesis

The medieval hermeneutical framework distinguishing literal, allegorical, tropological, and anagogical senses of a text. (Theology)

Genre

A category of literary or textual composition characterized by shared formal properties, audience, and function. (Literary theory)

Gospel Formalization

The systematic extraction of structural laws from Evangelic maxims, treating them as conditional transition operators across dual systems. (ISM methodology)

Gular pouch

The expandable throat sac of pelicans, used for catching fish. The source of the misobservation that the pelican feeds young with its blood. (Ornithology)

Hermeneutics

The theory and methodology of interpretation, especially of texts. (Philosophy / theology)

Homeostasis

The self-regulating process by which a biological system maintains internal stability. Corresponds to “closure” in the cross-domain table. (Biology)

Homiletic

Of or relating to the art of preaching. (Theology)

Horizontal gene transfer

The transfer of genetic material between organisms other than by vertical transmission from parent to offspring. Corresponds to the correction operator $\mathcal{R}$. (Molecular biology)

Incipit

The opening words of a text or chapter, often written in display script in medieval manuscripts. (Cod-icology)

In bono

“In the good [sense]” — an interpretation of a symbol that identifies it with a positive moral or theological quality. (Medieval exegesis)

In malo

“In the bad [sense]” — an interpretation of a symbol that identifies it with a negative moral or theological quality. (Medieval exegesis)

Information density

The ratio of pedagogically relevant content to total textual volume in a section of a didactic corpus. (This paper / information theory)

Interpretive Complexity Scale

See C_1 – C_4 . (This paper)

Invariant

A property or structure that remains unchanged under a specified class of transformations. (Mathematics / systems theory)

ISM (Intellectual Substrate Mining)

A methodology for extracting formal laws from pre-formal wisdom traditions. (ISM methodology)

Isomorphism

A structure-preserving mapping between two systems such that corresponding elements satisfy identical relations. (Mathematics)

 $\mathcal{K}$: Structural kernel

The invariant ordered triple (S, D, μ) defining a pedagogical genre’s identity. (This paper)

 $\mathcal{K}_T$: Kingdom-System

An open, non-finite evaluative system with non-conservation and non-local assessment. (This paper / Gospel Formalization)

Kernel (viability)

In viability theory, the largest set of initial states from which a trajectory can remain within a constraint set indefinitely. (Viability theory)

λ The parameter indexing an agent’s degree of orientation toward the World-System. $\lambda = 1$: pure $\mathcal{W}$ -optimization; $\lambda = 0$: pure $\mathcal{K}_T$ -orientation. (This paper)

Leonine hexameter

A verse form in which the final word of each line rhymes with a word at the caesura. Used for mnemonic bestiary verses. (Medieval Latin prosody)

Liber naturae

“The book of nature” — the theological doctrine that the created world is a second book, parallel to Scripture, through which God communicates. (Theology)

Literal sense

The first level of medieval exegesis, attending to the surface meaning of the text. (Theology)

Lossy compression

Compression that sacrifices detail to reduce volume while preserving essential information above a threshold. (Information theory)

 $\mathcal{M}$: Moral-interpretive space

The set of all available moral or allegorical interpretations. The range of μ . (This paper)

$\mathcal{M}_{\text{compatible}}(s)$

The subset of moral attributions archetypally congruent with the observable and symbolic properties of species s . (This paper)

Minimally counterintuitive

A cognitive property of representations that violate one or two ontological expectations while preserving others, making them attention-grabbing and memorable. Term from Boyer. (Cognitive science of religion)

Mnemonic

Of or relating to techniques for improving memory retention. (Cognitive psychology)

Modularity

The property of a system whose components can be independently modified without destroying the whole. (Systems theory)

Moralization

The moral or allegorical interpretation appended to a bestiary chapter's zoological description. (Bestiary studies)

Moralization density (ρ)

The proportion of symbolic nodes in a section that carry explicit moral interpretations. $\rho(A) = |\{s \in A : \mu(s) \neq \emptyset\}|/|A|$. (This paper)

μ : Moralization function

The mapping from each symbolic node to its moral or allegorical interpretation. $\mu : S \rightarrow \mathcal{M}$. (This paper)

μ^* : Expanded moralization

The set-valued moralization function after application of the expansion operator $\mathcal{E}$. $\mu^*(s) = \{m_1, \dots, m_n\}$. (This paper)

Mutualism

A biological interaction in which both species benefit. The crocodile–plover relationship documented in bestiaries is a real example. (Ecology)

Natura

“Nature” (of an animal) — the bestiary term for a distinctive behavior or property of a species serving as the basis for moralization. (Bestiary studies)

Noether's theorem

A result in physics linking continuous symmetries to conservation laws. Referenced in the Unified Structural Theory. (Theoretical physics)

Ontological asymmetry

The qualitative (not merely quantitative) difference between the World-System and the Kingdom-System. (This paper / Gospel Formalization)

Optimization structure

The formal identification of what a system maximizes and minimizes, independent of the specific numerical values. (This paper)

Ouroboros

The ancient symbol of a serpent eating its own tail, representing cyclical renewal. An archetypal configuration associated with the serpent. (Symbolic studies)

$\mathcal{P}(\mathcal{M})$

The power set of the moral-interpretive space: the set of all subsets of $\mathcal{M}$. The range of the expanded moralization function. (Set theory)

Paleography

The study of historical handwriting, its forms, techniques, and social contexts. (Medieval studies)

Parasitism (cultural)

The appropriation of pre-existing symbolic structures by a new cultural system, used here in the neutral ecological sense. (This paper)

Parchment

Animal skin (usually calf, sheep, or goat) prepared for writing. The material substrate of medieval manuscripts. (Codicology)

Pedagogical technology

Any organized system of instruction designed to transmit knowledge, skills, or values across time and populations. (Education theory / this paper)

Pelicanus in pietate sua

“The pelican in her piety” — the iconic image of the pelican feeding her young with her own blood, a widespread Eucharistic symbol. (Christian iconography)

Perceptual symbol systems

The theory that conceptual processing is grounded in modal sensory simulations, not amodal propositional representations. Term from Barsalou. (Cognitive psychology)

Persistence Triad

The three necessary conditions for genre persistence: closure, boundary maintenance, and resilience. (This paper)

Physiologus

“The Naturalist” — the Greek text (2nd–4th c.) from which the medieval Latin bestiary descends. Approximately 48 chapters of zoological description with moral interpretation. (Bestiary studies)

Power set

The set of all subsets of a given set. $\mathcal{P}(X) = \{A : A \subseteq X\}$. (Set theory)

Praelectio

A preparatory overview of material to be learned, used as an advance organizer. (Medieval pedagogy)

Prepared learning

The predisposition of cognitive systems to form certain associations more readily than others, due to evolutionary pressures. (Evolutionary psychology)

Quattuor Colores Caelestis

“The Four Colors of the Heavenly [Rainbow]” — a non-zoological chapter unique to KK 4.25, interpreting the rainbow’s colors as symbols of baptismal grace. (Bestiary studies)

Quattuor Modis Peccatum

“The Four Ways of Sin” — a chapter from Gregory the Great, integrated into KK 4.25 as part of a cause–disease–cure homiletic sequence. (Bestiary studies)

Rank inversion

A reversal of ordering between two evaluation systems: what is highest in $\mathcal{W}$ may be lowest in $\mathcal{K}_T$. “The last shall be first.” (This paper / Gospel Formalization)

Recension

A critical edition of a text established by comparison of multiple manuscript witnesses. (Textual criticism)

Repetitio

A retrospective review of material already learned, reinforcing retention. (Medieval pedagogy)

Requisite variety

Ashby’s law: a system’s regulatory capacity must match the variety of disturbances it faces. (Cybernetics)

Resilience

The capacity of a system to recover from perturbation without losing structural identity. One of the three conditions of the Persistence Triad. (Systems theory)

Rubric

A heading, instruction, or liturgical direction written in red ink in a manuscript. (Codicology)

S: Symbolic nodes

The finite set of subjects of instruction in a pedagogical genre. For the bestiary: the set of animal species. (This paper)

Scriptio continua

Writing without word separation, common in early medieval manuscripts. (Paleography)

Scriptorium

A room in a monastery or other institution dedicated to the copying of manuscripts. (Medieval studies)

Second Family

A group of bestiary manuscripts (subfamilies A–D) expanding the BIs core with additional Isidorian material. (Bestiary studies)

Self-organization

The spontaneous emergence of ordered structures in systems far from equilibrium, without external design. (Complexity theory)

Semantic fidelity

The degree to which a compressed or transformed text preserves the meaning of the original. $\text{sem}(v(s)) \approx \text{sem}(D(s), \mu(s))$. (This paper / information theory)

Shadow (Jungian)

The unconscious aspect of the personality containing repressed, denied, or undeveloped qualities. The fox in the bestiary activates the shadow archetype. (Analytical psychology)

Shannon entropy

The measure $H = -\sum p_i \log_2 p_i$ quantifying uncertainty or information content in a probability distribution. (Information theory)

Situational pedagogy

Pedagogical strategy that maximizes cognitive friction by using human agents in social scenarios. Optimized for immediate transformative encounter. Genre: parable. (This paper)

Species (bestiary)

The identification portion of a bestiary chapter: the naming of the animal being described and moralized. (Bestiary studies)

Splitting

In Kleinian psychoanalysis, the infant's earliest psychic operation dividing the object-world into idealized good and persecutory bad objects. Corresponds to the *in bono/in malo* structure. (Psychoanalysis)

Stemma

A diagram representing the genealogical relationships between manuscript copies of a text. (Textual criticism)

Structural Distortion Principle

The formal result that every act of perception is simultaneously an act of non-perception. (Unified Structural Theory)

Structural index

A quantity used to clarify logical structure and identify discontinuities, rather than to provide exact numerical values. The interpretive entropy H is a structural index. (This paper)

Structural kernel

See $\mathcal{K}$. (This paper)

Survivorship bias

The error of drawing conclusions from surviving cases while ignoring those that did not survive. (Statistics / methodology)

Teleological epistemology

A framework in which knowing *what something is for* is prior to knowing *what it does*. The bestiary's zoology operates teleologically. (Philosophy of science)

Textualis

A formal bookhand used for main text in medieval manuscripts, characterized by angular strokes and consistent letter forms. (Paleography)

Third Family

The most expansive bestiary textual family (over 150 chapters), integrating Second Family material with extensive *distinctiones*. (Bestiary studies)

Topological repair

The restoration of structural completeness in a corpus by importing material to fill gaps. (This paper)

Total Formalizability

The first Formalization Law: any phenomenon exhibiting determinate structure is formally representable under a sufficient constraint set. (ISM methodology)

Transition operator

A function that maps an observation in the World-System to an interpretation in the Kingdom-System. The bestiary moralization as a transition operator. (This paper / Gospel Formalization)

Tree-Top Meta-Method

The methodological principle: generalize formally until generalization is no longer possible, then derive domain-specific consequences from what remains true. (Unified Structural Theory)

Trickster

An archetypal figure embodying cunning, boundary-violation, and ambiguous morality. The fox is the bestiary's trickster. (Analytical psychology / anthropology)

Tropological sense

The third level of medieval exegesis, reading a text as moral guidance for individual conduct. (Theology)

Typology

A mode of biblical interpretation in which Old Testament events prefigure New Testament fulfillments. Extended in the bestiary to natural phenomena. (Theology)

Unified Structural Theory

The broader research program arguing that persistent systems across physical, biological, cognitive, and social domains satisfy the same formal conditions. (Kriger, 2025)

Univocal moralization

Moralization at C_1 level: each symbolic node maps to exactly one interpretation. $|\mu(s)| = 1$. (This paper)

$V_{\mathcal{K}_T}$ The value function of the Kingdom-System, potentially anti-correlated with $V_{\mathcal{W}}$. (This paper / Gospel Formalization)

$V_{\mathcal{W}}$ The value function of the World-System, monotonic with accumulation and security. (This paper / Gospel Formalization)

Viability theory

A mathematical framework modeling the conditions under which a system's state trajectory remains within a constraint set. Developed by Aubin. (Mathematics)

 $\mathcal{W}$: World-System

A closed, finite evaluative system characterized by scarcity, irreversible loss, and locally measurable outcomes. (This paper / Gospel Formalization)

Wear patterns

Physical evidence of repeated use on manuscript pages (rubbing, staining, dog-earing), indicating which sections were most frequently consulted. (Codicology)

William de Montibus

(c. 1140–1213) Chancellor of Lincoln Cathedral, author of *Distinctiones Theologicae* and *Numerale*, principal source of Third Family *distinctiones*. (Medieval studies)

Zoomorphic symbolism

The use of animal images as carriers of moral, spiritual, or abstract meaning. (Symbolic studies / this paper)

 ρ : Moralization density

See Moralization density. (This paper)

 ρ^* : Moralization threshold

The minimum moralization density below which a compiler perceives a section as didactically inadequate, triggering the correction operator. (This paper)

 ϕ : Cognitive friction

See Cognitive friction. (This paper)

Cross-system mapping

A function that translates observations or values from one evaluative system ($\mathcal{W}$) to another ($\mathcal{K}_T$). The bestiary moralization is a cross-system mapping. (This paper / Gospel Formalization)

Florilegium

(pl. *florilegia*) A medieval compilation of excerpts from authoritative sources, organized by topic. Shares modular architecture with the bestiary. (Medieval studies)

Glossator

A scholar who added explanatory notes (glosses) to a text, constituting a micro-scale adaptive operation on the manuscript. (Medieval studies)

Heterogeneous audience

A recipient population with varying levels of intellectual preparation, requiring materials adaptable across complexity levels C_1 – C_4 . (This paper / education theory)

Kalīla wa-Dimna

An Arabic collection of animal fables of Indian origin (from the Sanskrit *Pañcatantra*), representing an independent convergent instance of zoomorphic moral pedagogy. (Comparative literature)

Maneki-neko

“Beckoning cat” — a Japanese talisman of good fortune, illustrating culturally divergent symbolism of the cat (cf. witchcraft associations in medieval Europe). (Cross-cultural studies)

Overdetermination

A condition in which multiple independent factors each suffice to produce a given outcome, making that outcome structurally robust. The bestiary’s emergence is overdetermined. (Philosophy of science / this paper)

Pañcatantra

“Five Treatises” — a Sanskrit collection of animal fables (c. 3rd c. BCE), representing a non-Christian parallel to the bestiary’s pedagogical architecture. (Comparative literature)

Jātaka tales

Stories of the Buddha’s previous births, frequently in animal form, constituting a Buddhist analogue to zoomorphic moral pedagogy. (Comparative religion)

Selective omission

The principle that adaptive systems optimize by excluding information, not by including all of it. The cognitive foundation of the Structural Distortion Principle. (Cognitive science / this paper)

Ars dictaminis

“The art of letter-writing” — a medieval rhetorical discipline with a kernel-plus-modules structure analogous to the bestiary’s. Candidate for Persistence Triad testing. (Medieval studies)

Symbolic carrier

Any concrete image, narrative, or object used to convey abstract or moral content. Animals, human characters, and heraldic devices are all symbolic carriers with different cognitive properties. (This paper)

E Falsifiable Predictions and Monte Carlo Analysis of Manuscript Survival

This appendix derives a set of falsifiable predictions from the formal framework developed in the main text and proposes a Monte Carlo methodology for analyzing manuscript survival patterns in the Latin bestiary tradition. The central question is: given that approximately 150 bestiary manuscripts survive from a tradition spanning the 2nd through 16th centuries, what can we infer about the *total* population of manuscripts that once existed, about the survival process that selected the extant corpus, and about the likelihood of discovering new manuscripts?

E.1 The Survival Problem

Of the manuscripts containing Latin bestiaries produced between the 2nd and 16th centuries, approximately 150 survive. This number, established through cumulative cataloguing efforts over a century [9, 10, 11], represents an unknown fraction of the total production. Manuscripts were destroyed by fire, water, neglect, deliberate scraping for parchment reuse (*palimpsest*), dissolution of monastic libraries, and the simple accumulation of centuries. The survival rate for medieval manuscripts generally is estimated at 5–10% of total production for popular texts and below 1% for rare texts.

The question “how many bestiaries were originally produced?” has not, to our knowledge, been addressed with formal methods. Philological tradition has approached manuscript survival as a brute historical fact: what survives, survives; what is lost, is lost. But the formal framework of the main paper permits a more analytical approach, because the surviving manuscripts are not a random sample of the original population. They are the output of a *selection process* governed by identifiable structural, institutional, and material factors—factors that the Persistence Triad formalizes.

E.2 Estimating the Total Population

E.2.1 The Capture-Recapture Analogy

Ecologists estimate animal populations using capture-recapture methods: tag a sample, release, recapture a second sample, and estimate total population from the overlap. An analogous method applies to manuscripts. When independent cataloguing campaigns (James 1928, McCulloch 1962, Clark 2006, Dines 2008–2025) each discover some manuscripts not found by predecessors, the overlap between campaigns permits an estimate of the total population.

Formal specification: Let C_1, C_2, C_3, C_4 denote the sets of manuscripts identified by each of the four major cataloguing efforts. Define:

$$\hat{N} = \frac{|C_i| \cdot |C_j|}{|C_i \cap C_j|}$$

for each pair (i, j) . The Chapman estimator (bias-corrected):

$$\hat{N}_{\text{Chapman}} = \frac{(|C_i| + 1)(|C_j| + 1)}{|C_i \cap C_j| + 1} - 1$$

Data required: The number of manuscripts newly identified by each campaign that were unknown to all previous campaigns.

Expected result: If the discovery rate has been declining approximately exponentially (each new campaign finds fewer new manuscripts), the total population $\hat{N}$ can be estimated with confidence intervals. Preliminary informal estimates suggest $\hat{N}$ in the range 800–2000, implying a survival rate of 7.5–19%.

E.2.2 The Unseen Species Model

A more sophisticated approach borrows from biodiversity estimation. The Chao1 estimator, designed for species richness from abundance data, can be adapted:

$$\hat{N}_{\text{Chao1}} = N_{\text{obs}} + \frac{f_1^2}{2f_2}$$

where $N_{\text{obs}} = 150$ (observed manuscripts), f_1 = number of manuscripts known from only one cataloguing campaign (“singletons”), and f_2 = number known from exactly two campaigns (“doubletons”).

This estimator is robust to heterogeneous detection probabilities—exactly the situation with manuscripts, where survival probability varies with institutional setting, material quality, and geographic location.

E.3 Monte Carlo Simulation of Manuscript Survival

E.3.1 The Generative Model

We propose a Monte Carlo simulation that generates a synthetic population of N bestiary manuscripts, subjects them to a multi-factor survival process, and compares the properties of the surviving sample with the properties of the observed corpus.

Step 1: Generate the initial population.

- Assign each manuscript i a production date $t_i \sim$ distribution calibrated to historical estimates of manuscript production rates (peak: 1170–1280 CE for English bestiaries).
- Assign a textual family $F_i \in \{\text{BIs, 2A, 2B, 2C, 2D, 3F, other}\}$ with probabilities proportional to observed family frequencies.
- Assign a complexity level $C_i \in \{C_1, C_2, C_3, C_4\}$ determined by family assignment.
- Assign an institutional origin $O_i \in \{\text{monastic, cathedral, secular, mendicant}\}$ with century-dependent probabilities.
- Assign material quality $Q_i \sim \text{Beta}(\alpha_Q, \beta_Q)$, reflecting variation in parchment quality, binding, and storage conditions.

Step 2: Apply survival hazards. Each manuscript faces multiple hazards at each time step:

$$p_{\text{survive}}(i, t) = p_{\text{material}}(Q_i, t) \cdot p_{\text{institutional}}(O_i, t) \cdot p_{\text{use}}(C_i, t) \cdot p_{\text{catastrophe}}(t)$$

where:

- p_{material} depends on parchment quality and age (exponential decay with quality-dependent rate).
- $p_{\text{institutional}}$ depends on the survival of the host institution (monastic libraries suffered massive losses during the Dissolution of 1536–1541; cathedral libraries survived better).

- p_{use} depends on complexity level: more complex manuscripts (C_3 , C_4) may have higher use-related wear but also higher perceived value, creating opposing effects.
- $p_{\text{catastrophe}}$ captures discrete events: the Dissolution, the English Civil War, the Cotton Library fire of 1731.

Step 3: Observe the surviving sample. After running the survival process to the present, record the properties of surviving manuscripts and compare with the observed corpus.

E.3.2 Calibration Targets

The simulation is calibrated to match the following observed properties of the 150 surviving manuscripts:

1. **Family distribution:** Approximately 60% Second Family, 25% Third Family, 10% BIs, 5% other.
2. **Date distribution:** Peak production 1170–1280, with tails in the 12th and 14th centuries.
3. **Geographic distribution:** Overwhelmingly English, with a smaller number of Continental manuscripts.
4. **Institutional distribution:** Most surviving manuscripts are associated with monastic or cathedral libraries.
5. **Complexity distribution:** The observed $C_1:C_2:C_3:C_4$ ratio.

E.3.3 The Key Question: Anomaly or Regularity?

The Monte Carlo simulation answers a fundamental question: *is the surviving corpus an anomalous subset of the original population, or does it represent a statistically expected outcome of the survival process?*

Specifically, we test three hypotheses:

H_1 : Uniform survival. All manuscripts had equal survival probability regardless of family, complexity, or institutional origin. If the surviving corpus matches the predicted distribution under uniform survival, then the observed family ratios reflect production ratios.

H_2 : Quality-biased survival. More complex manuscripts (C_3 , C_4) had higher survival probability because they were perceived as more valuable and therefore better preserved. If confirmed, the observed corpus *overrepresents* complex manuscripts and *underrepresents* simple ones.

H_3 : Institution-biased survival. Manuscripts in institutional libraries (monastic, cathedral) survived at higher rates than those in private or secular hands. If confirmed, the observed corpus overrepresents institutional production.

Each hypothesis generates distinct predictions about the properties of as-yet-undiscovered manuscripts (Section E.4).

E.4 Falsifiable Predictions

The formal framework generates the following predictions, each testable against future manuscript discoveries or against existing data that has not yet been systematically analyzed.

E.4.1 Predictions About Undiscovered Manuscripts

Prediction 1: Family distribution of new discoveries. If H_2 (quality-biased survival) is correct, then newly discovered manuscripts should be disproportionately from the simpler families (BIs, Second A/B) rather than the Third Family. The formal argument: C_1 manuscripts were perceived as less valuable and therefore less carefully preserved, making them more likely to be the undiscovered remainder. Conversely, if H_1 (uniform survival) is correct, new discoveries should approximately match the existing family distribution.

Test: Track the family affiliations of the next 10–20 manuscript discoveries. A statistically significant overrepresentation of simple families would support H_2 .

Prediction 2: Existence of undiscovered C_4 manuscripts. Currently, KK 4.25 is the only known C_4 manuscript. The formal framework predicts that the correction operator $\mathcal{R}$ should have been triggered independently in other scriptoria with access to multiple exemplars. Therefore, additional C_4 manuscripts should exist (or have existed). If no further C_4 manuscripts are ever discovered despite extensive searching, this would weaken the claim that $\mathcal{R}$ is a general adaptive operation.

Test: Systematic re-examination of manuscripts currently classified as “hybrid” or “contaminated” in stemmatic analyses, using the formal criteria for $\mathcal{R}$ (ρ -deficit in specific sections, import from identifiable external sources, preservation of kernel identity).

Prediction 3: Moralization density regularities. The main paper argues that the correction operator is triggered by $\rho(A) < \rho^*$. This predicts that in the surviving corpus, sections with low ρ should show higher rates of cross-family borrowing than sections with high ρ . The prediction is that ρ and borrowing rate are negatively correlated across sections and manuscripts.

Test: Systematic computation of ρ per section across all classified manuscripts, correlated with documented cross-family borrowing.

Prediction 4: Convergent verse production. The compression operator $\mathcal{C}$ was deployed in Bodley 533 and confirmed independently in the Cambridge manuscript. The framework predicts that mnemonic versification of bestiary material should have occurred independently in other contexts. Unidentified verse bestiaries or bestiary verse fragments may exist in unexamined manuscripts or in misattributed sections of other texts.

Test: Search for Leonine hexameter verses with bestiary-related vocabulary in uncatalogued manuscript collections, particularly in fragments and flyleaves.

E.4.2 Predictions About the Survival Process

Prediction 5: Institutional survival gradient. Manuscripts from institutions that survived the Dissolution (cathedrals, colleges) should be overrepresented in the surviving corpus relative to their share of original production. Manuscripts from dissolved monasteries should be underrepresented.

Test: Compare the institutional provenance distribution of surviving bestiaries with the known distribution of medieval English scriptoria.

Prediction 6: The “use-wear paradox.” The most pedagogically useful manuscripts (those at C_3 with *distinctiones*, used heavily by preachers) should show the highest physical wear but also the highest survival rate. This paradox arises because high use both damages the manuscript and signals its value, prompting repair and preservation.

Test: Correlate physical condition (wear patterns, repair evidence) with complexity level and institutional provenance.

Prediction 7: Geographic clustering of losses. If $p_{\text{catastrophe}}$ is the dominant survival factor, then manuscript losses should be geographically clustered around known catastrophic events

(specific monastery dissolutions, fires, floods). The Monte Carlo simulation should reproduce the observed geographic distribution only when catastrophe effects are included.

Test: Map surviving manuscripts geographically and test whether gaps correspond to known catastrophic loss events.

E.4.3 Predictions About the Formal Framework

Prediction 8: The Persistence Triad in other genres. The main paper claims that the Persistence Triad is a necessary condition for genre persistence. This predicts that other long-lived pedagogical genres (Aesopian fable, catechism, *ars dictaminis*, Jātaka tales) should satisfy all three conditions, while shorter-lived genres should fail at least one.

Test: Apply the diagnostic scoring instrument (Model 12 in the main paper) to 20–30 historically documented pedagogical genres. Compute the correlation between score and documented lifespan.

Prediction 9: Cross-cultural kernel invariance. The kernel $\mathcal{K}$ (observation $\rightarrow$ description $\rightarrow$ interpretation) should be identifiable in non-Western zoomorphic traditions (*Pañcatantra*, *Kalīla wa-Dimna*, Jātaka, *Shanhaijing*), with culturally specific instantiations of D and μ but identical formal structure.

Test: Formal analysis of non-Western zoomorphic traditions using the vocabulary of the main paper, checking whether the kernel triple is preserved.

Prediction 10: The Formalization Laws in the surviving corpus. The Law of Plurality and Incompleteness predicts that no single bestiary family should exhaust the interpretive potential of any animal. Therefore, for any given species s , the union $\bigcup_F \mu_F^*(s)$ across all families F should be strictly larger than any individual $\mu_F^*(s)$.

Test: Compute $|\bigcup_F \mu_F^*(s)|$ vs. $\max_F |\mu_F^*(s)|$ for all species. The prediction is that the ratio exceeds 1 for all species with more than one attested moralization.

E.5 Monte Carlo Results: What to Expect

While the full Monte Carlo simulation requires the data collection described above, the formal framework permits several *a priori* predictions about what the simulation will reveal:

E.5.1 The Missing Manuscripts Are Mostly Simple

Under any reasonable survival model incorporating quality bias (H_2), the simulation should predict that the “missing” manuscripts (those that once existed but do not survive) are predominantly from the simpler families (C_1 , C_2). This is because:

- C_1 manuscripts were small, inexpensive, and produced for individual or classroom use. They were less likely to be bound into permanent volumes or deposited in institutional libraries.
- C_3 and C_4 manuscripts were large, expensive, often illustrated, and produced for institutional libraries. They were more likely to be preserved, repaired, and catalogued.

If this prediction is confirmed, the observed predominance of Second and Third Family manuscripts in the surviving corpus is a survival artifact, not a production ratio. The original production was likely more evenly distributed, with a larger proportion of simple C_1 texts than the surviving corpus suggests.

E.5.2 The “Dark Matter” of the Tradition

By analogy with cosmological dark matter—mass that must exist to account for observed gravitational effects but cannot be directly observed—the Monte Carlo simulation should identify a population of “dark manuscripts” whose former existence can be inferred from structural regularities in the surviving corpus.

Specifically: if the surviving corpus shows discontinuities in the family distribution (sudden jumps or gaps in the genealogical stemma), these discontinuities may be explained by the loss of intermediate manuscripts that once connected the surviving witnesses. The Monte Carlo simulation can test whether the observed stemmatic structure is consistent with uniform loss or requires structured loss (loss concentrated in specific families, periods, or regions).

E.5.3 The Discovery Frontier

The simulation should estimate the probability that additional manuscripts remain to be discovered, and predict their most likely properties:

- **Most likely family:** BIs or Second A (simple, small, easily overlooked).
- **Most likely location:** Continental European libraries (English collections have been more thoroughly catalogued), or as fragments bound into later codices.
- **Most likely condition:** Fragmentary (complete manuscripts are more likely to have been already identified).
- **Most likely complexity:** C_1 or C_2 (simple texts were more likely to escape cataloguing attention).

These predictions constitute a concrete guide for future manuscript discovery efforts: they tell researchers where to look, what to look for, and what properties the discoveries should have if the formal framework is correct.

E.6 Concluding Remarks

The companion paper demonstrates that the formal apparatus of “Adaptive Pedagogical Systems with Invariant Kernels” is not merely a descriptive vocabulary but a *predictive engine*. The Persistence Triad, the operators $\mathcal{C}/\mathcal{E}/\mathcal{R}$, the complexity scale C_1 – C_4 , and the moralization density ρ generate at least ten specific, falsifiable predictions about the manuscript tradition, the survival process, and the cross-cultural generality of the formal framework. The Monte Carlo methodology provides a computational instrument for testing these predictions against the empirical record.

The most provocative prediction is also the most practically useful: the simulation should generate a probability distribution over the properties of undiscovered manuscripts, converting the question “are there more bestiaries out there?” from an unanswerable speculation into a quantitative estimate with confidence intervals. If the estimate is positive—and preliminary reasoning suggests it should be—then the formal framework will have contributed not only to the theoretical understanding of the bestiary tradition but to the practical program of manuscript discovery.

The 150 surviving manuscripts are not the tradition. They are the visible part of a larger structure whose full extent can be estimated, whose properties can be predicted, and whose missing members can, in principle, be found.